Improved Statistical Inference for Expectile-Based Risk Measures

An Optimal Pooling Approach in the Heavy-Tailed Regime

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Abstract

[**TODO:** Write a $\sim$200-word abstract once Chapter 3 is in shape. Outline of the abstract: (i) expectiles as the only coherent and elicitable law-invariant risk measure; (ii) the problem of combining tail-index and extreme-quantile information from several samples (distributed inference, multi-study evidence synthesis); (iii) our pooled extreme-expectile estimator, its joint CLT via plug-in into the expectile–quantile asymptotic equivalence, and the variance-/AMSE-optimal weights tailored to the expectile target; (iv) Weissman-style extrapolation to very-extreme expectile levels; (v) outlook on relaxing the iid assumption.]

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Chapter 1

Introduction

1.1 Motivation

Quantitative risk management routinely needs to summarise the tail of a loss distribution by a single number. The industry-standard summary has long been the Value-at-Risk (VaR), defined as a high quantile of the loss distribution. VaR is intuitive and easy to compute, but it suffers from two well-known shortcomings that have shaped the search for alternatives. First, VaR is not subadditive: it can deem a diversified portfolio more risky than its components, contradicting the diversification principle and disqualifying VaR from the class of *coherent* risk measures in the sense of Artzner et al. (1999). Expected Shortfall (ES) corrects this defect and has progressively replaced VaR in regulatory practice. However, ES is not *elicitable* (Gneiting 2011): no scoring function has its expectation uniquely minimised at the true ES, so ES forecasts cannot be ranked or backtested directly.

Expectiles are the only law-invariant risk measure that is simultaneously coherent (for confidence levels $\tau \geq 1/2$) and elicitable (Bellini et al. 2014). Originally introduced by Newey and Powell (1987), the τ-expectile of a random loss X is defined as the minimiser of an asymmetric squared loss:

$$\xi_\tau(X) \;=\; \underset{\theta\in\mathbb{R}}{\arg\min}\; \mathbb{E}\Big[\, |\tau - \mathbb{1}\{X \leq \theta\}|\; (X-\theta)^2 \,\Big]\,.$$

Setting $\tau = 1/2$ recovers the mean; setting $\tau \to 1$ moves the expectile into the right tail. Because the asymmetric loss is squared rather than absolute, expectiles — unlike quantiles — are sensitive to the *magnitude* of tail losses, not just to their frequency. Ehm et al. (2016) place expectiles within a broader theory of consistent scoring functions and Choquet representations.

This thesis addresses inference for expectiles at levels τ close to one, in two situations that are increasingly relevant in practice:

(i) the loss distribution is heavy-tailed, in the sense of having an extreme-value index $\gamma > 0$, so that the empirical asymmetric-least-squares estimator becomes unstable as τ approaches one and extreme-value theory becomes the natural framework;

(ii) the data are spread across several independent samples — whether because of distributed computation, regulatory data-sharing restrictions, or natural heterogeneity between studies — and inference must be conducted by communicating only summary statistics between sources.

1.2 Extreme expectile inference

The central bridge between expectile estimation and classical extreme-value theory is the *expectile–quantile asymptotic equivalence*. Under heavy-tailed assumptions with extreme-value index $\gamma \in (0,1)$ and finite first moment,

$$\frac{\xi_\tau(X)}{Q_X(\tau)} \xrightarrow{\tau \uparrow 1} \psi(\gamma) := (\gamma^{-1} - 1)^{-\gamma}, \tag{1.1}$$

where $Q_X(\tau)$ is the τ-quantile of X. Two families of estimators arise naturally from (1.1). The *asymmetric-least-squares* (LAWS) estimator minimises the empirical asymmetric criterion over the data; the *quantile-based* (QB) estimator substitutes consistent estimators of γ and of an extreme quantile of X into the asymptotic-equivalence formula. Their asymptotic theory at intermediate levels and via Weissman-type extrapolation to very-extreme levels was developed for iid heavy-tailed data by Daouia, Girard, et al. (2018) and subsequent work, and extended to the dependent setting by Davison et al. (2023); to the multivariate setting by Padoan and Stupfler (2022). The relevant machinery is reviewed in Chapter 2.

1.3 Pooled inference across samples

When m samples are available, with heterogeneous sizes and possibly heterogeneous tails, the natural question is how best to combine them. Daouia, Padoan, et al. (2024) develop an optimal pooling framework for the tail index γ and for extreme quantiles. Their estimators take the form of weighted arithmetic combinations (for the tail index) and weighted *geometric* combinations (for extreme quantiles), the latter specifically adapted to the multiplicative-power structure of the Weissman extrapolation. They derive the joint asymptotic normality of the per-sample estimators across m samples, allowing for asymptotic dependence between samples driven by a tail-copula, and identify the variance-optimal and AMSE-optimal pooling weights in closed form. In the important special case of *distributed inference* — where data are iid both within and across machines — the cross-sample tail-copula vanishes, off-diagonal entries of the joint asymptotic covariance collapse to zero, and the pooled estimators recover the asymptotic efficiency of the unfeasible Hill / Weissman estimator built from the combined sample.

The framework of Daouia, Padoan, et al. (2024) is the methodological foundation of this thesis. The framework does not address expectiles; the question of how to combine m pairs $(\hat\gamma_j, \hat Q_j(p_j))$ into a coherent inference procedure for an extreme expectile ξ_{τ_n} has not been studied.

1.4 Research question and scope

The thesis answers the following question:

> *Given m independent samples of iid heavy-tailed observations, with each sample j contributing point estimates $(\hat{\gamma}_j, \hat{Q}_j(p_j))$ together with their joint asymptotic covariance, how should one combine these summary statistics to perform inference and Weissman-style extrapolation for an extreme expectile of the common underlying distribution?*

The scope is restricted as follows:

- the data within each sample are univariate iid (no time-series dependence), matching the independent common-distribution distributed setting of Daouia, Padoan, et al. (2024);
- the loss distribution is heavy-tailed with extreme-value index $\gamma \in (0, 1)$ and finite first moment, the natural range for the quantile-based expectile–quantile bridge used by the thesis; the stronger finite-variance range $\gamma \in (0, 1/2)$ is invoked only when direct LAWS expectile estimators are discussed for comparison;
- the input to the pooling procedure is summary statistics, not raw data, in line with practical distributed-inference constraints.

1.5 Contributions

1. We define a *pooled extreme-expectile estimator* at intermediate levels by feeding the pooled tail-index estimator $\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)$ and the pooled weighted-geometric Weissman extreme-quantile estimator $\widehat{q}_n^{\star}(\,\cdot\,; \boldsymbol{\omega}_q)$ of Daouia, Padoan, et al. (2024) into the asymptotic-equivalence formula (1.1).
2. We derive the joint asymptotic distribution of the pooled inputs, accounting for the cross-source structure (which is diagonal because the samples are independent) and for the within-source dependence between $\hat{\gamma}_j$ and $\hat{Q}_j$ induced by the use of overlapping top order statistics.
3. Applying the delta method through ψ to this joint pooled CLT, we obtain the asymptotic normality of the pooled intermediate-level expectile estimator together with explicit forms of its asymptotic bias and variance.
4. We characterise the *variance-optimal* and *AMSE-optimal* pooling weights tailored to the pooled expectile estimator. These need not coincide with the weights optimal for the tail index or for the extreme quantile alone, owing to the non-trivial dependence between the two pieces of information.
5. We extend the construction to very-extreme levels $\tau_n' \uparrow 1$ with $n(1 - \tau_n') \to c < \infty$ via a Weissman-style extrapolation of the pooled intermediate-level estimator, and derive the corresponding central limit theorem by feeding the joint intermediate-level limit of the

previous contribution into the plug-in CLT of Davison et al. (2023, Proposition C.6). As in the single-sample case, the extrapolated estimator inherits the asymptotic variance of the pooled tail-index estimator.

6. We construct asymptotic confidence intervals for extreme expectiles and a test for *expectile-tail-homogeneity* across the m samples, modelled on the tail-quantile-homogeneity test of Daouia, Padoan, et al. (2024) and on the GLS-weighted equality-of-means test of Padoan and Stupfler (2022, Section 3.3).

1.6 Outline of the thesis

Chapter 2 collects the technical background: regular variation and the second-order condition, the Hill and Weissman estimators, expectiles and their characterisation, single-sample iid extreme-expectile inference, and the optimal pooling framework of Daouia, Padoan, et al. (2024). Chapter 3 develops the pooled estimator at intermediate expectile levels and the associated asymptotic theory. Chapter 4 treats the Weissman-style extrapolation to very-extreme expectile levels. Chapter 6 discusses limitations, the link to Expected Shortfall, and avenues for future work, including extensions to time-series dependence and multivariate observations. Detailed proofs are gathered in Appendix A.

Chapter 2

Background

Outline of the chapter

This chapter collects the technical ingredients on which the thesis builds: regular variation and the second-order condition; the Hill and Weissman estimators for the tail index and extreme quantiles; expectiles and their coherence and elicitability properties as risk measures; single-sample iid extreme-expectile inference, with the plug-in central limit theorem that the contribution chapters take as their workhorse; and the optimal pooling framework of Daouia, Padoan, et al. (2024). Nothing in this chapter is original; the contributions begin in Chapter 3.

2.1 Regular variation and the second-order condition

Throughout the thesis we work with a real-valued random loss X that has continuous distribution function F and survival function $\overline{F} = 1 - F$. Given an iid sample $X_1, \dots, X_n$ from F, the corresponding order statistics are denoted by $X_{1:n} \leq X_{2:n} \leq \cdots \leq X_{n:n}$.

Heavy-tailed asymptotics are governed by the language of regular variation; the reference is Haan and Ferreira (2006, Appendix B.1).

Definition 2.1 (Regular variation)**.** A Lebesgue measurable function $f : (0, \infty) \to \mathbb{R}$ that is eventually positive is *regularly varying at infinity with index* $\alpha \in \mathbb{R}$, written $f \in \mathrm{RV}_\alpha$, if for every $y > 0$,

$$\lim_{t \to \infty} \frac{f(ty)}{f(t)} \;=\; y^\alpha.$$

The functions in RV_0 are called *slowly varying*.

The first-order condition under which all of our asymptotics are taken asserts that F has a Pareto-type tail.

Condition 2.2 (First-order condition)**.** There exists $\gamma > 0$ such that $\overline{F} \in \mathrm{RV}_{-1/\gamma}$, or equivalently the *tail quantile function*

$$U(t) \;\coloneqq\; \inf\Big\{x \in \mathbb{R} \,:\, \tfrac{1}{\overline{F}(x)} \geq t\Big\}, \qquad t > 1,$$

satisfies $U \in \mathrm{RV}_\gamma$.

Throughout, for a continuous distribution function F, we write

$$Q_X(\tau) \;\coloneqq\; F^{\leftarrow}(\tau) = \inf\{x \in \mathbb{R} : F(x) \geq \tau\}, \qquad 0 < \tau < 1.$$

With the tail quantile $U(t) = F^{\leftarrow}(1 - 1/t)$, $t > 1$, this gives

$$Q_X(\tau) = U((1-\tau)^{-1}), \qquad Q_X(1-p) = U(1/p).$$

Condition 2.2 is equivalent to saying that X lies in the Fréchet max-domain of attraction with extreme-value index γ (Haan and Ferreira 2006, Theorem 1.2.1). The parameter $\gamma > 0$ is the *extreme-value index* of F; it controls how slowly $\overline{F}$ decays at infinity. Standard heavy-tailed distributions encountered in risk management satisfy Condition 2.2, including the Pareto, Burr, Student-t, Fréchet, and F-distribution families (Haan and Ferreira 2006, §2.6).

For the asymptotic-normality statements that drive the inferential theory we need a quantitative refinement of Condition 2.2 controlling the rate of convergence in Definition 2.1. The standard refinement is the second-order condition.

Condition 2.3 (Second-order condition $\mathcal{C}_2(\gamma, \rho, A)$)**.** There exist $\gamma > 0$, a second-order parameter $\rho \leq 0$, and a measurable auxiliary function A of constant sign with $A(t) \to 0$ as $t \to \infty$, such that for every $y > 0$,

$$\lim_{t\to\infty} \frac{1}{A(t)} \left[\frac{U(ty)}{U(t)} - y^\gamma\right] \;=\; y^\gamma \, \frac{y^\rho - 1}{\rho},$$

where the right-hand side is read as $y^\gamma \log y$ when $\rho = 0$.

The function $|A|$ is itself regularly varying with index ρ (Haan and Ferreira 2006, Theorem 2.3.9). Since the auxiliary function A has constant sign eventually, all signed ratios $A(ty)/A(t)$ used below have the same limit y^ρ. Smaller $|\rho|$ means slower convergence and therefore larger second-order bias of the tail-index estimators we shall encounter in Section 2.2.

Condition 2.4 (Finite first-moment condition)**.** The extreme-value index satisfies $\gamma \in (0, 1)$ and the negative part $X^- = \max(-X, 0)$ satisfies

$$\mathbb{E}\, X^- < \infty.$$

Under Condition 2.2, the condition $\gamma < 1$ controls the positive tail and gives $\mathbb{E}\, X^+ < \infty$; the displayed integrability condition controls the lower tail. Together they are the natural finite-mean regime in which expectiles are well defined and the quantile-based expectile approximation is available.

Condition 2.5 (LAWS lower-tail moment condition)**.** For some $\delta > 0$, the negative part $X^- = \max(-X, 0)$ satisfies

$$\mathbb{E}(X^-)^{2+\delta} < \infty.$$

Daouia, Girard, et al. (2018) write the same assumption as $\mathbb{E}\,|Y_-|^{2+\delta} < \infty$ with $Y_- = \min(Y, 0)$; here X^- denotes the nonnegative negative part.

This stronger condition is kept for direct LAWS expectile estimation, whose asymptotic normality is built from tail averages and requires finite-variance-type control. It is not part of the quantile-based plug-in route used in the contribution chapters.

2.2 Tail-index and extreme-quantile estimation

Under the first-order condition, the extreme-value index $\gamma > 0$ is the primary parameter determining the behaviour of F in the tail. We collect here the two classical estimators that the thesis builds on: the Hill estimator of γ and the Weissman extrapolation for extreme quantiles. Both are standard; the textbook reference is Haan and Ferreira (2006, Ch. 3–4).

2.2.1 The Hill estimator

Given an iid sample $X_1, \ldots, X_n$ from F and an integer $k = k_n \in \{1, \ldots, n-1\}$, the *Hill estimator* (Hill 1975) of γ based on the $k+1$ top order statistics is

$$\hat{\gamma}_n(k) \;=\; \frac{1}{k}\sum_{i=1}^{k} \log\left(\frac{X_{n-i+1\,:\,n}}{X_{n-k\,:\,n}}\right). \tag{2.1}$$

Since $U(t) \to \infty$ under Condition 2.2, the event $X_{n-k:n} > 0$ has probability tending to one for intermediate k. The log-based tail estimators in this section are understood on that event; equivalently one may assume that the upper tail is eventually supported on $(0, \infty)$, or, for tail-index estimation only, apply a deterministic shift, which leaves the extreme-value index unchanged. For quantile or expectile estimation the shift must be undone after extrapolation; the large-quantile ratio asymptotics are unaffected because the upper endpoint is infinite. The interpretation is direct: if the upper tail of F were exactly Pareto with index $1/\gamma$, the logarithms of excess ratios above a fixed tail threshold would be exponential with mean γ; in Hill's order-statistic derivation, this appears through the scaled consecutive log-spacings $i\{\log X_{n-i+1:n} - \log X_{n-i:n}\}$, which are independent exponential variables with mean γ in the exact Pareto case and whose average is (2.1). Condition 2.2 asserts that the upper tail is *approximately* Pareto. The consistency statement below is Haan and Ferreira (2006, Theorem 3.2.2); the normal limit is Haan and Ferreira (2006, Theorem 3.2.5) when $\rho < 0$, with the $\rho = 0$ endpoint in the same $\mathcal{C}_2$ convention covered by the $m = 1$ specialisation of Daouia, Padoan, et al. (2024, Theorem 1).

Theorem 2.6. *Assume $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma > 0$. Let $k = k_n \to \infty$ and $k/n \to 0$ as $n \to \infty$. Then $\hat{\gamma}_n(k) \xrightarrow{\mathbb{P}} \gamma$. If, in addition, $\sqrt{k}\,A(n/k) \to \lambda \in \mathbb{R}$, then*

$$\sqrt{k}\,(\hat{\gamma}_n(k) - \gamma) \;\xrightarrow{d}\; \mathcal{N}\left(\frac{\lambda}{1-\rho},\, \gamma^2\right).$$

The choice of k controls the well-known bias–variance trade-off: too small a k inflates the variance, while too large a k injects second-order bias through the term $\lambda/(1-\rho)$. Adaptive choice of k is a research topic in its own right; we do not pursue it in this thesis and treat k as fixed by the data analyst.

2.2.2 Weissman extrapolation to extreme quantiles

The *Weissman extrapolation* (Weissman 1978) extrapolates from the intermediate-order statistic $X_{n-k:n}$ — which estimates the quantile $Q_X(1-k/n)$ at the intermediate level $1-k/n$ — to a quantile at a very-extreme level $1-p$ with $p = p_n \to 0$ as $n \to \infty$. Plugging the Hill estimator into the first-order identity $U(ty)/U(t) \to y^\gamma$ gives the Weissman estimator

$$\widehat{q}_n^\star(1-p \mid k) \;=\; \left(\frac{k}{np}\right)^{\hat{\gamma}_n(k)} X_{n-k\,:\,n}. \tag{2.2}$$

For $p = c/n$ and fixed k, (2.2) reduces to the large-quantile estimator proposed by Weissman (1978, §4) in the Fréchet case, written there with tail parameter $\alpha = 1/\gamma$ and descending order statistics. The theorem below covers the far-tail regime $np = o(k)$. The especially severe very-extreme case has $np = O(1)$, where the expected number of observations beyond $Q_X(1-p)$ remains bounded; when $p < 1/n$, the target is beyond the usual sample-maximum scale. It is the Hill-estimator specialisation of Haan and Ferreira (2006, Theorem 4.3.8); since $\log k = o(\sqrt{k})$, the source condition $\log(np_n) = o(\sqrt{k})$ is equivalently written here as $\log(k/(np_n)) = o(\sqrt{k})$.

Theorem 2.7. *Assume $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma > 0$ and $\rho < 0$. Let $k = k_n \to \infty$, $k/n \to 0$, $p = p_n \to 0$ with $k/(np) \to \infty$ and $\sqrt{k}/\log(k/(np)) \to \infty$. Suppose $\sqrt{k}\,A(n/k) \to \lambda \in \mathbb{R}$. Then*

$$\frac{\sqrt{k}}{\log(k/(np))}\left(\frac{\widehat{q}_n^\star(1-p \mid k)}{Q_X(1-p)} - 1\right) \;\xrightarrow{d}\; \mathcal{N}\!\left(\frac{\lambda}{1-\rho},\, \gamma^2\right).$$

The rate $\sqrt{k}/\log(k/(np))$ reflects the cost of extrapolation: the further into the tail one extrapolates, the larger the log-denominator and the slower the rate. The asymptotic variance is identical to that of the Hill estimator (Theorem 2.6); the extrapolation contributes only through the multiplicative log-scale factor. This is the structural fact that Daouia, Padoan, et al. (2024, §2.3) exploits to motivate the weighted-*geometric* pooling of Weissman estimators — rather than the arithmetic pooling natural for centred quantities — and that we shall in turn exploit in Chapter 4 for the extrapolation of pooled expectiles.

2.3 Expectiles

This section collects the definition and the basic properties of expectiles that we shall use in the rest of the thesis. Expectiles were introduced by Newey and Powell (1987) as an asymmetric-least-squares analogue of quantiles. Their role as risk measures, in particular their coherence properties, was clarified by Bellini et al. (2014) and Ziegel (2016); their elicitability and scoring-function representations are treated in Gneiting (2011) and Ehm et al. (2016).

2.3.1 Definition

Let X be a real-valued random variable with $\mathbb{E}\,|X| < \infty$, and fix a level $\tau \in (0,1)$. The τ-*expectile* of X is defined as

$$\xi_\tau(X) \;:=\; \operatorname*{arg\,min}_{\theta\in\mathbb{R}} \;\mathbb{E}[\,\eta_\tau(X-\theta) - \eta_\tau(X)\,], \qquad \eta_\tau(u) := |\tau - \mathbb{1}\{u \le 0\}|\; u^2. \tag{2.3}$$

The asymmetric squared check function η_τ penalises positive and negative deviations from θ with the asymmetric weights τ and $1-\tau$, respectively. Recovering the mean in the symmetric case is immediate: $\eta_{1/2}(u) = u^2/2$, so $\xi_{1/2}(X) = \mathbb{E}\,X$. The subtraction of $\eta_\tau(X)$ in (2.3) is a centering device that makes the criterion finite under the first-moment condition $\mathbb{E}\,|X| < \infty$; it does not affect the minimiser in θ.

The convex objective (2.3) has a unique minimiser, and its first-order condition reduces to the implicit equation

$$\tau\;\mathbb{E}[(X - \xi_\tau(X))_+] \;=\; (1-\tau)\;\mathbb{E}[(\xi_\tau(X) - X)_+], \tag{2.4}$$

where $u_+ = \max(u, 0)$. Thus $\xi_\tau(X)$ is the unique point where the upper and lower first partial moments, with asymmetric weights, balance. For a strictly increasing continuous distribution, $Q_X(\tau)$ is equivalently characterised by $F(Q_X(\tau)) = \tau$. This can be written in the formally balanced form

$$\tau\;\mathbb{P}(X > Q_X(\tau)) = (1-\tau)\;\mathbb{P}(X < Q_X(\tau)),$$

but, unlike (2.4), this identity expresses a probability balance rather than a balance of tail magnitudes.

2.3.2 Basic properties

The following properties are immediate from (2.3) or (2.4); we collect them without proof and refer to Newey and Powell (1987, §2) and Bellini et al. (2014, Sections 2–3) for details.

Proposition 2.8. *Let X and Y be real-valued random variables with $\mathbb{E}\,|X|, \mathbb{E}\,|Y| < \infty$. For each $\tau \in (0,1)$:*

(i) ***Existence and uniqueness.*** *$\xi_\tau(X)$ exists and is uniquely defined.*

(ii) ***Law invariance.*** *If X and Y have the same distribution, then $\xi_\tau(X) = \xi_\tau(Y)$.*

(iii) ***Mean recovery.*** *$\xi_{1/2}(X) = \mathbb{E}\,X$.*

(iv) ***Location equivariance.*** *$\xi_\tau(X + c) = \xi_\tau(X) + c$ for $c \in \mathbb{R}$.*

(v) ***Positive homogeneity.*** *$\xi_\tau(\lambda X) = \lambda\,\xi_\tau(X)$ for $\lambda \ge 0$.*

(vi) ***Sign reflection.*** *$\xi_\tau(-X) = -\xi_{1-\tau}(X)$.*

(vii) ***Monotonicity in τ.*** *The map $\tau \mapsto \xi_\tau(X)$ is nondecreasing on $(0,1)$, and is strictly increasing when X is nondegenerate, with the boundary limits $\xi_\tau(X) \to \operatorname{ess\,inf}(X)$ as $\tau \to 0$ and $\xi_\tau(X) \to \operatorname{ess\,sup}(X)$ as $\tau \to 1$ (infinite limits allowed).*

(viii) ***Monotonicity in the argument.*** *If $X \le Y$ almost surely, then $\xi_\tau(X) \le \xi_\tau(Y)$.*

For losses, expectiles are translation equivariant, positively homogeneous, and monotone. For positions, the induced risk functional $\rho_\tau(Y) = \xi_\tau(-Y)$ is cash-additive with the usual sign convention $\rho_\tau(Y + c) = \rho_\tau(Y) - c$ and is monotone decreasing in positions. The remaining coherence axiom in Artzner et al. (1999) is subadditivity, to which we turn next.

2.3.3 Coherence and elicitability

In Sections 2.1, 2.2 and 2.4 and in the contribution chapters, X denotes a loss and high risk is represented by the upper expectile $\xi_\tau(X)$ with $\tau \uparrow 1$. The coherence literature is often written for financial positions, where positive values are gains and larger values are better. In this subsection only, write Y for such a position and define the capital requirement

$$\rho_\tau(Y) \;=\; \xi_\tau(-Y).$$

Subadditivity of ρ_τ requires $\rho_\tau(Y + Z) \le \rho_\tau(Y) + \rho_\tau(Z)$ for all positions Y, Z.

Theorem 2.9 (Bellini et al. (2014))**.** *The functional $\rho_\tau(Y) = \xi_\tau(-Y)$ is a coherent risk measure if and only if $\tau \ge 1/2$.*

Recall that a real-valued functional T on a class of distributions is *elicitable* (Gneiting 2011) if there exists a strictly consistent scoring function $S : \mathbb{R} \times \mathbb{R} \to [0, \infty)$ whose expected score is uniquely minimised by reporting $T(P)$ for every P in the class. Gneiting (2011, §3.3) shows that the τ-expectile is elicitable on classes with finite first moment; the familiar asymmetric squared score $S_\tau(\theta, x) = \eta_\tau(x - \theta)$ is strictly consistent on classes with finite second moment, while the general characterisation uses Bregman-type scores with the corresponding integrability conditions. Thus competing expectile forecasts can be compared by empirical averages of strictly consistent scores. Taken together with Theorem 2.9, this places expectiles in a privileged position among risk measures. On suitable convex classes of distributions and under the usual monetary-risk-measure convention, Bellini et al. show that, within the class of M-quantiles, the coherent risk measures are exactly the expectiles, while Ziegel (2016) characterises elicitable law-invariant coherent risk measures as certain expectiles, with the expectation appearing as the special case.

2.3.4 Choquet representation

Ehm et al. (2016) place expectiles within a broader theory of consistent scoring functions. They show that regular scoring functions consistent for expectiles admit a mixture representation over elementary scores, analogous in spirit to the corresponding representation for quantile scores. After a normalisation that removes trivial dilations of the scoring class, these elementary scores are extreme points, yielding the advertised Choquet-type representation. This is relevant for forecast comparison and backtesting, but it is not used in the asymptotic theory below; all subsequent arguments rely only on the expectile first-order condition and the extreme expectile–quantile equivalence.

2.4 Extreme expectile inference: the iid heavy-tailed baseline

We now restrict to a heavy-tailed loss X satisfying Condition 2.2 with extreme-value index $\gamma \in (0,1)$, and consider inference for ξ_τ at levels τ approaching one. We distinguish, throughout, between two asymptotic regimes:

- *intermediate-level*: $\tau_n \to 1$ with $n(1-\tau_n) \to \infty$. There are still many observations above $Q_X(\tau_n)$, and central limit theorems hold at the rate $\sqrt{n(1-\tau_n)}$.
- *extreme-level*: $\tau'_n \to 1$ with $n(1-\tau'_n) \to c \in [0,\infty)$. There are at most finitely many observations above $Q_X(\tau'_n)$, and direct estimation is infeasible; extrapolation is required.

The results in this section are due, in the iid setting, to Daouia, Girard, et al. (2018), Daouia, Girard, et al. (2019), and Daouia, Girard, et al. (2020); the time-series extension that we have used to organise the exposition is Davison et al. (2023), and we shall adapt Proposition C.6 of that paper's supplement into the abstract plug-in lemma that the contribution chapters use.

2.4.1 The expectile–quantile asymptotic equivalence

The single most important fact for extreme expectile inference is that, in the heavy-tailed regime, the expectile is asymptotically proportional to the quantile of the same level.

Proposition 2.10 (Asymptotic equivalence; Bellini et al. (2014), Daouia, Girard, et al. (2019), Corollary 1 with $p = 2$)**.** *Suppose X satisfies Condition 2.2 with index $\gamma \in (0,1)$ and $\mathbb{E}\, X^- < \infty$; equivalently, suppose Condition 2.4 holds. Then*

$$\frac{\xi_\tau(X)}{Q_X(\tau)} \xrightarrow{\tau\uparrow 1} \psi(\gamma) \;:=\; (\gamma^{-1}-1)^{-\gamma}. \tag{2.5}$$

The factor $\psi(\gamma)$ is the linchpin of all extreme-expectile inference: it converts asymptotic statements about extreme quantiles into asymptotic statements about extreme expectiles, and vice versa. The first-order equivalence in Proposition 2.10 requires $\gamma < 1$ together with the integrability of the lower tail. The stronger range $\gamma < 1/2$ is not needed for this bridge; it enters only when one uses direct LAWS expectile CLTs. The function ψ is continuous on $(0,1)$, with $\psi(\gamma) \to 1$ as $\gamma \downarrow 0$ and $\psi(\gamma) \to \infty$ as $\gamma \uparrow 1$. It is *not* monotone: its logarithmic derivative

$$m(\gamma) \;:=\; \frac{d}{d\gamma} \log \psi(\gamma) \;=\; \frac{1}{1-\gamma} - \log\left(\gamma^{-1}-1\right) \tag{2.6}$$

changes sign at the value $\gamma^\star \approx 0.2178$ solving $m(\gamma^\star) = 0$, so ψ decreases on $(0,\gamma^\star)$ to a minimum $\psi(\gamma^\star) \approx 0.757$ and increases on $(\gamma^\star, 1)$; in particular it returns to $\psi = 1$ at $\gamma = 1/2$. Only for tail indices beyond $\gamma^\star$ does a heavier tail move the expectile closer to the same-level quantile; for $\gamma > 1/2$, the factor $\psi(\gamma)$ is larger than one.

Under the second-order condition $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma \in (0,1)$ and $\mathbb{E}\, X^- < \infty$, the first-order equivalence (2.5) sharpens to the second-order expansion (Daouia, Girard, et al. 2020, Proposition 1(i))

$$\frac{\xi_\tau(X)}{Q_X(\tau)} = \psi(\gamma)\Big[1 + c(\gamma,\rho)\, A((1-\tau)^{-1}) + \gamma(\gamma^{-1}-1)^{\gamma}\frac{\mathbb{E}\, X}{Q_X(\tau)} \\ + o(A((1-\tau)^{-1})) + o(Q_X(\tau)^{-1})\Big], \tag{2.7}$$

$$c(\gamma,\rho) := \frac{(\gamma^{-1}-1)^{-\rho}}{1-\gamma-\rho} + \frac{(\gamma^{-1}-1)^{-\rho}-1}{\rho}.$$

The displayed form is the version used in the thesis: it combines the second-order tail expansion with finite first-moment control of the non-tail contribution to the expectile equation. At the endpoint $\rho = 0$, the second term in $c(\gamma,\rho)$ is read by continuity as $\rho \to 0$, so that $c(\gamma, 0) = m(\gamma)$. The constant $c(\gamma,\rho)$ measures the deterministic gap between the extreme expectile and the same-level quantile beyond the leading factor $\psi(\gamma)$. The term $\gamma(\gamma^{-1}-1)^{\gamma}\,\mathbb{E}\, X/Q_X(\tau)$ is the explicit first-moment contribution in the expansion of Daouia, Girard, et al. (2020); it is often abbreviated below as an $O(Q_X(\tau)^{-1})$ contribution, because $\mathbb{E}\, X^- < \infty$ and $\gamma < 1$ ensure that it is well controlled. Under the moment condition $\sqrt{n(1-\tau_n)}\,/Q_X(\tau_n) \to 0$ adopted below it is $\sqrt{n(1-\tau_n)}$-negligible, so the A-driven term alone governs the asymptotic bias of the quantile-based estimator. This source is the version used here because it is stated under the $\rho \le 0$ convention of Condition 2.3; the earlier Daouia, Girard, et al. (2018, Corollary 1) gives the same constant under its stricter $\rho < 0$ formulation.

2.4.2 Two families of estimators at intermediate levels

Two natural estimators of ξ_{τ_n} at an intermediate level τ_n arise from the asymmetric-LS definition (2.3) on the one hand, and from the asymptotic equivalence (2.5) on the other.

LAWS direct estimator. The *asymmetric-least-squares* (LAWS) estimator replaces the population minimisation (2.3) with its empirical analogue:

$$\widetilde{\xi}_{\tau_n} \;=\; \arg\min_{\theta\in\mathbb{R}} \sum_{i=1}^{n} \eta_{\tau_n}(X_i - \theta). \tag{2.8}$$

The estimator $\widetilde{\xi}_{\tau_n}$ can be computed with standard expectile routines; Daouia, Girard, et al. (2018, §3.1.2) use the `expectile` function of the `expectreg` package.

Theorem 2.11 (Intermediate LAWS CLT; Daouia, Girard, et al. (2018), Theorem 2)**.** *Assume Condition 2.2 with $\gamma \in (0, 1/2)$, and assume Condition 2.5. Let $\tau_n \to 1$ with $n(1-\tau_n) \to \infty$. Then*

$$\sqrt{n(1-\tau_n)}\left(\frac{\widetilde{\xi}_{\tau_n}}{\xi_{\tau_n}} - 1\right) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{2\gamma^3}{1-2\gamma}\right).$$

Quantile-based plug-in estimator. A second family substitutes estimators of γ and of the intermediate quantile $Q_X(\tau_n)$ into the asymptotic-equivalence formula (2.5):

$$\widehat{\xi}_{\tau_n} \;=\; \psi(\hat{\gamma}_n)\, \widehat{Q}(\tau_n), \tag{2.9}$$

where $\widehat{Q}(\tau_n) = X_{n-\lfloor n(1-\tau_n)\rfloor:n}$ is the empirical quantile (an intermediate-order statistic) and $\hat{\gamma}_n$ is a tail-index estimator admitting the required $\sqrt{n(1-\tau_n)}$ limit theory. The natural choice is the Hill estimator of (2.1) with effective sample size $k = \lfloor n(1-\tau_n)\rfloor$.

Theorem 2.12 (Intermediate QB CLT; Daouia, Girard, et al. (2018), Corollary 2; Daouia, Girard, et al. (2020), Proposition 1)**.** *Assume $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma \in (0,1)$ and assume $\mathbb{E}\, X^- < \infty$. Let $\tau_n \to 1$ with $n(1-\tau_n) \to \infty$ and suppose*

$$\sqrt{n(1-\tau_n)}\, A\big((1-\tau_n)^{-1}\big) \to \lambda \in \mathbb{R}, \qquad \frac{\sqrt{n(1-\tau_n)}}{Q_X(\tau_n)} \to 0.$$

With $\hat{\gamma}_n$ the Hill estimator at effective sample size $k = \lfloor n(1-\tau_n)\rfloor$,

$$\sqrt{n(1-\tau_n)}\left(\frac{\widehat{\xi}_{\tau_n}}{\xi_{\tau_n}} - 1\right) \xrightarrow{d} \mathcal{N}\left(\Big(\frac{m(\gamma)}{1-\rho} - c(\gamma,\rho)\Big)\lambda,\ \gamma^2\big(1 + m(\gamma)^2\big)\right),$$

with $m(\gamma)$ as in (2.6) and $c(\gamma, \rho)$ the second-order expectile–quantile constant of (2.7). The asymptotic bias has two pieces: the Hill bias $\lambda/(1-\rho)$ propagated through the logarithmic derivative $m(\gamma)$ of ψ, and the deterministic expectile–quantile gap $-c(\gamma,\rho)\,\lambda$. The empirical quantile $\widehat{Q}(\tau_n)$ contributes no asymptotic mean term at the $\sqrt{n(1-\tau_n)}$ scale when centered at $Q_X(\tau_n)$; the condition $\sqrt{n(1-\tau_n)}\,/Q_X(\tau_n) \to 0$ inherited from the theorem ensures that the moment-term contribution to the expectile–quantile gap is negligible.

This is the zero-first-moment-bias specialisation of the iid quantile-based CLT of Daouia, Girard, et al. (2018, Corollary 2), combined with the second-order expectile–quantile expansion of Daouia, Girard, et al. (2020, Proposition 1). In the more general statement one allows

$$\lambda_2 = \lim_{n\to\infty} \frac{\sqrt{n(1-\tau_n)}}{Q_X(\tau_n)}.$$

In the general statement, the asymptotic mean contains an additional term of order λ_2 depending on the first moment of X. The present thesis imposes $\sqrt{n(1-\tau_n)}/Q_X(\tau_n) \to 0$ because this is the regime needed later to isolate the second-order tail bias. Padoan and Stupfler (2022, Corollary 2.3) states the analogous multivariate QB result under the same finite-mean range $0 < \gamma_j < 1$, and explicitly contrasts it with the stronger finite-variance assumptions required by direct LAWS estimation. This is why the contribution chapters impose only the finite-mean QB range $0 < \gamma < 1$: the restriction $\gamma < 1/2$ belongs to direct LAWS asymptotic normality, not to the quantile-based route. The older strict-monotonicity assumption in Daouia, Girard, et al. (2018) is not retained here; the generalised-inverse quantile convention used in Section 2.1 is sufficient for the QB formulation, in line with Daouia, Girard, et al. (2020, Proposition 1) and Padoan and Stupfler (2022, Corollary 2.3).

Comparison. The LAWS estimator has asymptotic variance $2\gamma^3/(1-2\gamma)$, which is independent of the choice of tail-index estimator. The QB estimator's variance $\gamma^2(1+m(\gamma)^2)$ depends on the tail-index estimator used and reflects the contribution of $\hat{\gamma}_n$ via the term $m(\gamma)^2$. Numerical comparison over $\gamma \in (0, 1/2)$ reveals that no single estimator dominates the other across the whole range: the two variances cross once, at $\gamma \approx 0.2623$. For lighter heavy tails, i.e. small γ, the LAWS variance is smaller. For sufficiently heavy tails, and especially as $\gamma \uparrow 1/2$, the QB variance is smaller because the LAWS variance diverges. The thesis works with the QB construction throughout not because it uniformly dominates LAWS, but because it composes directly with the pooled $(\hat{\gamma}, \hat{Q})$ inputs of the pooling framework. This comparison concerns direct intermediate-level expectile estimation; in the very-extreme extrapolation regime below, the first-order limit is instead driven by the tail-index estimator.

2.4.3 Weissman extrapolation to very-extreme expectile levels

For levels τ_n' so close to one that $n(1-\tau_n') \to c \in [0,\infty)$, neither $\widetilde{\xi}_{\tau_n'}$ nor $\widehat{\xi}_{\tau_n'}$ provides a consistent direct estimator of $\xi_{\tau_n'}$. Extrapolation from an intermediate level τ_n to the extreme level τ_n' is required.

The asymptotic equivalence (2.5) and the first-order identity $U(ty)/U(t) \to y^{\gamma}$ together yield the heuristic

$$\frac{\xi_{\tau_n'}}{\xi_{\tau_n}} \sim \frac{Q_X(\tau_n')}{Q_X(\tau_n)} \sim \left(\frac{1-\tau_n'}{1-\tau_n}\right)^{-\gamma},$$

which suggests the *Weissman-type* extrapolated estimator

$$\overline{\xi}^{\star}_{\tau_n'} = \left(\frac{1-\tau_n'}{1-\tau_n}\right)^{-\hat{\gamma}_n} \overline{\xi}_{\tau_n}, \tag{2.10}$$

where $\overline{\xi}_{\tau_n}$ is any consistent intermediate-level estimator satisfying a $\sqrt{n(1-\tau_n)}$-CLT, such as $\widetilde{\xi}_{\tau_n}$ or $\widehat{\xi}_{\tau_n}$ under the conditions above, and $\hat{\gamma}_n$ is a $\sqrt{n(1-\tau_n)}$-consistent tail-index estimator. We write $\tilde{\xi}^{\star}$ and $\hat{\xi}^{\star}$ for the LAWS-anchored and QB-anchored extrapolations, respectively.

Theorem 2.13 (Extreme-level CLT; Daouia, Girard, et al. (2018), Corollaries 3–4)**.** *Assume $\mathcal{C}_2(\gamma,\rho,A)$ with $\gamma \in (0,1)$ and $\rho < 0$, and assume $\mathbb{E}\, X^- < \infty$. Let $\tau_n \to 1$ satisfy $n(1-\tau_n) \to \infty$ and let $\tau_n' \to 1$ satisfy $n(1-\tau_n') \to c \in [0,\infty)$, $(1-\tau_n')/(1-\tau_n) \to 0$ and $\sqrt{n(1-\tau_n)}\,/\log((1-\tau_n)/(1-\tau_n')) \to \infty$. Suppose moreover that*

$$\sqrt{n(1-\tau_n)}\, A((1-\tau_n)^{-1}) \to \lambda \in \mathbb{R}, \qquad \frac{\sqrt{n(1-\tau_n)}}{Q_X(\tau_n)} \to 0.$$

If $\hat{\gamma}_n$ is the Hill estimator and $\overline{\xi}_{\tau_n}$ is the QB intermediate estimator of Theorem 2.12, then

$$\frac{\sqrt{n(1-\tau_n)}}{\log((1-\tau_n)/(1-\tau_n'))}\left(\frac{\overline{\xi}^{\star}_{\tau_n'}}{\xi_{\tau_n'}} - 1\right) \xrightarrow{d} \mathcal{N}\!\left(\frac{\lambda}{1-\rho},\, \gamma^2\right).$$

The same conclusion holds with $\overline{\xi}_{\tau_n}$ equal to the LAWS estimator of Theorem 2.11 under the additional direct-LAWS assumptions $\gamma \in (0,1/2)$ and Condition 2.5.

Two remarks are in order. First, the rate $\sqrt{n(1-\tau_n)}\,/\log((1-\tau_n)/(1-\tau_n'))$ is the classical Weissman rate of Theorem 2.7: the further into the tail one extrapolates, the larger the log-denominator and the slower the rate of convergence. Second, the asymptotic variance is γ^2 *regardless of whether the anchor estimator is LAWS or QB*: the contribution of $\overline{\xi}_{\tau_n}$ is dominated, at the extreme level, by that of $\hat{\gamma}_n$ entering the multiplicative power. This is the central structural feature that makes extreme-expectile inference tractable; at the extreme level, asymptotic inference for $\xi_{\tau_n'}$ reduces to asymptotic inference for the tail index.

2.4.4 The plug-in central limit theorem

Theorems 2.11 to 2.13 together specify a complete asymptotic theory for extreme expectiles in the iid heavy-tailed setting. For the pooled inference developed in Chapters 3 and 4, however, we require a more abstract formulation that decouples the asymptotic distribution of the inputs $(\hat{\gamma}_n, \overline{\xi}_{\tau_n})$ from the way in which they are obtained. We use the following thesis-level plug-in lemma, adapted from *Proposition C.6* in the supplement to Davison et al. (2023). The source proposition is stated at the natural rate $\sqrt{n(1-\tau_n)}$ under its high-level Condition C^*; the version below isolates the deterministic extrapolation error as an explicit assumption and writes the normalising rate as a generic sequence a_n, which is the form needed when the intermediate estimator is itself produced by pooling. Its short proof is given in Appendix A.8.

Proposition 2.14 (Plug-in CLT for extrapolated expectiles; adapted from Davison et al. (2023), Prop. C.6)**.** *Assume $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma \in (0,1)$ and $\rho < 0$, and assume $\mathbb{E}\, X^- < \infty$. Let $a_n \to \infty$, let $\hat{\gamma}_n$ be an estimator of γ, and let $\widehat{\xi}^I_{\tau_n}$ be an intermediate-level estimator of ξ_{τ_n} satisfying the joint asymptotic representation*

$$a_n \begin{pmatrix} \hat{\gamma}_n - \gamma \\ \widehat{\xi}^I_{\tau_n}/\xi_{\tau_n} - 1 \end{pmatrix} \xrightarrow{d} \begin{pmatrix} \Gamma \\ \Delta \end{pmatrix}, \tag{2.11}$$

for some random vector $(\Gamma, \Delta)^\top$. Let $\tau_n, \tau_n' \to 1$ with $n(1-\tau_n) \to \infty$, $n(1-\tau_n') \to c \in [0, \infty)$, $(1-\tau_n')/(1-\tau_n) \to 0$, put

$$\ell_n \;:=\; \log\!\left(\frac{1-\tau_n}{1-\tau_n'}\right),$$

and suppose

$$\frac{a_n}{\ell_n} \to \infty, \qquad a_n A((1-\tau_n)^{-1}) \to \lambda_1 \in \mathbb{R}, \qquad \frac{a_n}{Q_X(\tau_n)} \to \lambda_2 \in \mathbb{R}.$$

Assume moreover that $\mathbb{P}(\widehat{\xi}^I_{\tau_n} > 0) \to 1$ and that the deterministic extrapolation error satisfies

$$\frac{a_n}{\ell_n}\left\{\log \xi_{\tau_n'} - \log \xi_{\tau_n} - \gamma \ell_n\right\} \;\longrightarrow\; 0.$$

Then the Weissman-extrapolated expectile

$$\widehat{\xi}^\star_{\tau_n'} = \left(\frac{1-\tau_n'}{1-\tau_n}\right)^{-\hat{\gamma}_n} \widehat{\xi}^I_{\tau_n}$$

satisfies

$$\frac{a_n}{\log((1-\tau_n)/(1-\tau_n'))}\left(\frac{\widehat{\xi}^\star_{\tau_n'}}{\xi_{\tau_n'}} - 1\right) \xrightarrow{d} \Gamma.$$

Remark 2.15 (Deterministic ratio control)*.* The deterministic log-ratio condition in Proposition 2.14 is the high-level form of the extrapolation bias control. It is precisely the negligibility

$$\frac{a_n}{\ell_n}\left\{\log \xi_{\tau_n'} - \log \xi_{\tau_n} - \gamma \ell_n\right\} \longrightarrow 0.$$

In the original supplement of Davison et al. (2023), this control is part of the high-level condition C^*. In the iid heavy-tailed setting, this is supplied by the very-extreme expectile extrapolation argument of Daouia, Girard, et al. (2020, Theorem 5), which is the proof step invoked by Davison et al. (2023, Proposition C.6). The relevant quantile expansion is the sequential Weissman form along the diverging ratio $y_n = (1-\tau_n)/(1-\tau_n') \to \infty$, not merely the fixed-$y$ display in Condition 2.3. In applications of Proposition 2.14, this source-paper control combines with (2.7): the A-term contributes $O(a_n \left|A((1-\tau_n)^{-1})\right| / \ell_n) = o(1)$ and the lower-tail remainder contributes $O(a_n/\{Q_X(\tau_n)\ell_n\}) = o(1)$.

Proposition 2.14 is the technical workhorse of the thesis. The original supplement of Davison et al. (2023) states the result under broader high-level assumptions and a concrete root-rate normalisation. The version above is deliberately restricted to the heavy-tailed iid regime used throughout this thesis, while Remark 2.15 records the deterministic step that links this concrete regime to the abstract source formulation. It says that the asymptotic distribution of the extrapolated expectile at very-extreme levels is governed *only* by the asymptotic distribution of the tail-index estimator: the contribution of the intermediate-level expectile estimator $\widehat{\xi}^{I}_{\tau_n}$ washes out through the log-rate. In Chapter 4 we shall feed the pooled inputs $(\hat{\gamma}_n(\boldsymbol{\omega}), \overline{\xi}^{\mathrm{pool}}_{\tau_n})$ of Chapter 3 through this proposition to obtain the CLT for the pooled extrapolated expectile in essentially one step.

2.5 The optimal pooling framework

We now turn to the pooling framework of Daouia, Padoan, et al. (2024), which the rest of the thesis extends from quantiles to expectiles. We refer throughout to the final Bernoulli article version listed in the bibliography; the earlier arXiv preprint circulated under the same arXiv identifier but differs in some display notation. The framework concerns inference from m samples for the tail index and, under an additional tail-homoskedasticity condition, for the extreme quantile. The samples may have heterogeneous sizes, heterogeneous second-order properties, and non-trivial cross-sample tail dependence.

2.5.1 Setup

Throughout Section 2.5, unless explicitly stated otherwise, m is fixed while $n = \sum_{j=1}^{m} n_j \to \infty$. This is the finite-$m$ regime imported into Chapters 3 and 4. Let $\boldsymbol{X}_i = (X_{i,1}, \ldots, X_{i,m})^\top$, $i \geq 1$, be iid copies of a generic m-dimensional random vector $\boldsymbol{X} = (X_1, \ldots, X_m)^\top$. Margin j is observed for $i = 1, \ldots, n_j$, so if $n_j \neq n_\ell$ the two margins overlap only for the first $\min(n_j, n_\ell)$ vector observations. The jth sample therefore comprises n_j observations $X_{1,j}, X_{2,j}, \ldots, X_{n_j,j}$ with continuous, right-heavy-tailed marginal distribution function F_j, tail quantile function $U_j = (1/\overline{F}_j)^{\leftarrow}$, and extreme-value index $\gamma_j > 0$. The order statistics of the jth sample are denoted $X_{1:n_j,j} \leq \cdots \leq X_{n_j:n_j,j}$. The total sample size is $n = \sum_{j=1}^{m} n_j$.

Tail inference in sample j uses its top $k_j + 1$ order statistics, where the effective sample size $k_j = k_j(n)$ satisfies $k_j \to \infty$ and $k_j/n_j \to 0$ as $n \to \infty$. The total effective sample size is $k = \sum_{j=1}^{m} k_j$. Assume the asymptotic-proportionality conditions

$$\frac{n_1}{n_j} \to b_j \in (0, \infty), \qquad \frac{k_1}{k_j} \to c_j \in (0, \infty), \tag{2.12}$$

with $b_1 = c_1 = 1$. The marginal F_j satisfies $\mathcal{C}_2(\gamma_j, \rho_j, A_j)$ (Condition 2.3). The thesis works under the *common-tail-index* assumption

$$(H_\gamma) \qquad \gamma_1 \;=\; \cdots \;=\; \gamma_m \;=\; \gamma,$$

which is the natural setup when the m samples are believed to come from the same underlying tail behaviour. All common-γ vector limits below are expressed on the aggregate scale $\sqrt{k}$; the constants c_j translate the margin-specific $\sqrt{k_j}$ rates onto this common scale.

The cross-sample asymptotic dependence is captured by a pairwise *tail-copula* structure (Daouia, Padoan, et al. 2024, §2.1).

Condition $\mathcal{J}(\boldsymbol{R})$ (tail-copula condition). For each pair (j, ℓ) with $j \neq \ell$, there exists a function $R_{j,\ell} : [0, \infty]^2 \setminus \{(\infty, \infty)\} \to [0, \infty)$ such that, for every $(x_j, x_\ell) \in [0, \infty]^2 \setminus \{(\infty, \infty)\}$,

$$\lim_{s \to \infty} s\, \mathbb{P}\big(1 - F_j(X_j) \le x_j/s,\ 1 - F_\ell(X_\ell) \le x_\ell/s\big) \;=\; R_{j,\ell}(x_j, x_\ell).$$

The function $R_{j,\ell}$ is the bivariate tail copula of (X_j, X_ℓ) (Haan and Ferreira 2006, §8.2); it measures the rate at which the joint upper-tail probability of (X_j, X_ℓ) decays in comparison with the marginals. The case $R_{j,\ell} \equiv 0$ corresponds to *asymptotic tail independence* of X_j and X_ℓ, which is in particular implied by independence of the samples themselves — the setting of this thesis. This formulation covers the general pooling framework in which cross-sample extremal dependence may be represented through a tail copula. In the independent common-distribution distributed case used later in the thesis, the samples are independent marginal subsamples and all pairwise tail-copula terms vanish.

2.5.2 The pooled Hill estimator

Each sample contributes a Hill estimator $\hat{\gamma}_j(k_j)$ as in (2.1); we collect these into the random vector

$$\hat{\boldsymbol{\gamma}}_n \;=\; \big(\hat{\gamma}_1(k_1), \ldots, \hat{\gamma}_m(k_m)\big)^\top.$$

Given a weight vector $\boldsymbol{\omega} \in \mathbb{R}^m$ with $\boldsymbol{\omega}^\top \mathbf{1} = 1$, the *pooled Hill estimator* is the weighted arithmetic combination

$$\hat{\gamma}_n(\boldsymbol{\omega}) \;:=\; \boldsymbol{\omega}^\top \hat{\boldsymbol{\gamma}}_n \;=\; \sum_{j=1}^{m} \omega_j\, \hat{\gamma}_j(k_j). \tag{2.13}$$

The constraint $\boldsymbol{\omega}^\top \mathbf{1} = 1$ ensures consistency of $\hat{\gamma}_n(\boldsymbol{\omega})$ for the common γ under (H_γ). The naive choice $\boldsymbol{\omega} = m^{-1}\mathbf{1}$ recovers a simple arithmetic average; the choice $\boldsymbol{\omega} \propto (k_1, \ldots, k_m)$ recovers effective-sample-size weighting. The goal of the optimal-pooling literature is to identify weight choices that improve on these heuristics in terms of asymptotic variance or AMSE.

The fundamental asymptotic result is the joint CLT for the marginal Hill estimators.

Theorem 2.16 (Daouia, Padoan, et al. (2024), Theorem 1)**.** *Assume $\mathcal{C}_2(\gamma_j, \rho_j, A_j)$ for every j, the tail-copula condition $\mathcal{J}(\boldsymbol{R})$, the proportionality conditions (2.12), and $\sqrt{k_j}\, A_j(n_j/k_j) \to \lambda_j \in \mathbb{R}$ for every j. Then*

$$\left(\sqrt{k_1}(\hat{\gamma}_1(k_1) - \gamma_1), \ldots, \sqrt{k_m}(\hat{\gamma}_m(k_m) - \gamma_m)\right)^\top \xrightarrow{d} \mathcal{N}(\boldsymbol{B}, \boldsymbol{V}),$$

with $\boldsymbol{B} = (\lambda_j/(1-\rho_j))_{j=1}^m$ and $\boldsymbol{V}$ the symmetric matrix whose entries are

$$V_{j,\ell} \;=\; \begin{cases} \gamma_j^2, & j = \ell, \\ \dfrac{\gamma_j \gamma_\ell}{\max(b_j, b_\ell)\sqrt{c_j c_\ell}}\, R_{j,\ell}(b_j c_\ell,\, b_\ell c_j), & j < \ell. \end{cases}$$

The matrix is symmetric, so $V_{\ell,j} = V_{j,\ell}$ for $\ell > j$. Suppose, in addition, the common-γ condition (H_γ). Then the same marginal estimators, now viewed on the common $\sqrt{k}$ scale, satisfy

$$\sqrt{k}\,(\hat{\boldsymbol{\gamma}}_n - \gamma\mathbf{1}) \;\xrightarrow{d}\; \mathcal{N}(\boldsymbol{B_c}, \boldsymbol{V_c}),$$

where

$$(\boldsymbol{B_c})_j = \sqrt{c_j \sum_{i=1}^m c_i^{-1}}\, \frac{\lambda_j}{1-\rho_j} \qquad j = 1, \ldots, m.$$

The covariance matrix $\boldsymbol{V_c}$ is given by

$$(\boldsymbol{V_c})_{j,\ell} = \left(\sum_{i=1}^m c_i^{-1}\right) \begin{cases} \gamma^2 c_j, & j = \ell, \\ \dfrac{\gamma^2}{\max(b_j, b_\ell)}\, R_{j,\ell}(b_j c_\ell, b_\ell c_j), & j < \ell, \end{cases}$$

with $(\boldsymbol{V_c})_{\ell,j} = (\boldsymbol{V_c})_{j,\ell}$ for $\ell > j$. Consequently, for any $\boldsymbol{\omega}$ with $\boldsymbol{\omega}^\top \mathbf{1} = 1$,

$$\sqrt{k}\,(\hat{\gamma}_n(\boldsymbol{\omega}) - \gamma) \;\xrightarrow{d}\; \mathcal{N}(\boldsymbol{\omega}^\top \boldsymbol{B_c},\; \boldsymbol{\omega}^\top \boldsymbol{V_c} \boldsymbol{\omega}).$$

In the case of *independent* samples ($R_{j,\ell} \equiv 0$ for every $j \neq \ell$), the off-diagonal entries of $\boldsymbol{V_c}$ vanish identically and $\boldsymbol{V_c}$ becomes the diagonal matrix

$$\boldsymbol{V_c} \;=\; \gamma^2 \left(\sum_{i=1}^m c_i^{-1}\right) \operatorname{diag}(c_1, \ldots, c_m). \tag{2.14}$$

Equation (2.14) is the form of $\boldsymbol{V_c}$ that the thesis will exploit in Chapters 3 and 4.

2.5.3 Variance- and AMSE-optimal weights

Given Theorem 2.16, two natural choices of $\boldsymbol{\omega}$ emerge as solutions to convex quadratic programmes. Unless explicitly stated otherwise, weights are allowed to be real and are constrained only by $\boldsymbol{\omega}^\top \mathbf{1} = 1$; the optimal weights therefore need not be convex weights.

Proposition 2.17 (Optimal weights; Daouia, Padoan, et al. (2024), Theorem 1 and §2.2)**.** *Suppose $\boldsymbol{V_c}$ is positive definite. Under the common-γ condition (H_γ):*

(i) The variance-optimal weights, minimising $\boldsymbol{\omega}^\top \boldsymbol{V_c} \boldsymbol{\omega}$ subject to $\boldsymbol{\omega}^\top \mathbf{1} = 1$, are

$$\boldsymbol{\omega}^{\mathrm{Var}} = \frac{\boldsymbol{V_c}^{-1}\mathbf{1}}{\mathbf{1}^\top \boldsymbol{V_c}^{-1}\mathbf{1}},$$

with corresponding asymptotic variance $\boldsymbol{\omega}^{\mathrm{Var},\top} \boldsymbol{V_c} \boldsymbol{\omega}^{\mathrm{Var}} = (\mathbf{1}^\top \boldsymbol{V_c}^{-1}\mathbf{1})^{-1}$.

(ii) The AMSE-optimal weights, minimising $(\boldsymbol{\omega}^\top \boldsymbol{B_c})^2 + \boldsymbol{\omega}^\top \boldsymbol{V_c}\boldsymbol{\omega}$ subject to $\boldsymbol{\omega}^\top \mathbf{1} = 1$, are

$$\boldsymbol{\omega}^{\mathrm{AMSE}} = \frac{(1 + \boldsymbol{B_c}^\top \boldsymbol{V_c}^{-1}\boldsymbol{B_c})\, \boldsymbol{V_c}^{-1}\mathbf{1} - (\mathbf{1}^\top \boldsymbol{V_c}^{-1}\boldsymbol{B_c})\, \boldsymbol{V_c}^{-1}\boldsymbol{B_c}}{(1 + \boldsymbol{B_c}^\top \boldsymbol{V_c}^{-1}\boldsymbol{B_c})(\mathbf{1}^\top \boldsymbol{V_c}^{-1}\mathbf{1}) - (\mathbf{1}^\top \boldsymbol{V_c}^{-1}\boldsymbol{B_c})^2}.$$

In the independent-samples case, $\boldsymbol{V_c}$ is diagonal (Equation (2.14)) and the variance-optimal weights reduce to

$$\omega_j^{\mathrm{Var}} = \left(\sum_{i=1}^{m} c_i^{-1}\right)^{-1} c_j^{-1} \sim \frac{k_j}{k}, \tag{2.15}$$

where the equivalence uses $c_i^{-1} = \lim k_i/k_1$. Thus the samples contribute to the pooled tail index in proportion to their effective sample size, the same rule that one would expect from the unfeasible Hill estimator built from the entire combined sample.

In practice $\boldsymbol{V_c}$ and $\boldsymbol{B_c}$ must be replaced by consistent estimators $\widehat{\boldsymbol{V_c}}$ and $\widehat{\boldsymbol{B_c}}$ to render the optimal weights data-driven. The estimators are built from the marginal $\hat{\gamma}_j$, estimators of the second-order quantities entering A_j and ρ_j, and a plug-in estimator $\widehat{R_{j,\ell}}$ of the bivariate tail copula (Daouia, Padoan, et al. 2024, §2.2). The composite estimator $\hat{\gamma}_n(\hat{\boldsymbol{\omega}})$ is then $\sqrt{k}$-asymptotically equivalent to $\hat{\gamma}_n(\boldsymbol{\omega})$ provided $\hat{\boldsymbol{\omega}}^\top \mathbf{1} = 1$ exactly and $\hat{\boldsymbol{\omega}} \xrightarrow{\mathbb{P}} \boldsymbol{\omega}$. If exact normalisation is not imposed, one must instead require $\sqrt{k}\left|\hat{\boldsymbol{\omega}}^\top \mathbf{1} - 1\right| \xrightarrow{\mathbb{P}} 0$. In applications, the main additional difficulty is not the quadratic minimisation itself but the stable estimation of $\boldsymbol{V_c}$, especially its off-diagonal tail-copula entries.

2.5.4 Weighted geometric pooling of extreme quantiles

For estimation of an extreme quantile $q(1-p)$ with $p = p(n) \to 0$, each sample contributes a Weissman estimator

$$\hat{q}_j^\star(1 - p \mid k_j) = \left(\frac{k_j}{n_j\, p}\right)^{\hat{\gamma}_j(k_j)} X_{n_j - k_j : n_j,\, j}.$$

The aggregation of these per-sample Weissman estimators must be adapted to their multiplicative-power structure (Equation (2.2)), and is done by *weighted geometric* — rather than arithmetic —

averaging. Under the additional assumption that the samples have asymptotically equivalent extreme quantiles, formalised as the asymptotic-equivalence condition

Condition 2.18 (Tail homoskedasticity ($\mathcal{H}$))**.** For every $j \neq \ell$, $U_j(t)/U_\ell(t) \to 1$ as $t \to \infty$.

The terminology reflects the fact that the marginal tail scales are asymptotically equal. Equivalently, $Q_j(1-p)/Q_\ell(1-p) \to 1$ as $p \downarrow 0$. The notation in Daouia, Padoan, et al. (2024) writes this population extreme quantile as $q_j(1-p)$; here we keep the thesis-wide notation $Q_j(1-p)$. The pooled extreme-quantile estimator of Daouia, Padoan, et al. (2024, §2.3) is

$$\widehat{q}_n^{\star}(1-p \mid \boldsymbol{\omega}) \;\coloneqq\; \prod_{j=1}^{m} \left[\hat{q}_j^{\star}(1-p \mid k_j)\right]^{\omega_j}. \tag{2.16}$$

The reason for the geometric form is visible from the log-scale identity $\log \hat{q}_j^{\star}(1-p \mid k_j) = \log(k_j/(n_j p))\,\hat{\gamma}_j(k_j) + \log X_{n_j-k_j:n_j,j}$: a weighted sum on the log scale puts the per-sample $\hat{\gamma}_j(k_j)$ on the standard linear scale, which is what the joint CLT of Theorem 2.16 addresses. Under the proportionality conditions (2.12), $\log(k_j/(n_j p)) = \log(k/(np)) + O(1)$, so the per-sample logarithmic normalisers are asymptotically equivalent to the common normaliser in the diverging-log regime.

Theorem 2.19 (Daouia, Padoan, et al. (2024), Theorem 2)**.** *Assume the hypotheses of Theorem 2.16, the common-γ condition (H_γ), the tail-homoskedasticity condition $(\mathcal{H})$, and, in addition, $\rho_j < 0$ for every $j = 1, \ldots, m$. Let $p = p(n) \to 0$ with $k/(np) \to \infty$ and $\sqrt{k}/\log(k/(np)) \to \infty$. Then for any $\boldsymbol{\omega}$ with $\boldsymbol{\omega}^\top \mathbf{1} = 1$ and any $j \in \{1, \ldots, m\}$,*

$$\frac{\sqrt{k}}{\log(k/(np))}\left(\frac{\widehat{q}_n^{\star}(1-p \mid \boldsymbol{\omega})}{Q_j(1-p)} - 1\right) \xrightarrow{d} \mathcal{N}(\boldsymbol{\omega}^\top \boldsymbol{B_c},\, \boldsymbol{\omega}^\top \boldsymbol{V_c}\boldsymbol{\omega}).$$

The strict condition $\rho_j < 0$ is part of the Weissman extrapolation theorem and is stronger than the $\rho_j \leq 0$ condition needed for the marginal Hill CLT. Condition ($\mathcal{H}$) itself is intentionally only the first-order tail-homoskedasticity condition of Daouia, Padoan, et al. (2024). The recentering by any marginal $Q_j(1-p)$ in Theorem 2.19 is valid because the strict second-order assumption $\rho_j < 0$ supplies the missing rate: by Daouia, Padoan, et al. (2024, Supplement, Lemma B.2), for any j, ℓ,

$$\log Q_j(1-p) - \log Q_\ell(1-p) = O(\,|A_j(1/p)|\,) + O(\,|A_\ell(1/p)|\,) = o(k^{-1/2}),$$

along the Weissman sequence. After multiplication by $\sqrt{k}/\log(k/(np))$, this deterministic centering difference is negligible.

The asymptotic distribution of the pooled Weissman estimator is *identical*, up to the log-rate factor, to that of the pooled Hill estimator. Two structural consequences follow. First, the variance- and AMSE-optimal weights of Proposition 2.17 are also optimal for the pooled extreme-quantile estimator. Second, the entire information needed for inference at very-extreme quantile levels is contained in the joint distribution of the per-sample Hill estimators; the extreme-quantile inputs contribute only through the deterministic log-scale factor. This structural fact is the analogue, in the multi-sample setting, of the dominance of the tail-index estimator at the extreme level that we have already encountered in Theorem 2.13 and Proposition 2.14. In the geometric quantile pool, real weights remain algebraically well-defined because the Weissman estimators are positive with probability tending to one under the positivity convention of Section 2.2.1.

2.5.5 Tail-homogeneity testing and confidence intervals

The covariance structure $\boldsymbol{V_c}$ also yields a Wald-type chi-square test for the common-γ hypothesis (H_γ), sometimes called *tail homogeneity* across samples.

Theorem 2.20 (Tail-homogeneity test; Daouia, Padoan, et al. (2024), Corollary 3)**.** *Assume the hypotheses of Theorem 2.16, the null $H_0 : \gamma_1 = \cdots = \gamma_m$, and the bias-free calibration $\lambda_j = 0$ for every j. The statistic*

$$\Lambda_n \;=\; k\,(\hat{\boldsymbol{\gamma}}_n - \bar{\mu}_n\mathbf{1})^\top \widehat{\boldsymbol{V}}_{\boldsymbol{c}}^{-1}(\hat{\boldsymbol{\gamma}}_n - \bar{\mu}_n\mathbf{1}), \qquad \bar{\mu}_n \;=\; \frac{\mathbf{1}^\top \widehat{\boldsymbol{V}}_{\boldsymbol{c}}^{-1}\hat{\boldsymbol{\gamma}}_n}{\mathbf{1}^\top \widehat{\boldsymbol{V}}_{\boldsymbol{c}}^{-1}\mathbf{1}},$$

where $\widehat{\boldsymbol{V}}_{\boldsymbol{c}}$ is a consistent estimator of the covariance matrix of $\sqrt{k}(\hat{\boldsymbol{\gamma}}_n - \gamma\mathbf{1})$ under H_0. Then $\Lambda_n \xrightarrow{d} \chi^2_{m-1}$ under H_0 and $\Lambda_n \xrightarrow{\mathbb{P}} \infty$ under fixed alternatives. A test of asymptotic size α rejects H_0 when $\Lambda_n > \chi^2_{m-1,\,1-\alpha}$.

Daouia, Padoan, et al. (2024) state the bias-free version above. The same Gaussian GLS calculation also shows that, if the asymptotic bias is retained rather than removed by undersmoothing or bias correction, the deviance has the noncentral limit

$$\Lambda_n \xrightarrow{d} \chi^2_{m-1}(\delta_\gamma), \qquad \delta_\gamma = \boldsymbol{B_c}^\top \boldsymbol{V_c}^{-1}\boldsymbol{B_c} - \frac{(\mathbf{1}^\top \boldsymbol{V_c}^{-1}\boldsymbol{B_c})^2}{\mathbf{1}^\top \boldsymbol{V_c}^{-1}\mathbf{1}}.$$

Thus the central chi-square limit holds whenever $\boldsymbol{B_c} \in \operatorname{span}\{\mathbf{1}\}$, in particular under $\lambda_j = 0$ for every j. Under contiguous alternatives of order $k^{-1/2}$, the limit is likewise noncentral rather than divergent.

Under tail homogeneity, Proposition 2.17 furnishes asymptotic confidence intervals for the common γ on the $\sqrt{k}$ scale; log-scale confidence intervals for the extrapolated extreme quantile follow from Theorem 2.19. These variance-only intervals are centered correctly under undersmoothing or after valid bias correction; otherwise the asymptotic bias term must be included in the centering.

2.5.6 Distributed inference as a special case

The independent common-distribution distributed setting is the special case in which the m samples are obtained as iid subdivisions of one common underlying iid sample: data points are iid both within and across samples, with a common marginal distribution. The framework simplifies considerably and matches the assumptions of this thesis.

Under these independent common-distribution distributed assumptions:

- the common-γ condition (H_γ) and the tail-homoskedasticity condition ($\mathcal{H}$) are trivially satisfied;
- the tail copula $R_{j,\ell}$ between any pair of distinct samples vanishes identically (independence implies tail independence);
- $\boldsymbol{V_c}$ in Theorem 2.16 becomes the diagonal matrix (2.14);
- the variance-optimal weights of Proposition 2.17 reduce to $\omega_j^{\mathrm{Var}} \propto k_j$ as in (2.15).

Corollary 2.21 (Distributed oracle equivalence; Daouia, Padoan, et al. (2024), Corollary 6, Theorem 3 and Corollary 8)**.** *Under the distributed-inference assumptions (iid across machines) of Corollary 5 in Daouia, Padoan, et al. (2024), suppose the sample fractions are asymptotically equal in the stronger sense $k_j/n_j = (k/n)(1+O(k^{-1/2}))$ for every j. Then, under the conditions of their Theorem 3, the variance-optimally-pooled Hill estimator $\hat{\gamma}_n(\boldsymbol{\omega}^{\mathrm{Var}})$ is $\sqrt{k}$-asymptotically equivalent to the unfeasible benchmark Hill estimator $\hat{\gamma}_n^{(\mathrm{Hill})}(k)$ built from the entire combined sample. If, in addition, $\rho < 0$ and $p = p(n)$ satisfies $k/(np) \to \infty$ and $\sqrt{k}/\log(k/(np)) \to \infty$, then the variance-optimally-pooled Weissman estimator $\widehat{q}_n^{\star}(1-p \mid \boldsymbol{\omega}^{\mathrm{Var}})$ is asymptotically equivalent to the unfeasible benchmark Weissman estimator built from the combined sample.*

Corollary 2.21 is the central efficiency result of the optimal pooling framework: under the independent common-distribution distributed assumptions, equal sample fractions, and variance-optimal weighting, the pooled estimators pay no first-order asymptotic efficiency price relative to the infeasible pooled-sample benchmark. It is this property that we shall seek to preserve in the pooled-expectile estimators of Chapters 3 and 4.

Chapter 3

Pooled inference for extreme expectiles at intermediate levels

Overview

This chapter introduces the pooled extreme-expectile estimator at intermediate levels and develops its asymptotic theory. The construction plugs the pooled tail-index estimator $\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)$ and the pooled weighted-geometric Weissman extreme-quantile estimator $\widehat{q}_n^\star(\cdot \mid \boldsymbol{\omega}_q)$ of Daouia, Padoan, et al. (2024) into the expectile–quantile asymptotic equivalence $\xi_\tau \approx \psi(\gamma)\, Q_X(\tau)$. The delta method applied through ψ yields a joint Gaussian limit, from which we read off the variance- and AMSE-optimal pooling weights tailored to the pooled expectile target.

3.1 Setup and notation

The contribution of this chapter rests on the pooling framework of Section 2.5, specialised to independent samples with a common marginal distribution — the common-distribution distributed setting of Section 2.5.6. This is deliberately stronger than assuming only a common tail index: it gives one common population target ξ_{τ_n} and one common quantile function Q_X, whereas heterogeneous margins would require sample-specific targets and additional tail-scale alignment conditions before pooled expectile inference could be interpreted. Thus the chapter is intentionally narrower than the full heterogeneous-tail-copula scope of Daouia, Padoan, et al. (2024): the restriction is what makes the pooled object a single common expectile rather than a collection of sample-specific targets.

The equal effective sample fraction condition $k_j/n_j = (k/n)(1 + o(1))$ is *not* part of the standing setup; it is invoked only in Sections 3.5 to 3.7 where it simplifies the variance-optimal weights and the limiting bias structure. When oracle equivalence to the unfeasible pooled-sample benchmark is cited, we use the stronger $k_j/n_j = (k/n)(1 + O(k^{-1/2}))$ condition of Corollary 2.21.

We restate below only those assumptions used throughout. The pooled-Hill CLT of Theorem 2.16 is imported directly; the weighted-geometric Weissman construction of Section 2.5 is kept as the quantile primitive, while its finite-log intermediate limit is derived inside this chapter rather than read off from the diverging-log Weissman CLT Theorem 2.19.

The m samples and the pooling primitives. We have m *independent* samples $\{X_{i,j}\}_{i=1,\dots,n_j}$, $j = 1, \dots, m$, with common marginal distribution function F and common extreme-value index γ. Let $n := \sum_{j=1}^{m} n_j$. All asymptotics are taken along a triangular array with fixed m, $n_j \to \infty$, $k_j \to \infty$, and $k_j/n_j \to 0$ for every j; all CLTs below are therefore finite-m statements. Independence of the samples implies asymptotic tail independence, hence the pairwise tail-copula functions in Condition $\mathcal{J}(\boldsymbol{R})$ vanish:

$$R_{j,\ell} \;\equiv\; 0 \qquad \text{for every } j \neq \ell. \tag{3.1}$$

The proportionality conditions of (2.12) are in force:

$$\frac{n_1}{n_j} \to b_j \in (0, \infty), \qquad \frac{k_1}{k_j} \to c_j \in (0, \infty), \qquad j = 1, \dots, m,$$

with $b_1 = c_1 = 1$, and the total effective sample size is $k = \sum_{j=1}^{m} k_j$. Consequently $k_j \asymp k$ uniformly in j and, with $S := \sum_{i=1}^{m} c_i^{-1}$, $k/k_j \to c_j S$. Each sample j supplies the two pooling primitives of Section 2.5: a local Hill estimator $\hat{\gamma}_j(k_j)$ as in (2.1), and a local Weissman extreme-quantile estimator $\hat{q}_j^{\star}(\,\cdot \mid k_j)$ as in (2.2). The dot does not represent an additional communicated function; it denotes the algebraic Weissman rule recoverable from the communicated Weissman value together with $(\hat{\gamma}_j(k_j), k_j, n_j)$. At any fixed target level this rule contributes one quantile scalar per sample. These values, together with their joint asymptotic covariance, constitute the entire information shared between samples; no raw observation and no expectile-specific summary statistic crosses between machines.

Heavy-tail regime. The common marginal F satisfies the second-order condition $\mathcal{C}_2(\gamma, \rho, A)$ of Condition 2.3 with

$$\gamma \in (0, 1), \qquad \mathbb{E}\, X^{-} < \infty, \tag{3.2}$$

the per-sample bias-rate condition $\sqrt{k_j}\, A(n_j/k_j) \to \lambda_j \in \mathbb{R}$ holds for every j, and the moment-negligibility condition

$$\frac{\sqrt{k}}{Q_X(\tau_n)} \;\to\; 0 \tag{3.3}$$

is in force. The range $\gamma < 1$ and the integrability of X^{-} place the analysis in the finite-mean regime in which expectiles are well defined and the quantile-based expectile–quantile bridge applies. The stronger finite-variance restriction $\gamma < 1/2$ is needed only for the direct LAWS estimator reviewed in Theorem 2.11; it is not an assumption of the pooled QB results below. Condition (3.3) is the multi-sample counterpart of the $\sqrt{n(1-\tau_n)}\,/Q_X(\tau_n) \to 0$ hypothesis of the QB intermediate theorem and the second-order expectile–quantile expansion: it ensures that the moment term in the expectile–quantile expansion (2.7) is asymptotically negligible, so that the deterministic bias of the pooled estimator is governed by the single second-order rate carried by A. This is an additional rate restriction linking the target level and the effective sample size, not an automatic consequence of heavy-tailedness. Under the equal-fraction scaling $1 - \tau_n \asymp k/n$, and suppressing slowly varying factors, $Q_X(\tau_n) \asymp (n/k)^\gamma$, so (3.3) requires $k^{1/2+\gamma}/n^\gamma \to 0$. The endpoint $\rho = 0$ is retained in this intermediate-level chapter: the expectile–quantile expansion and the QB CLT used below are stated under the same $\rho \leq 0$ convention as Condition 2.3. The stricter $\rho < 0$ assumption re-enters only for the very-extreme extrapolation step in Chapter 4, through Proposition 2.14 and Theorem 2.19.

The target. We target the common population expectile ξ_{τ_n} at an *intermediate* level $\tau_n \to 1$ with $n(1-\tau_n) \to \infty$, in the regime sense of Section 2.4. In addition, the per-sample log-rate factor

$$L_j(\tau_n) \ :=\ \log \frac{k_j}{n_j(1-\tau_n)}$$

is assumed to admit a finite limit $L_j^\star \in \mathbb{R}$ for every j; equivalently, $1-\tau_n$ has the same order as k_j/n_j for each sample. This condition controls the Hill–Weissman cross-covariance in Proposition 3.2 and is the chapter's link between the chosen target level and the local effective sample fractions. The very-extreme regime $n(1-\tau_n') \to c \in [0,\infty)$ is deferred to Chapter 4, where the pooled intermediate-level result proved here serves as the anchor for Weissman-style extrapolation through Proposition 2.14.

Per-sample QB plug-in: a forced consequence of the pooling protocol. Because each sample shares the pooling primitives $(\hat\gamma_j(k_j),\, \hat q_j^\star(\cdot \mid k_j))$ and no expectile-specific summary statistic, the only one of the two standard intermediate-expectile constructions considered in the thesis that can be assembled *locally* on each sample, before pooling, under this communication protocol is the QB plug-in (2.9) obtained by factoring through the expectile–quantile equivalence of Proposition 2.10:

$$\widehat{\xi}_{\tau,j} \ =\ \psi(\hat\gamma_j(k_j))\, \hat q_j^\star(\tau \mid k_j).$$

The LAWS direct estimator (2.8) would require sharing $\widetilde{\xi}_{\tau_n,j}$ across samples, because its asymmetric-least-squares objective is solved as an expectile estimation problem rather than assembled from the Hill and Weissman quantile primitives; this lies outside the pooling protocol of Section 2.5. The comparison of Theorems 2.11 and 2.12 shows that neither LAWS nor QB uniformly dominates the other on the direct-LAWS range $\gamma \in (0,1/2)$; the QB construction is therefore adopted not because it is uniformly more efficient, but because it is the available member of this pair under the communicated pooling primitives. The choice is a derived consequence of the Paper-3 communication protocol, not a claim that QB is intrinsically optimal among all possible expectile procedures.

Roadmap. Section 3.2 defines the pooled QB intermediate expectile estimator from the pooled Hill and pooled Weissman inputs. Section 3.3 derives the joint $\sqrt{k}$-CLT for the pooled Hill and log-Weissman inputs, combining the pooled-Hill limit of Theorem 2.16 and the weighted-geometric pooling map with the local Hill / intermediate-threshold joint CLT, specialised to the independent-samples case (3.1). The output is the Gaussian input to which the delta method is then applied. Section 3.4 applies the delta method through ψ and states the main asymptotic-normality theorem of the chapter. Section 3.5 derives the variance- and AMSE-optimal pooling weights for the expectile target and compares them with the corresponding optima for the quantile target. Section 3.6 provides feasible confidence intervals built from estimated weights, and Section 3.7 adapts the tail-homogeneity test of Theorem 2.20 to the expectile-tail-homogeneity null hypothesis.

3.2 The pooled QB expectile estimator

The shared primitives $\big(\hat{\gamma}_j(k_j),\, \hat{q}^{\star}_j(\tau_n \mid k_j)\big)_{j=1,\dots,m}$ admit two natural *single-weight* constructions of a pooled QB intermediate-expectile estimator. The first, denoted (A), applies the QB plug-in once, at the centre, after pooling the Hill and Weissman primitives separately; the second, denoted (B), computes a per-sample QB expectile on each machine and then geometric-pools the m local estimates. At matched single weights — the only setting in which the two constructions are directly comparable — Proposition 3.1 below shows they share the same $\sqrt{k}$-asymptotic distribution, so on this common domain the choice between them is asymptotically inert. We adopt (A) by parsimony, since it is the direct lift of the iid QB pipeline of Theorem 2.12 — one tail-index input, one quantile input, one application of ψ — to the pooled setting.

Crucially, (A) exposes an intrinsic *two-weight* structure: the pooled Hill and pooled Weissman primitives can in principle receive distinct weight vectors $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$. The literal local-QB construction (B), understood as first compressing each sample to one QB scalar and then pooling those scalars with a single weight vector, does not preserve this separation. One could define a componentwise two-weight geometric variant by pooling the $\psi(\hat{\gamma}_j)$ factors with $\boldsymbol{\omega}_\gamma$ and the Weissman factors with $\boldsymbol{\omega}_q$; by the same Taylor argument as Proposition 3.1, that variant is again $\sqrt{k}$-equivalent to (A) for deterministic affine weights. We therefore adopt (A) because it is algebraically simpler and keeps the two primitive weight vectors visible. Section 3.5 will return to the question of whether this two-weight freedom is ever genuinely exploited, and show that under the equal-fraction distributed-inference specialisation the variance-optimal pair lies on the matched-weight diagonal, apart from the non-identifiability of the Hill weights when $m(\gamma) = 0$.

Pooled Hill and pooled Weissman primitives. For weight vectors $\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q \in \mathbb{R}^m$ satisfying $\boldsymbol{\omega}_\gamma^\top \mathbf{1} = \boldsymbol{\omega}_q^\top \mathbf{1} = 1$, the linear pooled Hill estimator (2.13) and the weighted-geometric pooled Weissman estimator (2.16) read

$$\hat{\gamma}_n(\boldsymbol{\omega}_\gamma) \;=\; \boldsymbol{\omega}_\gamma^\top \hat{\boldsymbol{\gamma}}_n \;=\; \sum_{j=1}^m \omega_{\gamma,j}\, \hat{\gamma}_j(k_j), \tag{3.4}$$

$$\widehat{q}^{\star}_n(\tau_n \mid \boldsymbol{\omega}_q) \;=\; \prod_{j=1}^m \big[\hat{q}^{\star}_j(\tau_n \mid k_j)\big]^{\omega_{q,j}}, \tag{3.5}$$

both evaluated at the chapter's intermediate level τ_n. The weights are affine: only the sum-to-one constraint is imposed here, and the coordinates need not be nonnegative unless a later special case says so. The geometric products in (3.5) and below are therefore interpreted on the log scale and require positive local Weissman factors. Under the right-heavy-tailed setting with $\gamma > 0$ and $k_j/n_j \to 0$, the intermediate order statistic $X_{n_j-k_j:n_j,\,j}$ diverges to the right in probability by regular variation of the tail quantile; hence $\mathbb{P}(\hat{q}^{\star}_j(\tau_n \mid k_j) > 0,\ j = 1,\dots,m) \to 1$. All asymptotic log-scale statements below are understood on this high-probability event, intersected with the event on which the local and pooled Hill estimators entering $\log \psi$ lie in $(0, 1)$. Arbitrary definitions on the complementary event do not affect the weak limits.

The two candidate estimators. The first construction, which we adopt as canonical, pools the primitives separately and applies the QB plug-in once at the centre,

$$\widehat{\xi}^{(A)}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \;:=\; \psi(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma))\, \widehat{q}^{\star}_n(\tau_n \mid \boldsymbol{\omega}_q). \tag{3.6}$$

The second computes a per-sample QB expectile on each machine and geometric-pools the resulting m local estimates,

$$\widehat{\xi}^{(B)}_{\tau_n}(\boldsymbol{\omega}) \;:=\; \prod_{j=1}^{m} \left[\psi(\hat{\gamma}_j(k_j))\, \hat{q}^{\star}_j(\tau_n \mid k_j)\right]^{\omega_j} \;=\; \Big(\prod_{j=1}^{m} \psi(\hat{\gamma}_j(k_j))^{\omega_j}\Big) \cdot \widehat{q}^{\star}_n(\tau_n \mid \boldsymbol{\omega}). \tag{3.7}$$

When construction (A) is restricted to matched weights $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) = (\boldsymbol{\omega}, \boldsymbol{\omega})$, the Weissman factor is identical in the two formulae; the constructions differ only in whether the nonlinearity ψ acts once on the pooled $\hat{\gamma}_n$ (construction (A)) or componentwise on the m per-sample $\hat{\gamma}_j$'s followed by weighted-geometric averaging (construction (B)). The iid QB plug-in of Theorem 2.12 ($m = 1$) reduces both formulae to $\psi(\hat{\gamma})\, \hat{q}^{\star}$; construction (A) is the direct $m \geq 2$ generalisation of that structure, whereas (B) introduces m separate applications of ψ that have no counterpart in the iid baseline.

Proposition 3.1 ($\sqrt{k}$-asymptotic equivalence of (A) and (B))**.** *Under the assumptions of Section 3.1, for fixed $m \in \mathbb{N}$ and every deterministic weight vector $\boldsymbol{\omega} \in \mathbb{R}^m$ with $\boldsymbol{\omega}^\top \mathbf{1} = 1$,*

$$\sqrt{k}\, \log \frac{\widehat{\xi}^{(A)}_{\tau_n}(\boldsymbol{\omega}, \boldsymbol{\omega})}{\widehat{\xi}^{(B)}_{\tau_n}(\boldsymbol{\omega})} \;\xrightarrow{\mathbb{P}}\; 0.$$

In fact the unscaled log-ratio is itself $O_p(k^{-1})$; multiplying by $\sqrt{k}$ therefore gives $O_p(k^{-1/2}) = o_p(1)$, so the discrepancy is negligible at the CLT scale. Consequently, when construction (A) is evaluated at matched weights $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) = (\boldsymbol{\omega}, \boldsymbol{\omega})$, the two estimators share the same $\sqrt{k}$-asymptotic distribution and Theorem 3.3 applies to either without modification. The same conclusion extends to estimated weights $\hat{\boldsymbol{\omega}}$ satisfying $\hat{\boldsymbol{\omega}}^\top \mathbf{1} = 1$ exactly and $\hat{\boldsymbol{\omega}} \xrightarrow{\mathbb{P}} \boldsymbol{\omega}$. If exact normalisation is relaxed to $\sqrt{k}\,\left|\hat{\boldsymbol{\omega}}^\top \mathbf{1} - 1\right| \xrightarrow{\mathbb{P}} 0$, the $\sqrt{k}$-equivalence still holds, although the unscaled $O_p(k^{-1})$ refinement need not.

Sketch. A first-order Taylor expansion of $\log \psi$ around γ gives $\log \psi(\hat{\gamma}_j) - \log \psi(\gamma) = m(\gamma)(\hat{\gamma}_j - \gamma) + O_p((\hat{\gamma}_j - \gamma)^2)$, and analogously for $\hat{\gamma}_n(\boldsymbol{\omega})$. The linear terms in $\log \widehat{\xi}^{(A)}$ and $\log \widehat{\xi}^{(B)}$ coincide because $\sum_j \omega_j(\hat{\gamma}_j - \gamma) = \hat{\gamma}_n(\boldsymbol{\omega}) - \gamma$, so the discrepancy is the difference of quadratic-and-higher remainders, which is $O_p(1/k)$. On the $\sqrt{k}$-scale this is $O_p(k^{-1/2}) = o_p(1)$. Full proof in Appendix A.1. □

Canonical choice. We adopt

$$\widehat{\xi}^{\text{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \;:=\; \psi(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma))\, \widehat{q}^{\star}_n(\tau_n \mid \boldsymbol{\omega}_q), \tag{3.8}$$

identical to (3.6), as the canonical pooled QB intermediate-expectile estimator for the remainder of this chapter. The choice is justified by parsimony: (3.6) reproduces the iid QB pipeline of Theorem 2.12 — one tail-index input, one quantile input, one application of ψ — directly at the pooled level, whereas (3.7) introduces m applications of ψ which, by Proposition 3.1, contribute no additional asymptotic information.[1]

Two weight vectors, not one. Definition (3.8) carries two free weight vectors rather than a single one. In the quantile-only analysis of (2.16), the weighted-geometric quantile pool is formed with a single $\boldsymbol{\omega}$, and the extreme-quantile limit of Theorem 2.19 is governed by that same vector. The variance- and AMSE-optima of Proposition 2.17 therefore apply to the quantile target through one set of weights. The expectile target combines the Hill and Weissman primitives through the nonlinear map ψ, and the asymptotic variance of $\widehat{\xi}^{\text{pool}}_{\tau_n}$ acquires a contribution from the within-sample cross-covariance of $\hat{\gamma}_j$ and $\hat{q}^{\star}_j$ that is absent in the quantile-only analysis. This contribution can in principle pull the variance-optimal $\boldsymbol{\omega}_\gamma$ apart from the variance-optimal $\boldsymbol{\omega}_q$; Section 3.5 identifies the exact discrepancy and the sufficient conditions under which the two optima coincide. Keeping the weights separate in the definition leaves that section free to specify their relationship rather than imposing it by hand.

Why the pooled Weissman at the intermediate level. At the intermediate level τ_n, the per-sample Weissman estimator $\hat{q}^{\star}_j(\tau_n \mid k_j) = (k_j/(n_j(1-\tau_n)))^{\hat{\gamma}_j(k_j)}\, X_{n_j-k_j:n_j,j}$ is the device that moves sample j from its own local intermediate level to the common chapter target. When τ_n matches the sample-specific intermediate level $\tau_{n_j} := 1 - k_j/n_j$, the multiplicative power factor collapses to one and $\hat{q}^{\star}_j$ reduces to the bare (k_j+1)-th upper order statistic of sample j — the empirical intermediate quantile. Otherwise, $\hat{q}^{\star}_j$ interpolates or mildly extrapolates across intermediate levels, and remains $\sqrt{k_j}$-consistent by the within-sample joint CLT for the Hill estimator and the intermediate log-order statistic invoked in Proposition 3.2 below — this is the finite-log-rate analogue of the diverging-log-rate Weissman CLT Theorem 2.7, not a direct consequence of it. This bridge across local intermediate levels is why we use the Weissman primitive uniformly, instead of switching to bare order statistics at intermediate levels: it is already the primitive shared by the pooling protocol of Section 2.5, and it carries the same form to the extreme regime treated in Chapter 4, keeping the two chapters on a single notational footing.

[1] At the second order, the two constructions differ through a Jensen-type term of order k^{-1} involving $m'(\gamma)$ and the dispersion of the local Hill estimators across samples; its sign depends on the weight vector and need not favour either construction in general. This term sits below the order of the $\sqrt{k}$-asymptotic theory developed here and below the standard EVI bias control $\sqrt{k_j}\, A(n_j/k_j) \to \lambda_j$ of Theorem 2.6, so we do not use it.

3.3 Joint asymptotic distribution of the pooled inputs

This section establishes the joint $\sqrt{k}$-Gaussian limit of $(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma), \log \widehat{q}_n^\star(\tau_n \mid \boldsymbol{\omega}_q))$ at intermediate level τ_n. The result is the input that Section 3.4 feeds through the delta method to obtain the asymptotic distribution of the pooled QB expectile estimator (3.8).

Within-sample decomposition. For each sample j, the log of the Weissman estimator decomposes exactly as

$$\log \hat{q}_j^\star(\tau_n \mid k_j) \;=\; \log X_{n_j - k_j : n_j, j} \;+\; L_j(\tau_n)\, \hat{\gamma}_j(k_j), \qquad L_j(\tau_n) \coloneqq \log \frac{k_j}{n_j(1-\tau_n)}. \tag{3.9}$$

The log-rate factor $L_j(\tau_n)$ records the extent to which sample j extrapolates from its own intermediate level $\tau_{n_j} \coloneqq 1 - k_j/n_j$ to the chapter target τ_n: $L_j = 0$ when the two coincide, $L_j > 0$ for extrapolation, $L_j < 0$ for interpolation. The only stochastic input to the proof of Proposition 3.2 is the within-sample joint asymptotic behaviour of the Hill estimator and its companion intermediate log-order statistic at effective sample size k_j. By the tail-empirical-process representation (Haan and Ferreira 2006, Theorem 2.4.8 and §5.1), the two statistics are jointly Gaussian on the $\sqrt{k_j}$-scale, *asymptotically independent*, and each has asymptotic variance γ^2. The zero off-diagonal entry is structural: the Hill estimator is a smooth functional of the centred log-spacings above $X_{n_j - k_j : n_j, j}$, while the intermediate-threshold fluctuation is the centred $(k_j + 1)$-th upper log-order statistic itself, and the two decouple in the tail-empirical-process limit. This same joint behaviour underlies the iid QB plug-in CLT of Daouia, Girard, et al. (2020) reproduced in Theorem 2.12. Combining it with (3.9), while using the second-order tail-quantile expansion for the deterministic centering, yields the within-sample joint central limit theorem

$$\sqrt{k_j} \begin{pmatrix} \hat{\gamma}_j(k_j) - \gamma \\ \log \hat{q}_j^\star(\tau_n \mid k_j) - \log Q_X(\tau_n) \end{pmatrix} \xrightarrow{d} \mathcal{N}\left(\boldsymbol{\mu}_j^\star,\, \gamma^2 \begin{pmatrix} 1 & L_j^\star \\ L_j^\star & 1 + (L_j^\star)^2 \end{pmatrix} \right), \tag{3.10}$$

under the common-marginal assumption of Section 3.1 and the condition $L_j(\tau_n) \to L_j^\star \in \mathbb{R}$. With Ψ_ρ the second-order kernel defined in (3.14), the limiting mean is

$$\boldsymbol{\mu}_j^\star = \begin{pmatrix} \lambda_j/(1-\rho) \\ \lambda_j \Big\{ L_j^\star/(1-\rho) - \Psi_\rho(e^{L_j^\star}) \Big\} \end{pmatrix}.$$

The first component is the usual Hill bias, while the second combines that Hill bias, transported through the log-rate $L_j^\star$, with the deterministic second-order Weissman remainder. The 2×2 covariance reveals the within-sample Hill–Weissman correlation induced by the log-rate factor: cross-covariance $\gamma^2 L_j^\star$, and a Weissman variance $\gamma^2(1 + (L_j^\star)^2)$ obtained from the asymptotically independent order-statistic and rescaled-Hill contributions.

Pooling under independence. Independence of the m samples implies that the cross-sample covariance blocks of the $2m$-vector $\big(\hat{\gamma}_j - \gamma,\, \log \hat{q}_j^\star - \log Q_X(\tau_n)\big)_{j=1,\ldots,m}$ vanish identically. Linear pooling with weight vectors $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$ then collapses this $2m$-vector onto a 2-vector whose covariance is assembled block-by-block from the within-sample blocks of (3.10), each rescaled to the common $\sqrt{k}$-scale via the proportionality conditions (2.12).

Proposition 3.2 (Joint pooled CLT at intermediate level)**.** *Under the assumptions of Section 3.1 and the convergence $L_j(\tau_n) \to L_j^\star \in \mathbb{R}$ for every j, for any weight vectors $\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q \in \mathbb{R}^m$ satisfying $\boldsymbol{\omega}_\gamma^\top \mathbf{1} = \boldsymbol{\omega}_q^\top \mathbf{1} = 1$,*

$$\sqrt{k}\begin{pmatrix} \hat{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma \\ \log \widehat{q}_n^\star(\tau_n \mid \boldsymbol{\omega}_q) - \log Q_X(\tau_n) \end{pmatrix} \xrightarrow{d} \mathcal{N}(\boldsymbol{\mu}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q),\, \boldsymbol{\Sigma}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)), \tag{3.11}$$

with covariance

$$\begin{aligned} \boldsymbol{D}^\star &:= \operatorname{diag}(\boldsymbol{L}^\star), \\ \boldsymbol{\Sigma}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \;&=\; \begin{pmatrix} \boldsymbol{\omega}_\gamma^\top \boldsymbol{V_c} \boldsymbol{\omega}_\gamma & \boldsymbol{\omega}_\gamma^\top \boldsymbol{D}^\star \boldsymbol{V_c}\, \boldsymbol{\omega}_q \\ \boldsymbol{\omega}_q^\top \boldsymbol{V_c} \boldsymbol{D}^\star \boldsymbol{\omega}_\gamma & \boldsymbol{\omega}_q^\top (\boldsymbol{V_c} + \boldsymbol{D}^\star \boldsymbol{V_c} \boldsymbol{D}^\star)\, \boldsymbol{\omega}_q \end{pmatrix}. \end{aligned} \tag{3.12}$$

Here $\boldsymbol{V_c}$ is the diagonal matrix (2.14), so $\boldsymbol{D}^\star \boldsymbol{V_c} \boldsymbol{D}^\star = (\boldsymbol{D}^\star)^2 \boldsymbol{V_c}$, and $\boldsymbol{L}^\star = (L_1^\star, \ldots, L_m^\star)^\top$. The mean

$$\boldsymbol{\mu}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \;=\; \begin{pmatrix} \boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c} \\ \boldsymbol{\omega}_q^\top \boldsymbol{B_c}^\star \end{pmatrix}, \tag{3.13}$$

where $\boldsymbol{B_c}$ is the pooled-Hill bias vector of Theorem 2.16 and the limiting pooled-Weissman bias vector at the intermediate level has entries

$$(\boldsymbol{B_c}^\star)_j \;=\; \sqrt{c_j \textstyle\sum_{i=1}^m c_i^{-1}}\; \lambda_j \Big[\frac{L_j^\star}{1-\rho} \;-\; \Psi_\rho(e^{L_j^\star})\Big], \qquad \Psi_\rho(y) \;=\; \begin{cases} (y^\rho - 1)/\rho, & \rho < 0, \\ \log y, & \rho = 0, \end{cases} \tag{3.14}$$

combining the Hill bias $b_j^H = \lambda_j/(1-\rho)$, weighted by the log-rate $L_j^\star$, with the limiting second-order Weissman remainder $-\lambda_j \Psi_\rho(e^{L_j^\star})$. The intermediate log-order statistic contributes no bias term of its own: centred at $\log Q_X(\tau_{n_j})$ it has zero asymptotic bias on the $\sqrt{k_j}$ scale (Haan and Ferreira 2006, Theorem 2.4.8), as established in Appendix A.2. The covariance (3.12) does not depend on $\boldsymbol{B_c}^\star$. The proposition is stated for deterministic affine weights. The same limit holds for estimated weights satisfying $\hat{\boldsymbol{\omega}}_\gamma^\top \mathbf{1} = \hat{\boldsymbol{\omega}}_q^\top \mathbf{1} = 1$ exactly and $\hat{\boldsymbol{\omega}}_\gamma \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_\gamma$, $\hat{\boldsymbol{\omega}}_q \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_q$; this estimated-weight lift is deferred to Section 3.6.

Proof. See Appendix A.2. □

Proposition 3.2 is the joint counterpart, at the intermediate finite-log-rate level $L_j(\tau_n) \to L_j^\star \in \mathbb{R}$, of the pooled-Hill limit and weighted-geometric Weissman construction recalled in Section 2.5. The extreme-level, diverging-log-rate analogue is the subject of Chapter 4. The new feature for the expectile target is the cross-covariance $\boldsymbol{\omega}_\gamma^\top \operatorname{diag}(\boldsymbol{L}^\star) \boldsymbol{V_c} \boldsymbol{\omega}_q$: it is the structural reason why the variance-optimal weights $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$ for the pooled expectile estimator need not coincide, an observation that Section 3.5 develops quantitatively.

3.4 Asymptotic normality of the pooled expectile estimator

The main result of this chapter is the central limit theorem for the pooled QB intermediate expectile estimator (3.8). The result is obtained by feeding the joint pooled CLT of Proposition 3.2 through the smooth log-linear map $(g, x) \mapsto \log\psi(g) + x$ via a first-order Taylor expansion, combined with the second-order expectile–quantile expansion that controls the deterministic gap between $\log\xi_{\tau_n}$ and $\log\psi(\gamma) + \log Q_X(\tau_n)$.

Derivative of $\log\psi$. A short calculation gives

$$\frac{d}{dg}\log\psi(g) \;=\; \frac{1}{1-g} - \log(g^{-1}-1) \;=:\; m(g), \tag{3.15}$$

which is the same $m(\gamma)$ that governs the variance of the QB intermediate-level estimator in the iid baseline Theorem 2.12. The function $m(\,\cdot\,)$ is continuous on $(0, 1)$, vanishes at the value $\gamma^\star \approx 0.2178$ solving $(1-\gamma)^{-1} = \log(\gamma^{-1}-1)$, and is positive (resp. negative) for γ above (resp. below) $\gamma^\star$.

The deterministic expectile–quantile gap. The first-order asymptotic equivalence of Proposition 2.10,

$$\xi_\tau/Q_X(\tau) \to \psi(\gamma), \qquad \tau \uparrow 1,$$

sharpens, under Condition 2.3, through the second-order expansion (2.7) (Daouia, Girard, et al. 2020, Proposition 1(i)),

$$\begin{aligned}\log\xi_\tau - \log\psi(\gamma) - \log Q_X(\tau) = c(\gamma,\rho)\,A\big((1-\tau)^{-1}\big) + o\big(A((1-\tau)^{-1})\big)\\ + O\big(Q_X(\tau)^{-1}\big), \qquad \tau \uparrow 1.\end{aligned} \tag{3.16}$$

Here the constant $c(\gamma, \rho)$ is given explicitly in (2.7). The final term in (3.16) is the first-moment contribution in the DGS expansion; it is not, in general, an $o(A)$ term. What is needed for the CLT is only its sequence-level disappearance after multiplication by $\sqrt{k}$. Along the chapter target τ_n, (3.3) gives $\sqrt{k}/Q_X(\tau_n) \to 0$, while the bias-rate assumptions below give $\sqrt{k}\,A((1-\tau_n)^{-1}) = O(1)$. Hence

$$\sqrt{k}\Big[\log\xi_{\tau_n} - \log\psi(\gamma) - \log Q_X(\tau_n)\Big] = c(\gamma,\rho)\sqrt{k}\,A\big((1-\tau_n)^{-1}\big) + o(1). \tag{3.17}$$

The chapter setup of Section 3.1 already *implies* a finite limit

$$\Lambda^\bullet \;:=\; \lim_{n\to\infty}\sqrt{k}\,A\big((1-\tau_n)^{-1}\big) \;\in\; \mathbb{R}, \tag{3.18}$$

without any additional hypothesis: starting from the per-sample bias rate $\sqrt{k_j}\,A(n_j/k_j) \to \lambda_j$, the log-rate convergence $L_j(\tau_n) \to L_j^\star$, and the signed ratio convergence for the constant-sign auxiliary function A under Condition 2.3, one obtains $\sqrt{k_j}\,A((1-\tau_n)^{-1}) \to \lambda_j\,(e^{L_j^\star})^\rho$, and a further rescaling by $\sqrt{k/k_j} \to \sqrt{c_j\sum_i c_i^{-1}}$ delivers (3.18). Equivalently, with $S = \sum_i c_i^{-1}$, the constants already introduced obey the compatibility identity

$$\sqrt{c_j S}\,\lambda_j e^{\rho L_j^\star} = \sqrt{c_\ell S}\,\lambda_\ell e^{\rho L_\ell^\star} \qquad \text{for all } j, \ell,$$

because both sides are limits of the same sequence $\sqrt{k}\,A((1-\tau_n)^{-1})$; see Appendix A.3 for the explicit verification. The finite asymptotic constant

$$b^{\xi/Q} \;:=\; \lim_{n\to\infty} \sqrt{k}\,[\log\psi(\gamma) + \log Q_X(\tau_n) - \log\xi_{\tau_n}] \;=\; -\,c(\gamma,\rho)\,\Lambda^\bullet \tag{3.19}$$

follows. The sign convention in (3.19) is chosen so that $b^{\xi/Q}$ enters the asymptotic mean of the pooled estimator with a + sign in (3.22) below; that is, $b^{\xi/Q}$ is the deterministic *bias contribution* of the gap, not the gap itself. The contribution $b^{\xi/Q}$ is *absent* from the analogue quantile-target pooling theorem of Daouia, Padoan, et al. (2024): it is the deterministic correction that arises when the quantile target is replaced by the expectile target through the QB plug-in.

Theorem 3.3 (Pooled QB intermediate expectile CLT)**.** *Under the setup and heavy-tail regime of Section 3.1 (including Equations (3.2) and (3.3)), the positivity/log-domain convention of Section 3.2, the assumptions of Proposition 3.2, and the finite expectile–quantile bias limit (3.18) implied by these conditions, for any weight vectors $\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q \in \mathbb{R}^m$ satisfying $\boldsymbol{\omega}_\gamma^\top \mathbf{1} = \boldsymbol{\omega}_q^\top \mathbf{1} = 1$,*

$$\sqrt{k}\left(\frac{\widehat{\xi}^{\,\mathrm{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)}{\xi_{\tau_n}} - 1\right) \;\xrightarrow{d}\; \mathcal{N}(\mu^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q),\, \sigma^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2), \tag{3.20}$$

with asymptotic variance

$$\sigma^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2 \;=\; m(\gamma)^2\,\boldsymbol{\omega}_\gamma^\top \boldsymbol{V_c}\boldsymbol{\omega}_\gamma \;+\; 2\,m(\gamma)\,\boldsymbol{\omega}_\gamma^\top \mathrm{diag}(\boldsymbol{L}^\star)\boldsymbol{V_c}\boldsymbol{\omega}_q \;+\; \boldsymbol{\omega}_q^\top(\boldsymbol{V_c} + \mathrm{diag}(\boldsymbol{L}^\star)^2\boldsymbol{V_c})\boldsymbol{\omega}_q, \tag{3.21}$$

and asymptotic bias

$$\mu^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \;=\; m(\gamma)\,\boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c} \;+\; \boldsymbol{\omega}_q^\top \boldsymbol{B}^\star_{\boldsymbol{c}} \;+\; b^{\xi/Q}, \tag{3.22}$$

where $\boldsymbol{B_c}$ and $\boldsymbol{B}^\star_{\boldsymbol{c}}$ are as in Proposition 3.2 (the pooled-Hill and limiting intermediate-level pooled-Weissman bias vectors) and $b^{\xi/Q}$ is the asymptotic expectile–quantile bias contribution of (3.19).

Proof. See Appendix A.3. $\square$

Writing $\boldsymbol{D}^\star = \mathrm{diag}(\boldsymbol{L}^\star)$, the variance in (3.21) factorises as

$$\sigma^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2 = \left(m(\gamma)\boldsymbol{\omega}_\gamma + \boldsymbol{D}^\star\boldsymbol{\omega}_q\right)^\top \boldsymbol{V_c}\left(m(\gamma)\boldsymbol{\omega}_\gamma + \boldsymbol{D}^\star\boldsymbol{\omega}_q\right) + \boldsymbol{\omega}_q^\top \boldsymbol{V_c}\boldsymbol{\omega}_q.$$

This form makes the non-negativity immediate and shows how the log-rate vector $\boldsymbol{L}^\star$ couples the Hill and Weissman inputs. The weight-optimisation section exploits exactly this factorisation.

Reduction to the iid baseline. Theorem 3.3 contains Theorem 2.12 as the special case $m = 1$ with the chapter target aligned to the single sample's intermediate level. The weight vectors collapse to scalars $\omega_\gamma = \omega_q = 1$; the alignment forces $L_1^\star = 0$, $\Psi_\rho(e^{L_1^\star}) = 0$, and $\boldsymbol{V_c} = \gamma^2$. The asymptotic variance (3.21) collapses to $\gamma^2(1+m(\gamma)^2)$, in exact agreement with Theorem 2.12. The bias (3.22) collapses likewise. The pooled-Weissman bias vanishes, $\boldsymbol{B}_{\boldsymbol{c}}^\star = 0$: at $L_1^\star = 0$ the bracket in (3.14) reduces to $L_1^\star/(1-\rho) - \Psi_\rho(1) = 0$, since the intermediate order statistic has zero asymptotic bias on the $\sqrt{k}$ scale. Hence

$$\mu^\xi \;=\; m(\gamma)\,\boldsymbol{B}_{\boldsymbol{c}} + \boldsymbol{B}_{\boldsymbol{c}}^\star + b^{\xi/Q} \;=\; \frac{m(\gamma)}{1-\rho}\lambda \;-\; c(\gamma,\rho)\,\lambda \;=\; \Big(\frac{m(\gamma)}{1-\rho} - c(\gamma,\rho)\Big)\lambda, \tag{3.23}$$

the Hill bias $\lambda/(1-\rho)$ propagated through $m(\gamma)$, minus the deterministic expectile–quantile gap $c(\gamma,\rho)\lambda$ of (2.7). This is precisely the bias of Theorem 2.12: there is no cancellation between an order-statistic bias and the gap, because — as Appendix A.2 establishes — the intermediate order statistic has zero asymptotic bias on the $\sqrt{k}$ scale and contributes no bias term at all.

Equal-fraction distributed simplification. Under the equal-effective-fraction distributed specialisation, $k_j/n_j = (k/n)(1+o(1))$ for every j, the log-rate factor $L_j(\tau_n) = \log(k/(n(1-\tau_n))) + o(1)$ becomes a sample-free quantity. Writing $L^\star$ for its common limit, (3.21) simplifies to

$$\sigma^\xi(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)^2 \;=\; m(\gamma)^2\,\boldsymbol{\omega}_\gamma^\top \boldsymbol{V}_{\boldsymbol{c}}\boldsymbol{\omega}_\gamma \;+\; 2\,m(\gamma)\,L^\star\,\boldsymbol{\omega}_\gamma^\top \boldsymbol{V}_{\boldsymbol{c}}\boldsymbol{\omega}_q \;+\; (1+(L^\star)^2)\,\boldsymbol{\omega}_q^\top \boldsymbol{V}_{\boldsymbol{c}}\boldsymbol{\omega}_q. \tag{3.24}$$

With matched weights $\boldsymbol{\omega}_\gamma = \boldsymbol{\omega}_q = \boldsymbol{\omega}$ and the chapter target aligned to the common per-sample intermediate level ($L^\star = 0$), (3.24) collapses to $(1+m(\gamma)^2)\,\boldsymbol{\omega}^\top \boldsymbol{V}_{\boldsymbol{c}}\boldsymbol{\omega}$ — the natural pooled analogue of the iid intermediate QB variance $\gamma^2(1+m(\gamma)^2)$ of Theorem 2.12.

Why two weight vectors. The separate weight pair $(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)$ is needed to expose the full variance structure of the pooled QB expectile estimator: the cross term in (3.21) couples the two pooled inputs through $\operatorname{diag}(\boldsymbol{L}^\star)$ and is a genuine structural feature of the expectile target. As Section 3.5 shows, however, the additional freedom is not exploited by the variance optimum in the equal-fraction distributed case: under the additional equal-fraction specialisation of Section 3.1, the variance-optimal quantile weights remain $\boldsymbol{\omega}_q^{\xi,\mathrm{Var}} = \boldsymbol{\omega}^{\mathrm{Var}}$ (Proposition 3.4). When $m(\gamma)\neq 0$ the unique variance-optimal pair is matched, $\boldsymbol{\omega}_\gamma^{\xi,\mathrm{Var}} = \boldsymbol{\omega}_q^{\xi,\mathrm{Var}} = \boldsymbol{\omega}^{\mathrm{Var}}$; when $m(\gamma)=0$, the γ-weight is variance-unidentified and this matched vector is only a canonical representative.

3.5 Variance- and AMSE-optimal weights

We now identify the weight vectors $(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)$ that minimise the asymptotic variance (3.21) and the AMSE $(\mu^\xi)^2 + \sigma^{\xi,2}$ of the pooled QB expectile estimator, subject to the affine sum-to-one constraints $\boldsymbol{\omega}_\gamma^\top\mathbf{1} = \boldsymbol{\omega}_q^\top\mathbf{1} = 1$. The qualitative question driving this section is whether the expectile-optimal weights coincide with those for the quantile target of Proposition 2.17, or whether the expectile target requires its own weight scheme.

Variance-optimal weights

The variance (3.21) admits the identity

$$\sigma^\xi(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)^2 \;=\; \boldsymbol{u}^\top \boldsymbol{V}_{\boldsymbol{c}}\boldsymbol{u} \;+\; \boldsymbol{\omega}_q^\top \boldsymbol{V}_{\boldsymbol{c}}\boldsymbol{\omega}_q, \qquad \boldsymbol{u} \;:=\; m(\gamma)\,\boldsymbol{\omega}_\gamma + \operatorname{diag}(\boldsymbol{L}^\star)\,\boldsymbol{\omega}_q, \tag{3.25}$$

which one verifies by direct expansion using the diagonality of $\boldsymbol{V_c}$ (and hence its commutativity with $\operatorname{diag}(\boldsymbol{L}^\star)$). For a quadratic form $\boldsymbol{x}^\top \boldsymbol{V_c} \boldsymbol{x}$ under a constraint $\boldsymbol{x}^\top \mathbf{1} = d$, the minimiser is proportional to $\boldsymbol{V_c}^{-1}\mathbf{1}$. Whether both terms in (3.25) can be minimised simultaneously by one pair $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$ depends on whether the linear constraint induced on $\boldsymbol{u}$ by the sum-to-one constraints is sample-free. This happens when $\boldsymbol{L}^\star$ is a scalar multiple of $\mathbf{1}$; in the independent common-distribution distributed specialisation of Section 3.1, this algebraic collapse is induced by the equal-fraction condition $k_j/n_j = (k/n)(1+o(1))$.

Proposition 3.4 (Variance-optimal weights, equal-fraction distributed inference)**.** *Under the assumptions of Theorem 3.3 and the equal-effective-sample-fraction condition $k_j/n_j = (k/n)(1 + o(1))$ described in Section 3.1, the variance-optimal weights for the pooled QB expectile estimator satisfy*

$$\boldsymbol{\omega}_q^{\xi,\mathrm{Var}} \;=\; \boldsymbol{\omega}^{\mathrm{Var}} \;=\; \frac{\boldsymbol{V_c}^{-1}\mathbf{1}}{\mathbf{1}^\top \boldsymbol{V_c}^{-1}\mathbf{1}}, \tag{3.26}$$

with corresponding asymptotic variance

$$\sigma^\xi\big(\boldsymbol{\omega}_\gamma^{\xi,\mathrm{Var}}, \boldsymbol{\omega}_q^{\xi,\mathrm{Var}}\big)^2 \;=\; \left[(m(\gamma) + L^\star)^2 + 1\right] \big(\mathbf{1}^\top \boldsymbol{V_c}^{-1}\mathbf{1}\big)^{-1}. \tag{3.27}$$

If $m(\gamma) \neq 0$, the variance-optimal pair is unique and $\boldsymbol{\omega}_\gamma^{\xi,\mathrm{Var}} = \boldsymbol{\omega}^{\mathrm{Var}}$ as well, so both pooled inputs receive the quantile-pooling weights of Daouia, Padoan, et al. (2024). If $m(\gamma) = 0$, the asymptotic variance does not depend on $\boldsymbol{\omega}_\gamma$: any admissible vector is then variance-optimal, and the matched representative $\boldsymbol{\omega}_\gamma^{\xi,\mathrm{Var}} = \boldsymbol{\omega}^{\mathrm{Var}}$ is one canonical choice.

Proof. The proposition is the specialisation $\boldsymbol{L}^\star = L^\star \mathbf{1}$ of Proposition 3.5; see Appendix A.4, which discharges both the $m(\gamma) \neq 0$ closed form and the reduced $m(\gamma) = 0$ case. □

Proposition 3.4 is the central variance-efficiency statement of this chapter: under equal-fraction distributed inference, *the variance-optimal pooling weights are invariant under the change of target from extreme quantile to extreme expectile.* The two targets share the same data-fusion recipe; the only change in the asymptotic variance is the multiplicative factor $(m(\gamma) + L^\star)^2 + 1$ replacing the corresponding Hill/extreme-quantile factor of 1 under the corresponding normalisation. Under the stronger oracle-rate equal-fraction condition of Corollary 2.21, this identifies the same pooling rule that would enter the corresponding unfeasible pooled-sample comparison for an expectile target; the present chapter uses only the variance-optimality statement proved above, rather than a separate oracle-equivalence theorem for a newly defined pooled-sample expectile benchmark. The AMSE criterion requires separate bookkeeping because the bias term is target-specific; as the second subsection below explains, however, the equal-fraction distributed setting is also a collapse regime for AMSE, while genuinely target-dependent AMSE weights can appear once the effective sample fractions are unequal.

Beyond the equal-fraction setting. When the effective sample fractions k_j/n_j vary across j, $\operatorname{diag}(\boldsymbol{L}^\star)$ ceases to be a scalar multiple of the identity, the constraint $\boldsymbol{u}^\top \mathbf{1} = m(\gamma) + \boldsymbol{\omega}_q^\top \boldsymbol{L}^\star$ acquires explicit dependence on $\boldsymbol{\omega}_q$, and the decoupling argument used in Proposition 3.4 breaks. The variance-optimal pair then lifts off the matched-weight diagonal and admits the explicit closed form recorded below. Set $\boldsymbol{D} := \operatorname{diag}(\boldsymbol{L}^\star)$ and abbreviate the three scalar invariants of $(\boldsymbol{V_c}, \boldsymbol{L}^\star)$:

$$a := \mathbf{1}^\top \boldsymbol{V_c}^{-1}\mathbf{1}, \qquad b := \mathbf{1}^\top \boldsymbol{V_c}^{-1}\boldsymbol{L}^\star, \qquad c := (\boldsymbol{L}^\star)^\top \boldsymbol{V_c}^{-1}\boldsymbol{L}^\star, \qquad \Delta := a(a+c) - b^2. \tag{3.28}$$

Proposition 3.5 (Variance-optimal weights, general case)**.** *Under the assumptions of Theorem 3.3 and $m(\gamma) \neq 0$, the Cauchy–Schwarz inequality in the $\boldsymbol{V_c}^{-1}$-inner product gives $ac - b^2 \geq 0$, with equality iff $\boldsymbol{L}^\star$ is a scalar multiple of $\mathbf{1}$. This is the limiting collapse induced by the equal-effective-sample-fraction condition in the distributed specialisation. Consequently*

$$\Delta \;=\; a^2 + (ac - b^2) \;\geq\; a^2 \;>\; 0,$$

with $\Delta = a^2$ exactly under the equal-fraction distributed specialisation of Proposition 3.4. The variance-optimal pooling weights for the pooled QB intermediate expectile estimator are unique and given by

$$\boldsymbol{\omega}_q^{\xi,\mathrm{Var}} \;=\; \frac{(a + c + m(\gamma)\, b)\, \boldsymbol{V_c}^{-1}\mathbf{1} \;-\; (m(\gamma)\, a + b)\, \boldsymbol{V_c}^{-1}\boldsymbol{L}^\star}{\Delta}, \tag{3.29}$$

$$\boldsymbol{\omega}_\gamma^{\xi,\mathrm{Var}} \;=\; \frac{(m(\gamma)\, a + b)\, \boldsymbol{V_c}^{-1}(\mathbf{1} + \boldsymbol{D}^2\mathbf{1}) \;-\; (a + c + m(\gamma)\, b)\, \boldsymbol{V_c}^{-1}\boldsymbol{L}^\star}{m(\gamma)\, \Delta}, \tag{3.30}$$

and the associated minimum asymptotic variance is

$$\sigma^\xi\big(\boldsymbol{\omega}_\gamma^{\xi,\mathrm{Var}}, \boldsymbol{\omega}_q^{\xi,\mathrm{Var}}\big)^2 \;=\; \frac{(m(\gamma)^2 + 1)\, a \;+\; 2\, m(\gamma)\, b \;+\; c}{\Delta}. \tag{3.31}$$

Under the equal-effective-fraction condition $\boldsymbol{L}^\star = L^\star \mathbf{1}$ one has $b = L^\star a$, $c = (L^\star)^2 a$, $\Delta = a^2$; formulas (3.29)–(3.31) then collapse to (3.26) and (3.27), recovering Proposition 3.4.

Proof. See Appendix A.4, where the change of variables $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \mapsto (\boldsymbol{u}, \boldsymbol{\omega}_q)$ with $\boldsymbol{u} = m(\gamma)\boldsymbol{\omega}_\gamma + \boldsymbol{D}\boldsymbol{\omega}_q$, positivity of Δ from Cauchy–Schwarz, and the explicit Lagrangian dual system are spelled out step-by-step. ☐

The assumption $m(\gamma) \neq 0$ is essential for the two-vector closed form: it is the condition that makes the change of variables $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \mapsto (\boldsymbol{u}, \boldsymbol{\omega}_q)$ a bijection, and it appears in the denominator of (3.30). When $m(\gamma) = 0$ the asymptotic variance does not identify $\boldsymbol{\omega}_\gamma$, and the optimisation collapses to the single-vector problem in $\boldsymbol{\omega}_q$ recorded in Appendix A.4.

Departure from the matched-weight diagonal. The discrepancy between $(\boldsymbol{\omega}_\gamma^{\xi,\mathrm{Var}}, \boldsymbol{\omega}_q^{\xi,\mathrm{Var}})$ and the Daouia–Padoan–Stupfler matched-weight pair $(\boldsymbol{\omega}^{\mathrm{Var}}, \boldsymbol{\omega}^{\mathrm{Var}})$ is controlled by the $\boldsymbol{V_c}^{-1}$-Cauchy–Schwarz defect $ac - b^2 \geq 0$, which vanishes exactly when $\boldsymbol{L}^\star$ is a scalar multiple of $\mathbf{1}$. Writing $\bar{L}^\star := b/a$ for the $\boldsymbol{V_c}^{-1}\mathbf{1}$-weighted mean of $\boldsymbol{L}^\star$, a direct expansion of (3.29) delivers the closed-form discrepancy

$$\boldsymbol{\omega}_q^{\xi,\mathrm{Var}} - \boldsymbol{\omega}^{\mathrm{Var}} \;=\; -\,\frac{m(\gamma)\, a + b}{\Delta}\, \boldsymbol{V_c}^{-1}(\boldsymbol{L}^\star - \bar{L}^\star \mathbf{1}), \tag{3.32}$$

verified in Appendix A.4 (Step (v)), with an analogous identity for $\boldsymbol{\omega}_\gamma^{\xi,\mathrm{Var}}$ obtained by inversion of the $\boldsymbol{u}$-definition. The correction lies entirely along the $\boldsymbol{V_c}^{-1}$-centred direction $\boldsymbol{V_c}^{-1}(\boldsymbol{L}^\star - \bar{L}^\star \mathbf{1})$. Writing $\Delta = a^2 + (ac - b^2)$ exhibits its scalar prefactor as

$$\frac{m(\gamma)\,a + b}{\Delta} \;=\; \frac{m(\gamma) + \bar{L}^\star}{a} \cdot \left[1 \;+\; O\!\left(\frac{ac - b^2}{a^2}\right)\right]$$

in the small Cauchy–Schwarz defect regime $ac - b^2 \ll a^2$. The quantile-pooling weights therefore remain near-optimal whenever the log-rate vector is nearly sample-free ($ac - b^2 \approx 0$, i.e. $\boldsymbol{L}^\star$ nearly constant in the $\boldsymbol{V_c}^{-1}\mathbf{1}$-weighted sense). The quantile-weight component $\boldsymbol{\omega}_q^{\xi,\mathrm{Var}}$ also remains close to $\boldsymbol{\omega}^{\mathrm{Var}}$ when $m(\gamma) + \bar{L}^\star \approx 0$, which combines the expectile-specific zero $\gamma^\star \approx 0.218$ of $m(\,\cdot\,)$ with the log-rate weighted mean; this second condition by itself controls the displayed $\boldsymbol{\omega}_q$ correction, not the full two-vector pair. The degenerate case $m(\gamma) = 0$ leaves $\boldsymbol{\omega}_\gamma$ indeterminate and reduces the optimisation to the single-vector problem $\min \boldsymbol{\omega}_q^\top \boldsymbol{V_c}(\boldsymbol{I} + \boldsymbol{D}^2)\boldsymbol{\omega}_q$ subject to $\boldsymbol{\omega}_q^\top \mathbf{1} = 1$, whose solution is recorded in Appendix A.4; it coincides with $\boldsymbol{\omega}^{\mathrm{Var}}$ in the equal-fraction setting and deviates from it otherwise.

AMSE-optimal weights

For the closed-form stacked optimisation in this subsection, assume $m(\gamma) \neq 0$. The degenerate case $m(\gamma) = 0$ requires a reduced optimisation because the variance term no longer identifies the Hill weights; we return to that boundary case whenever it affects a stated collapse result.

The AMSE-optimal pair $(\boldsymbol{\omega}_\gamma^{\xi,\mathrm{AMSE}}, \boldsymbol{\omega}_q^{\xi,\mathrm{AMSE}})$ minimises the asymptotic mean squared error

$$\mathrm{AMSE}^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \;:=\; \big(\mu^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)\big)^2 + \sigma^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2 \tag{3.33}$$

subject to the affine sum-to-one constraints $\boldsymbol{\omega}_\gamma^\top \mathbf{1} = \boldsymbol{\omega}_q^\top \mathbf{1} = 1$, with μ^ξ and σ^ξ as in (3.22) and (3.21). The objective parallels Proposition 2.17(ii), but the expectile target reshapes the bias landscape away from the equal-fraction collapse regime. In the two-weight problem the bias is a linear functional of the stacked vector $(\boldsymbol{\omega}_\gamma^\top, \boldsymbol{\omega}_q^\top)^\top$ with *stacked gradient* $(m(\gamma)\boldsymbol{B_c}, \boldsymbol{B_c^\star})^\top$: the pooled-Hill bias enters rescaled by the QB Jacobian $m(\gamma)$, while the pooled-Weissman bias enters unrescaled. Only after restricting to the matched-weight diagonal $\boldsymbol{\omega}_\gamma = \boldsymbol{\omega}_q$ does one see the composite direction $m(\gamma)\boldsymbol{B_c} + \boldsymbol{B_c^\star}$, which is the expectile analogue of the single quantile-pooling bias vector $\boldsymbol{B_c}$. In addition, μ^ξ carries the additive offset $b^{\xi/Q}$ of (3.19). This offset has no effect on the variance problem, but it enters the AMSE first-order conditions whenever the remaining bias part depends on the weights.

Stacked notation. To state the optimum compactly, stack the two pooling weight vectors into $\boldsymbol{w} := (\boldsymbol{\omega}_\gamma^\top, \boldsymbol{\omega}_q^\top)^\top \in \mathbb{R}^{2m}$ and assemble the bias gradient and variance Hessian implied by (3.22)–(3.21):

$$\boldsymbol{h} \;:=\; \begin{pmatrix} m(\gamma)\,\boldsymbol{B_c} \\ \boldsymbol{B_c^\star} \end{pmatrix} \in \mathbb{R}^{2m}, \qquad \boldsymbol{Q} \;:=\; \begin{pmatrix} m(\gamma)^2\,\boldsymbol{V_c} & m(\gamma)\,\boldsymbol{D}\,\boldsymbol{V_c} \\ m(\gamma)\,\boldsymbol{D}\,\boldsymbol{V_c} & \boldsymbol{V_c} + \boldsymbol{D}^2\boldsymbol{V_c} \end{pmatrix} \in \mathbb{R}^{2m\times 2m}, \tag{3.34}$$

so that $\mu^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) = \boldsymbol{h}^\top \boldsymbol{w} + b^{\xi/Q}$ and $\sigma^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2 = \boldsymbol{w}^\top \boldsymbol{Q}\boldsymbol{w}$. Write $\boldsymbol{A} \in \mathbb{R}^{2m\times 2}$ for the matrix with columns $\mathbf{1}_g := (\mathbf{1}^\top, \mathbf{0}^\top)^\top$ and $\mathbf{1}_q := (\mathbf{0}^\top, \mathbf{1}^\top)^\top$, so that the sum-to-one constraints read $\boldsymbol{A}^\top \boldsymbol{w} = \mathbf{1}_2$.

Proposition 3.6 (AMSE-optimal weights)**.** *Under the assumptions of Theorem 3.3 and $m(\gamma) \neq 0$, the matrix $\boldsymbol{M} := \boldsymbol{Q} + \boldsymbol{h}\boldsymbol{h}^\top$ is positive definite, and the AMSE objective (3.33) is strictly convex on the affine constraint set $\{\boldsymbol{w} \in \mathbb{R}^{2m} : \boldsymbol{A}^\top \boldsymbol{w} = \mathbf{1}_2\}$. The unique minimiser admits the closed form*

$$\boldsymbol{w}^{\xi,\mathrm{AMSE}} = \boldsymbol{M}^{-1}\Big[\boldsymbol{A}\,(\boldsymbol{A}^\top \boldsymbol{M}^{-1}\boldsymbol{A})^{-1}(\mathbf{1}_2 + b^{\xi/Q}\,\boldsymbol{A}^\top \boldsymbol{M}^{-1}\boldsymbol{h}) - b^{\xi/Q}\,\boldsymbol{h}\Big], \tag{3.35}$$

and is equivalently characterised by the first-order conditions

$$\begin{aligned} m(\gamma)^2\, \boldsymbol{V_c}\,\boldsymbol{\omega}_\gamma + m(\gamma)\,\boldsymbol{D}\,\boldsymbol{V_c}\,\boldsymbol{\omega}_q &= \tfrac{1}{2}\lambda_\gamma \mathbf{1} - m(\gamma)\,\mu^\xi\,\boldsymbol{B_c}, \\ m(\gamma)\,\boldsymbol{D}\,\boldsymbol{V_c}\,\boldsymbol{\omega}_\gamma + (\boldsymbol{V_c} + \boldsymbol{D}^2\boldsymbol{V_c})\,\boldsymbol{\omega}_q &= \tfrac{1}{2}\lambda_q \mathbf{1} - \mu^\xi\,\boldsymbol{B}^\star_{\boldsymbol{c}}, \end{aligned} \tag{3.36}$$

together with the sum-to-one constraints $\boldsymbol{\omega}_\gamma^\top \mathbf{1} = \boldsymbol{\omega}_q^\top \mathbf{1} = 1$ and the self-referential identity $\mu^\xi = m(\gamma)\,\boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c} + \boldsymbol{\omega}_q^\top \boldsymbol{B}^\star_{\boldsymbol{c}} + b^{\xi/Q}$. The Lagrange multipliers $\lambda_\gamma, \lambda_q \in \mathbb{R}$ are uniquely determined by these constraints.

Proof. See Appendix A.5, where positive-definiteness of $\boldsymbol{Q}$ (and hence of $\boldsymbol{M} = \boldsymbol{Q} + \boldsymbol{h}\boldsymbol{h}^\top$) is verified in Step (ii) via the variance factorisation, the Lagrangian and explicit FOC normalisation are recorded in Step (iii), and the block decomposition matching (3.36) is in Step (iv). ☐

Equal-fraction AMSE collapse. Under the equal-effective-sample-fraction condition $k_j/n_j = (k/n)(1+o(1))$, the log-rate vector is $\boldsymbol{L}^\star = L^\star \mathbf{1}$. The compatibility identity following (3.18) then gives

$$\sqrt{c_j S}\,\lambda_j = \Lambda^\bullet (e^{L^\star})^{-\rho}, \qquad S = \sum_{i=1}^m c_i^{-1},$$

which is sample-free. Combining this with (3.14) yields

$$\boldsymbol{B_c} = \beta_H \mathbf{1}, \qquad \boldsymbol{B}^\star_{\boldsymbol{c}} = \beta_W \mathbf{1}$$

for constants β_H, β_W depending on $(\gamma, \rho, L^\star, \Lambda^\bullet)$ but not on j. Hence, for all admissible weight pairs,

$$\mu^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) = m(\gamma)\beta_H + \beta_W + b^{\xi/Q},$$

which is constant on the constraint set. In this regime the AMSE criterion is therefore the variance criterion plus a constant. If $m(\gamma) \neq 0$, the unique AMSE-optimal pair is the same matched pair as in Proposition 3.4,

$$\boldsymbol{\omega}_\gamma^{\xi,\mathrm{AMSE}} = \boldsymbol{\omega}_q^{\xi,\mathrm{AMSE}} = \boldsymbol{\omega}^{\mathrm{Var}},$$

and when $m(\gamma) = 0$ the same indeterminacy of $\boldsymbol{\omega}_\gamma$ described in Proposition 3.4 remains. This is exactly parallel to the equal-fraction distributed setting of Daouia, Padoan, et al. (2024): there too the quantile-pooling bias vector $\boldsymbol{B_c}$ is collinear with $\mathbf{1}$, so the AMSE criterion collapses to the variance criterion under the affine constraint.

Comparison with the Daouia–Padoan–Stupfler AMSE optimum away from equal fractions. Outside the equal-fraction collapse regime, the expectile AMSE-optimum generally does *not* reduce to the Daouia–Padoan–Stupfler AMSE-optimum of Proposition 2.17(ii). Three structural differences are responsible:

(i) the bias gradient is the stacked vector $\boldsymbol{h} = (m(\gamma)\boldsymbol{B_c}, \boldsymbol{B_c^\star})^\top$ rather than the single-block $\boldsymbol{B_c}$ of Proposition 2.17;

(ii) μ^ξ carries the additive offset $b^{\xi/Q}$, which enters the AMSE first-order conditions through the self-consistent mean μ^ξ;

(iii) the expectile minimiser is genuinely a pair $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$, and the diagonal $\boldsymbol{\omega}_\gamma = \boldsymbol{\omega}_q$ is generically not a solution to the FOC (3.36) because the Hill and Weissman blocks of $\boldsymbol{h}$ enter asymmetrically through $m(\gamma)\boldsymbol{B_c}$ and $\boldsymbol{B_c^\star}$.

Exact coincidence with the Daouia–Padoan–Stupfler AMSE weights can occur, but it is a compatibility condition on the full FOC system rather than a simple proportionality of $\boldsymbol{B_c}$ and $\boldsymbol{B_c^\star}$. One would have to insert the Daouia–Padoan–Stupfler candidate weight into (3.36), impose the two affine constraints, and solve the resulting block equations in the quantities $(\boldsymbol{V_c}, \boldsymbol{D}, \boldsymbol{B_c}, \boldsymbol{B_c^\star}, b^{\xi/Q})$. We do not characterise that exceptional locus here. The robust qualitative point is narrower and sharper: the equal-fraction distributed specialisation preserves the Daouia–Padoan–Stupfler weights for both variance and AMSE, whereas unequal effective sample fractions generically make the expectile AMSE calibration target-dependent.

3.6 Estimated weights and confidence intervals

The optimal weights of Propositions 3.4 to 3.6 are unknown in practice: they depend on the population covariance $\boldsymbol{V_c}$, on the log-rate vector $\boldsymbol{L}^\star$, on $m(\gamma)$, and — for the AMSE optimum — on the bias quantities $\boldsymbol{B_c}, \boldsymbol{B_c^\star}$ and the expectile–quantile offset $b^{\xi/Q}$. We follow the plug-in scheme of Daouia, Padoan, et al. (2024, §2.2) recalled in Section 2.5: each population quantity is replaced by a consistent empirical analogue, and the resulting plug-in weights are substituted into the pooled estimator (3.8). This section organises the plug-in inputs by what each optimum needs, states the resulting confidence interval under undersmoothing (Corollary 3.7) with the variance-optimal plug-in as the default implementation, and contrasts it with the bias-corrected variant required in the AMSE-optimal regime (Remark 3.8).

Feasible plug-in weights

The three optima of Section 3.5 have different plug-in demands. Listing them in order of increasing complexity:

(a) *Equal-fraction variance-optimal weights* (Proposition 3.4). Under $k_j/n_j = (k/n)(1+o(1))$ the optimum collapses to $\boldsymbol{\omega}^{\mathrm{Var}} = \boldsymbol{V_c}^{-1}\mathbf{1}/(\mathbf{1}^\top \boldsymbol{V_c}^{-1}\mathbf{1})$. Since the independent-samples covariance

(2.14) is diagonal with a common multiplicative factor, this population vector has coordinates $\omega_j^{\mathrm{Var}} \propto c_j^{-1}$. The canonical distributed plug-in is therefore the effective-sample-size weight used by Daouia, Padoan, et al. (2024, §3.2),

$$\hat{\omega}_j^{\mathrm{Var}} = \frac{k_j}{k}, \qquad j = 1, \ldots, m. \tag{3.37}$$

Equivalently, any consistent diagonal covariance plug-in $\widehat{\boldsymbol{V}}_{\boldsymbol{c}} \xrightarrow{\mathbb{P}} \boldsymbol{V}_{\boldsymbol{c}}$ may be inserted into the generic formula of Daouia, Padoan, et al. (2024), $\widehat{\boldsymbol{V}}_{\boldsymbol{c}}^{-1}\mathbf{1}/(\mathbf{1}^\top \widehat{\boldsymbol{V}}_{\boldsymbol{c}}^{-1}\mathbf{1})$; it has the same limit. These are two asymptotically equivalent implementations, not the same finite-sample rule unless $\widehat{\boldsymbol{V}}_{\boldsymbol{c}}$ has the distributed diagonal shape proportional to $\mathrm{diag}(k/k_1, \ldots, k/k_m)$. Below the default interval uses the non-random vector (3.37) and evaluates the variance by direct substitution into (3.21); if the inverse-$\widehat{\boldsymbol{V}}_{\boldsymbol{c}}$ rule is used instead, the denominator form $(\mathbf{1}^\top \widehat{\boldsymbol{V}}_{\boldsymbol{c}}^{-1}\mathbf{1})^{-1}$ is exact for that alternative plug-in.

(b) *General variance-optimal weights* (Proposition 3.5). When k_j/n_j varies across j, the closed forms (3.29)–(3.30) also depend on $m(\gamma)$ and on the log-rate vector $\boldsymbol{L}^\star$. The plug-in therefore additionally requires $\hat{m} \xrightarrow{\mathbb{P}} m(\gamma)$ and $L_{j,n} \to L_j^\star$; the latter is deterministic ($L_{j,n} := L_j(\tau_n) = \log(k_j/(n_j(1-\tau_n)))$ converges by the chapter setup of Section 3.1), so the only random plug-in beyond $\widehat{\boldsymbol{V}}_{\boldsymbol{c}}$ is $\hat{m}$. To avoid circularity — the general formula for $\hat{\boldsymbol{\omega}}_\gamma$ would otherwise depend on $\hat{m}$, which would itself depend on $\hat{\boldsymbol{\omega}}_\gamma$ — we seed $\hat{m}$ from a preliminary consistent pooled Hill estimator that needs no weight optimisation, for instance the effective-sample-size pool

$$\hat{m}^{(0)} := m\Big(\sum_{j=1}^{m}(k_j/k)\,\hat{\gamma}_j(k_j)\Big), \qquad \hat{m}^{(0)} \xrightarrow{\mathbb{P}} m(\gamma), \tag{3.38}$$

which is consistent under Theorem 2.6 and avoids the self-reference. One may optionally iterate after computing the weights — updating $\hat{m} \leftarrow m(\hat{\gamma}_n(\hat{\boldsymbol{\omega}}_\gamma))$ once the weights have been formed — but this is asymptotically immaterial. The closed forms (3.29)–(3.30) are used only away from the interior root of $m(\,\cdot\,)$: formally, one may assume $|m(\gamma)| > \varepsilon$ and compute the plug-in on the event $|\hat{m}| > \varepsilon/2$, whose probability tends to one. If $|\hat{m}|$ is numerically close to zero, the variance problem should be treated through the $m(\gamma) = 0$ single-vector solution displayed in (A.31), with $\boldsymbol{\omega}_\gamma$ supplied by a fixed canonical affine choice such as (3.37).

(c) *AMSE-optimal weights* (Proposition 3.6). The closed form (3.35) depends on all of $\boldsymbol{V}_{\boldsymbol{c}}, \boldsymbol{L}^\star, m(\gamma), \boldsymbol{B}_{\boldsymbol{c}}, \boldsymbol{B}_{\boldsymbol{c}}^\star, b^{\xi/Q}$ through the stacked vector $\boldsymbol{h}$ and Hessian $\boldsymbol{Q}$ of (3.34). The plug-in therefore requires, on top of $\widehat{\boldsymbol{V}}_{\boldsymbol{c}}$ and $\hat{m}$, consistent estimators of the second-order quantities

$$\widehat{\boldsymbol{B}}_{\boldsymbol{c}} \xrightarrow{\mathbb{P}} \boldsymbol{B}_{\boldsymbol{c}}, \qquad \widehat{\boldsymbol{B}}_{\boldsymbol{c}}^\star \xrightarrow{\mathbb{P}} \boldsymbol{B}_{\boldsymbol{c}}^\star, \qquad \hat{b}^{\xi/Q} \xrightarrow{\mathbb{P}} b^{\xi/Q}. \tag{3.39}$$

These convergences are not automatic consequences of the abstract second-order condition $\mathcal{C}_2(\gamma, \rho, A)$: that condition specifies the limiting auxiliary function A, not a directly estimable finite-dimensional parameter. Operationally one needs an additional second-order working model, such as the Hall-type parametrisation $A(t) = \gamma\beta t^{\rho}$ used by Daouia, Padoan, et al. (2024, §2.2). Under such a parametrisation, machinewise estimators $\hat{\beta}_j, \hat{\rho}_j$ of the common second-order parameters can be inserted into (3.14)–(3.19) and into the formula for $\boldsymbol{B_c}$ from Theorem 2.16. Without this extra model, (3.39) should be read as a high-level assumption that consistent estimators of the limiting bias constants are available, not as a consequence of $\mathcal{C}_2(\gamma, \rho, A)$ alone. In the absence of such a model and of estimators satisfying (3.39), the AMSE-optimal formula is best read as an oracle or sensitivity-analysis device rather than as a fully feasible procedure. Thus a practical plug-in AMSE implementation is exploratory unless the application supplies, and defends, a stable second-order calibration. The closed form in Proposition 3.6 is also not claimed at the root $m(\gamma) = 0$; confidence intervals at that root require an externally supplied consistent affine weight pair.

The consistency argument is a local one. At the population value, $\mathbf{1}^\top \boldsymbol{V_c}^{-1}\mathbf{1} > 0$; in the general variance formula $\Delta > 0$ and, on the region where it is used, $m(\gamma) \neq 0$; in the AMSE formula $\boldsymbol{M}$ is positive definite and $\det(\boldsymbol{A}^\top \boldsymbol{M}^{-1}\boldsymbol{A}) > 0$. Continuity therefore gives a neighbourhood of the population arguments on which the relevant denominators remain non-zero. The assumed plug-in convergences place the estimated arguments in this neighbourhood with probability tending one, and the continuous-mapping theorem then transfers the convergences above into consistency $\hat{\boldsymbol{\omega}}_\gamma \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_\gamma, \hat{\boldsymbol{\omega}}_q \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_q$ of the optimal-weight plug-ins, and from there into consistency of the plug-in variance $\hat{\sigma}^{\xi}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q) \xrightarrow{\mathbb{P}} \sigma^{\xi}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$ via (3.21). By construction the closed forms (3.37), (3.29)–(3.30) and (3.35) are normalised against the affine sum-to-one constraints, so $\hat{\boldsymbol{\omega}}_\gamma^\top \mathbf{1} = \hat{\boldsymbol{\omega}}_q^\top \mathbf{1} = 1$ holds *exactly*, not just in probability — a property that the estimated-weight lift in Corollary 3.7 below relies on.

Practical recommendation. The complexity gradient (a) → (b) → (c) above is also a finite-sample stability gradient. Estimators of the second-order parameter ρ and the rate function $A(\,\cdot\,)$ that (3.39) relies on are notoriously delicate in the heavy-tailed regime; see the bias-correction discussion of Daouia, Girard, et al. (2018, §3.1.1) and Daouia, Padoan, et al. (2024, Remark 4). Because such second-order estimators are often unstable in finite samples, the variance-optimal plug-ins (a)–(b) are the conservative default unless the application supplies reliable second-order calibration for ρ and A.

Confidence interval under undersmoothing

The asymptotic bias (3.22) is the linear combination

$$\mu^{\xi}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \;=\; m(\gamma)\, \boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c} \;+\; \boldsymbol{\omega}_q^\top \boldsymbol{B_c^{\star}} \;+\; b^{\xi/Q}.$$

The formulas (3.14) and Theorem 2.16 show that $\boldsymbol{B_c}$ and $\boldsymbol{B_c^{\star}}$ are continuous linear combinations of the per-sample bias rates $\lambda_j := \lim \sqrt{k_j}\, A(n_j/k_j)$. The offset $b^{\xi/Q}$ is controlled by the compatibility identity following (3.18),

$$\Lambda^{\bullet} = \sqrt{c_j S}\, \lambda_j e^{\rho L_j^{\star}}, \qquad S = \sum_{i=1}^{m} c_i^{-1}, \qquad j = 1, \ldots, m,$$

which identifies the common limit $\sqrt{k}A((1-\tau_n)^{-1})$ through any sample j. The *undersmoothing calibration*

$$\sqrt{k_j}\, A(n_j/k_j) \;\to\; 0 \qquad \text{for every } j = 1, \ldots, m, \tag{3.40}$$

therefore forces $\boldsymbol{B_c} = \mathbf{0}$, $\boldsymbol{B}_{\boldsymbol{c}}^{\star} = \mathbf{0}$, $\Lambda^{\bullet} = 0$, $b^{\xi/Q} = 0$, and consequently $\mu^{\xi}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) = 0$ for every admissible pair $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$. This is the hypothesis under which the variance-only confidence interval below is asymptotically valid. Without (3.40) the limit-bias is generically non-zero and the interval must be re-centred on a bias-corrected point estimator (Remark 3.8).

Corollary 3.7 (Asymptotic confidence interval for ξ_{τ_n})**.** *Assume the setting of Theorem 3.3 together with the undersmoothing condition* (3.40)*, and let* $(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q)$ *be estimated weight pairs satisfying the affine sum-to-one constraints* $\hat{\boldsymbol{\omega}}_\gamma^\top \mathbf{1} = \hat{\boldsymbol{\omega}}_q^\top \mathbf{1} = 1$ *exactly and the consistency conditions*

$$\hat{\boldsymbol{\omega}}_\gamma \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_\gamma, \qquad \hat{\boldsymbol{\omega}}_q \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_q, \qquad \widehat{\boldsymbol{V}}_{\boldsymbol{c}} \xrightarrow{\mathbb{P}} \boldsymbol{V}_{\boldsymbol{c}}, \qquad \hat{m} \xrightarrow{\mathbb{P}} m(\gamma), \tag{3.41}$$

for some deterministic limit pair $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$ *on the affine constraint set. By the right-heavy-tailed setup and the expectile–quantile equivalence,* $\xi_{\tau_n} > 0$ *eventually; the local Weissman factors, and hence the pooled estimator, are positive with probability tending one. Thus the logarithms below are well defined on an event whose probability tends to one. Then, for every* $\alpha \in (0,1)$*, the random log-scale interval*

$$I_n^{\log} \;:=\; \log \widehat{\xi}^{\text{pool}}_{\tau_n}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q) \;\pm\; z_{1-\alpha/2}\, \frac{\hat{\sigma}^{\xi}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q)}{\sqrt{k}} \tag{3.42}$$

and the associated original-scale interval

$$I_n := \exp(I_n^{\log}) = [\exp(a_n), \exp(b_n)] \quad \textit{when } I_n^{\log} = [a_n, b_n],$$

are asymptotic $(1-\alpha)$ *confidence intervals for* $\log \xi_{\tau_n}$ *and* ξ_{τ_n} *respectively:*

$$\mathbb{P}\big(\log \xi_{\tau_n} \in I_n^{\log}\big) \;\to\; 1-\alpha, \qquad \mathbb{P}\big(\xi_{\tau_n} \in I_n\big) \;\to\; 1-\alpha,$$

as $n \to \infty$*, where* $z_{1-\alpha/2}$ *is the standard-normal quantile of order* $1-\alpha/2$ *and* $\hat{\sigma}^{\xi}$ *is the plug-in version of* (3.21)*.*

Proof. See Appendix A.6, where the four-step argument is spelled out: (i) the log-scale recasting of Theorem 3.3; (ii) the lift to estimated weights via the exact sum-to-one constraint and Slutsky; (iii) self-normalisation through the plug-in variance under undersmoothing; (iv) transport of coverage to the original scale by strict monotonicity of exp. □

Practical remarks. First, the log-scale interval (3.42) is symmetric in $\log \xi_{\tau_n}$ while its exponential is not; the latter is the natural form for applications in which ξ_{τ_n} is interpreted as a monetary risk measure. Second, under the equal-fraction distributed specialisation of Proposition 3.4 the variance-optimal weights are identical across the two pooled inputs. With the canonical effective-sample-size weights $\hat{\boldsymbol{\omega}}^k = (k_1/k, \ldots, k_m/k)^\top$ of (3.37), the finite-sample plug-in variance is the direct substitution

$$\hat{\sigma}^{\xi}(\hat{\boldsymbol{\omega}}^k, \hat{\boldsymbol{\omega}}^k)^2 \;=\; \left[(\hat{m} + L_n)^2 + 1\right] (\hat{\boldsymbol{\omega}}^k)^\top \widehat{\boldsymbol{V}}_{\boldsymbol{c}}\, \hat{\boldsymbol{\omega}}^k, \qquad L_n \;=\; \log\big(k/(n(1-\tau_n))\big).$$

If one instead uses the inverse-covariance plug-in $\hat{\boldsymbol{\omega}}^V = \widehat{\boldsymbol{V}}_{\boldsymbol{c}}^{-1}\mathbf{1}/(\mathbf{1}^\top \widehat{\boldsymbol{V}}_{\boldsymbol{c}}^{-1}\mathbf{1})$, the corresponding variance simplifies exactly to

$$\hat{\sigma}^{\xi}(\hat{\boldsymbol{\omega}}^V, \hat{\boldsymbol{\omega}}^V)^2 \;=\; \left[(\hat{m} + L_n)^2 + 1\right] / (\mathbf{1}^\top \widehat{\boldsymbol{V}}_{\boldsymbol{c}}^{-1}\mathbf{1}),$$

the direct plug-in image of the limit (3.27). The two variance estimates have the same limit, and they agree exactly when $\widehat{\boldsymbol{V}}_{\boldsymbol{c}}$ has the distributed diagonal structure described above.

CI validity at $m(\gamma) = 0$. The hypotheses of Corollary 3.7 do *not* require $m(\gamma) \neq 0$: the proof of Appendix A.6 uses only (i) consistency of the estimated weights toward a deterministic affine weight pair, (ii) consistency of the plug-in variance, and (iii) positivity of the limit variance $\sigma^{\xi}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2$. The restriction $m(\gamma) \neq 0$ enters elsewhere: it is needed for the variance-optimal closed form of Proposition 3.5 (because the change of variables $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \mapsto (\boldsymbol{u}, \boldsymbol{\omega}_q)$ ceases to be a bijection at $m(\gamma) = 0$) and for the AMSE positive-definiteness of Proposition 3.6. At the root $\gamma = \gamma^\star \approx 0.2178$, the equal-fraction variance-optimal $\hat{\boldsymbol{\omega}}^k$ of (3.37) remains well defined and consistent, and the CI (3.42) retains its $(1-\alpha)$ asymptotic coverage; the variance simply reduces to $\hat{\sigma}^{\xi,2} = [L_n^2 + 1]\,(\hat{\boldsymbol{\omega}}^k)^\top \widehat{\boldsymbol{V}}_{\boldsymbol{c}}\, \hat{\boldsymbol{\omega}}^k$. For the inverse-covariance plug-in $\hat{\boldsymbol{\omega}}^V$, this same statement uses $[L_n^2 + 1]/(\mathbf{1}^\top \widehat{\boldsymbol{V}}_{\boldsymbol{c}}^{-1}\mathbf{1})$.

Remark 3.8 (Bias-corrected confidence interval)*.* The undersmoothing condition (3.40) is the natural companion of the variance-only interval (3.42). It is a different asymptotic calibration from the second-order regime in which the bias component of (3.22) is retained and the AMSE weight criterion of Proposition 3.6 is meaningful. In that regime the limit-bias $\mu^{\xi}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$ need not vanish. When $\mu^{\xi} \neq 0$, (3.42) does not have nominal asymptotic coverage. The standard remedy is to use the log-scale bias-corrected interval

$$I_{n,\mathrm{bc}}^{\log} \;\coloneqq\; \log \widehat{\xi}_{\tau_n}^{\mathrm{pool}}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q) \;-\; \frac{\hat{\mu}^{\xi}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q)}{\sqrt{k}} \;\pm\; z_{1-\alpha/2}\frac{\hat{\sigma}^{\xi}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q)}{\sqrt{k}},$$

where $\hat{\mu}^{\xi}$ substitutes $\widehat{\boldsymbol{B}}_{\boldsymbol{c}}, \widehat{\boldsymbol{B}}_{\boldsymbol{c}}^{\star}, \hat{b}^{\xi/Q}$ and $\hat{m}$ into (3.22). This interval has the same asymptotic coverage argument as Corollary 3.7 provided all ingredients used there remain valid and, in addition,

$$\hat{\mu}^{\xi}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q) \xrightarrow{\mathbb{P}} \mu^{\xi}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q), \qquad \hat{\sigma}^{\xi}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q) \xrightarrow{\mathbb{P}} \sigma^{\xi}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) > 0,$$

with $\hat{\boldsymbol{\omega}}_\gamma \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_\gamma$ and $\hat{\boldsymbol{\omega}}_q \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_q$ under exact affine normalisation and the same high-probability well-definedness conditions for logarithms and square roots. If the interval is paired with AMSE-optimal weights, these requirements include consistency of the AMSE plug-in weights and the second-order bias estimators in (3.39), a hypothesis substantially more demanding than (3.41) because it requires consistent estimation of the second-order parameter ρ and the rate function A (Haan and Ferreira 2006, §5.2).

Standard practice in the extreme-value literature accordingly favours the variance-only interval

(3.42) under (3.40) over the bias-corrected variant of Remark 3.8: undersmoothing trades a small amount of asymptotic efficiency for the absence of a hard-to-estimate correction and places inference in a variance-dominated regime. This trade-off is the operational reason why Proposition 3.4 — not Proposition 3.6 — is the natural default companion to Corollary 3.7 in applications.

3.7 A root-k diagnostic for high-expectile homogeneity

The variance-optimal pooling of Proposition 3.4 rests on the common-γ assumption (H_γ) and, in the distributed-inference interpretation, on a common high-expectile target across the m samples. Although the chapter assumes a common marginal F throughout, no finite-dimensional statistic can validate equality of the entire marginal distribution. The statistic below should therefore be read more narrowly: it is a root-k diagnostic for equality of the high-expectile targets being pooled. It can reveal tail-index or tail-scale incompatibilities that affect the expectile level, but failure to reject does not validate the full common-marginal model. This mirrors the chi-square test of Theorem 2.20, which diagnoses tail-index homogeneity rather than equality of complete distributions.

Let

$$\widehat{\xi}_{\tau_n,j} \;\coloneqq\; \psi(\hat{\gamma}_j(k_j))\, \hat{q}^{\star}_j(\tau_n \mid k_j), \qquad j = 1, \ldots, m, \tag{3.43}$$

denote the per-sample QB intermediate-expectile estimators of Section 3.1 — one per machine, each assembled from that machine's own pooling primitives $(\hat{\gamma}_j(k_j),\, \hat{q}^{\star}_j)$ — and collect their logarithms into the vector

$$\hat{\boldsymbol{\Xi}}_n \;\coloneqq\; \big(\log \widehat{\xi}_{\tau_n,1}, \ldots, \log \widehat{\xi}_{\tau_n,m}\big)^{\top}.$$

As in Section 3.2, all logarithms are understood on the high-probability event on which the local Weissman factors are positive and the local Hill estimators entering $\log \psi$ lie in $(0,1)$; arbitrary definitions on the complementary event do not change the weak limits.

Joint limit of the per-sample log-expectiles. Applying the expectile-specific Jacobian row $(m(\gamma),\, 1)$ — the same linearisation of $\log \psi$ that produced Theorem 3.3 — to the within-sample 2-vector (3.10) of each sample, and aggregating across the m independent samples on the common $\sqrt{k}$ scale, yields the joint central limit theorem

$$\sqrt{k}\,\big(\hat{\boldsymbol{\Xi}}_n - \log \xi_{\tau_n}\, \mathbf{1}\big) \;\xrightarrow{d}\; \mathcal{N}(\boldsymbol{\beta},\, \boldsymbol{\Sigma}_{\Xi}), \tag{3.44}$$

with diagonal covariance matrix

$$\boldsymbol{\Sigma}_{\Xi} \;=\; \operatorname{diag}\!\big((m(\gamma) + L^{\star}_j)^2 + 1\big)_{j=1}^{m}\, \boldsymbol{V_c}, \qquad (\boldsymbol{\Sigma}_{\Xi})_{j,j} \;=\; \big[(m(\gamma) + L^{\star}_j)^2 + 1\big]\, (\boldsymbol{V_c})_{j,j}, \tag{3.45}$$

and asymptotic mean vector

$$\boldsymbol{\beta} \;=\; m(\gamma)\, \boldsymbol{B_c} \;+\; \boldsymbol{B}^{\star}_{\boldsymbol{c}} \;+\; b^{\xi/Q}\, \mathbf{1}, \tag{3.46}$$

where $\boldsymbol{V_c}$, $\boldsymbol{B_c}$, $\boldsymbol{B}^\star_{\boldsymbol{c}}$, $b^{\xi/Q}$ and the log-rate limits $L^\star_j$ are exactly the quantities of Proposition 3.2 and Theorem 3.3. The covariance (3.45) is diagonal because the m samples are independent (3.1); its jth diagonal entry is the $\sqrt{k}$-rescaled per-sample log-expectile variance $\gamma^2\big[(m(\gamma)+L^\star_j)^2+1\big]\,(k/k_j)$, read off from (3.10) through the row $(m(\gamma),1)$ and recognising $\gamma^2\,(k/k_j)\to(\boldsymbol{V_c})_{j,j}$. The mean (3.46) is the per-sample analogue of the pooled bias (3.22), the deterministic expectile–quantile offset $b^{\xi/Q}$ of (3.19) entering every coordinate equally (it is sample-free, hence collinear with $\mathbf{1}$). The derivation of (3.44) is recorded in Appendix A.7.

The GLS deviance statistic. Under the common-marginal assumption all $\widehat{\xi}_{\tau_n,j}$ estimate the same population ξ_{τ_n}, and their log-scale dispersion about the generalised-least-squares (GLS) weighted mean is captured by the deviance statistic

$$\Lambda^\xi_n \;:=\; k\,\big(\hat{\boldsymbol{\Xi}}_n-\hat{\nu}_n\mathbf{1}\big)^\top\hat{\boldsymbol{\Sigma}}^{-1}_{\Xi,n}\big(\hat{\boldsymbol{\Xi}}_n-\hat{\nu}_n\mathbf{1}\big),\qquad \hat{\nu}_n \;=\; \frac{\mathbf{1}^\top\hat{\boldsymbol{\Sigma}}^{-1}_{\Xi,n}\hat{\boldsymbol{\Xi}}_n}{\mathbf{1}^\top\hat{\boldsymbol{\Sigma}}^{-1}_{\Xi,n}\mathbf{1}}, \tag{3.47}$$

in which $\hat{\nu}_n$ is the GLS estimator of the common log-expectile under (3.44) and $\hat{\boldsymbol{\Sigma}}_{\Xi,n}$ is the plug-in image of (3.45) assembled from the consistent estimators of Section 3.6,

$$\hat{\boldsymbol{\Sigma}}_{\Xi,n} \;=\; \operatorname{diag}\!\big((\hat{m}+L_{j,n})^2+1\big)_{j=1}^m\,\widehat{\boldsymbol{V_c}},\qquad L_{j,n} \;=\; \log\!\big(k_j/(n_j(1-\tau_n))\big), \tag{3.48}$$

with $\hat{m}=\hat{m}^{(0)}$ the preliminary effective-sample-size seed (3.38) and $\widehat{\boldsymbol{V_c}}$ any consistent diagonal covariance plug-in for (2.14), as in Section 3.6. The residual $\hat{\boldsymbol{\Xi}}_n-\hat{\nu}_n\mathbf{1}$ is the vector of GLS residuals about $\hat{\nu}_n$; by construction it annihilates the direction $\mathbf{1}$, which is the structural reason the statistic is insensitive both to the common centring $\log\xi_{\tau_n}$ and — as the theorem makes precise — to any asymptotic bias collinear with $\mathbf{1}$.

Because (3.47) is scaled by k, the formal no-separation null behind the chi-square calibration is a *root-k* log-expectile homogeneity condition. If

$$\boldsymbol{\Xi}^\star_n \;:=\; \big(\log\xi_{\tau_n,1},\ldots,\log\xi_{\tau_n,m}\big)^\top$$

denotes the vector of population per-sample log-expectiles in a diagnostic reading beyond the common-marginal model, this condition is

$$H^\xi_{0,k}:\qquad \sqrt{k}\,\min_{c\in\mathbb{R}}\big\|\boldsymbol{\Xi}^\star_n-c\,\mathbf{1}\big\| \;\to\; 0.$$

It is automatic under the chapter's standing common-marginal assumption, for which $\boldsymbol{\Xi}^\star_n=\log\xi_{\tau_n}\mathbf{1}$. The weaker first-order condition $\xi_{\tau_n,j}/\xi_{\tau_n,\ell}\to 1$ is useful for interpreting the target of the diagnostic, but by itself it is not strong enough to justify a root-k chi-square calibration.

It is useful to keep three objects separate. The *model null* is the common-marginal/common-tail setup under which (3.44) supplies the covariance $\boldsymbol{\Sigma}_\Xi$ and bias vector $\boldsymbol{\beta}$. The *target-homogeneity null* is the root-k condition $H^\xi_{0,k}$ above. The *central-calibration null* requires, in addition, that the asymptotic bias be invisible to the GLS residual projection, namely $\boldsymbol{\beta}\in\operatorname{span}\{\mathbf{1}\}$. The theorem gives the general noncentral limit; the following corollary extracts the ordinary central chi-square rejection rule.

Theorem 3.9 (Noncentral high-expectile homogeneity diagnostic). *Adopt the assumptions of Theorem 3.3, assume $m \geq 2$, and work on the high-probability logarithmic domain described above. Suppose $\hat{\mathbf{\Sigma}}_{\Xi,n}$ is symmetric positive definite with probability tending one and $\hat{\mathbf{\Sigma}}_{\Xi,n} \xrightarrow{\mathbb{P}} \mathbf{\Sigma}_{\Xi}$, with $\mathbf{\Sigma}_{\Xi}$ positive definite. Under the chapter's common-marginal null model, which implies the common-target centring $\mathbf{\Xi}_n^\star = \log \xi_{\tau_n} \mathbf{1}$, the statistic (3.47) converges to a noncentral chi-square,*

$$\Lambda_n^\xi \xrightarrow{d} \chi^2_{m-1}(\delta), \qquad \delta = \boldsymbol{\beta}^\top \mathbf{\Sigma}_\Xi^{-1} \boldsymbol{\beta} - \frac{\left(\mathbf{1}^\top \mathbf{\Sigma}_\Xi^{-1} \boldsymbol{\beta}\right)^2}{\mathbf{1}^\top \mathbf{\Sigma}_\Xi^{-1} \mathbf{1}} \geq 0, \tag{3.49}$$

with $\boldsymbol{\beta}$, $\mathbf{\Sigma}_\Xi$ as in (3.45)–(3.46). The noncentrality δ is the squared $\mathbf{\Sigma}_\Xi^{-1}$-norm of $\boldsymbol{\beta}$ projected off the direction $\mathbf{1}$, and vanishes — so that the limit is the central χ^2_{m-1} *— exactly when $\boldsymbol{\beta} \in \operatorname{span}\{\mathbf{1}\}$. Moreover, if the statistic is used diagnostically outside the common-marginal model in a setting where $\hat{\mathbf{\Xi}}_n - \mathbf{\Xi}_n^\star = O_p(k^{-1/2})$ and the plug-in precision matrices $\hat{\mathbf{\Sigma}}_{\Xi,n}^{-1}$ remain bounded and uniformly positive definite in probability, then under any separated alternative for which the population log-expectiles are separated faster than $k^{-1/2}$ — formally, $\sqrt{k}\, \min_{c \in \mathbb{R}} \|(\log \xi_{\tau_n,j})_{j=1}^m - c\,\mathbf{1}\| \to \infty$ — the statistic diverges, $\Lambda_n^\xi \xrightarrow{\mathbb{P}} \infty$. This alternative-side statement is only a divergence result. It does not supply a central null calibration for heterogeneous margins satisfying root-k target equality; such a calibration would require the corresponding heterogeneous covariance and bias structure.*

Proof. See Appendix A.7. ☐

Corollary 3.10 (Central chi-square calibration). *Under the assumptions of Theorem 3.9, the central chi-square rejection rule*

$$\Lambda_n^\xi > \chi^2_{m-1,\,1-\alpha},$$

with $\chi^2_{m-1,\,1-\alpha}$ the $(1-\alpha)$-quantile of the central χ^2_{m-1} law, has asymptotic size α whenever $\boldsymbol{\beta} \in \operatorname{span}\{\mathbf{1}\}$. In particular this holds in two standing regimes of the chapter:

(i) the undersmoothing calibration (3.40), under which $\boldsymbol{B_c} = \boldsymbol{B_c^\star} = \mathbf{0}$ and $b^{\xi/Q} = 0$, hence $\boldsymbol{\beta} = \mathbf{0}$;

(ii) the equal-fraction distributed specialisation $k_j/n_j = (k/n)(1+o(1))$ described in Section 3.1, under which every coordinate of $\boldsymbol{\beta}$ collapses to one common value.

If $\delta > 0$ under the common-marginal model, the same central critical value has limiting rejection probability $\mathbb{P}\{\chi^2_{m-1}(\delta) > \chi^2_{m-1,\,1-\alpha}\}$, which is larger than α.

Proof. The sufficient conditions in (i)–(ii) are verified in Appendix A.7. The size statement follows immediately from Theorem 3.9 and the continuity of the central chi-square distribution function at its quantiles. ☐

Discussion. Three comments situate the test. First, under the chapter's standing assumption of a common marginal F the root-k null $H_{0,k}^\xi$ holds automatically, so the test is a specification check on the pooling hypothesis rather than a test of a substantive scientific null. A non-rejection is consistent with using the pooling recipes of Sections 3.5 and 3.6 at the tested level, while a rejection warns that the samples should not be pooled without modelling heterogeneity. Second, the central calibration in Corollary 3.10 is the same variance-dominated calibration that underlies the confidence interval of Corollary 3.7: case (i) is exactly the undersmoothing condition (3.40) adopted there, and case (ii) is the equal-fraction distributed regime in which the variance-optimal weights of Proposition 3.4 are used. Outside both regimes the asymptotic bias $\boldsymbol{\beta}$ is generically not collinear with $\mathbf{1}$, the limit is the noncentral $\chi^2_{m-1}(\delta)$ of (3.49), and the central critical value over-rejects by an amount governed by δ; this is the expectile-level counterpart of the identical bias caveat carried, but left implicit, by the tail-index test of Theorem 2.20.

Third, the first-order expectile null $H^{\xi}_{0,0}\colon \xi_{\tau_n,j}/\xi_{\tau_n,\ell} \to 1$ and the tail-index null $H_0\colon \gamma_1 = \cdots = \gamma_m$ of Theorem 2.20 are related but not identical. Per-sample expectile–quantile equivalence (Proposition 2.10) gives $\xi_{\tau_n,j} \sim \psi(\gamma_j)\, Q_{X_j}(\tau_n)$, so $H^{\xi}_{0,0}$ requires the products $\psi(\gamma_j)\, Q_{X_j}(\tau_n)$ to agree across samples in the limit. The non-injectivity of ψ on $(0,1)$ should not be read in isolation: under regular variation, different tail indices force the quantile ratio $Q_{X_j}(\tau)/Q_{X_\ell}(\tau)$ to tend to 0 or ∞, regardless of possible coincidences among the scalar values $\psi(\gamma_j)$. Thus fixed alternatives with unequal tail indices are not hidden by equal ψ values once the quantile factor is included. The complementary strength of Λ^{ξ}_n is instead scale sensitivity: when the tail indices are equal but the tail-homoskedasticity condition $(\mathcal{H})$ of Condition 2.18 fails, the tail-index test can fail to reject H_0 while $H^{\xi}_{0,0}$ fails. In that case the expectile test registers departures in the actual inferential target — the expectile — to which a test of the tail index alone is by construction blind. The two diagnostics are therefore complementary: one targets tail-index equality, the other targets equality of the high-expectile levels being pooled.

Chapter 4

Extrapolation to very-extreme expectile levels

Overview

This chapter extrapolates the pooled intermediate-level expectile estimator of Chapter 3 to very-extreme levels $\tau_n' \uparrow 1$ with $n(1-\tau_n') \to c \in [0,\infty)$. The argument is deliberately short: the pooled intermediate result of Theorem 3.3 is fed into the plug-in extrapolation lemma Proposition 2.14, which is the thesis-level adaptation of Davison et al. (2023, Proposition C.6) proved in Appendix A.8. In the independent common-margin distributed setting of Chapter 3, the resulting limit is the first-order expectile analogue of the pooled Weissman extreme-quantile limit in Theorem 2.19: after division by the diverging log-rate, only the pooled Hill estimator remains visible at first order.

4.1 Motivation

The intermediate-level theory of Chapter 3 estimates ξ_{τ_n} at levels for which the expected number of upper-tail observations diverges. This chapter moves to the very-extreme regime

$$\tau_n' \uparrow 1, \qquad n(1-\tau_n') \to c \in [0,\infty),$$

where only finitely many observations are expected beyond the target level, so the direct intermediate-level asymptotics of Chapter 3 no longer apply. As recalled in Section 2.4.3, the iid extreme-expectile theory of Daouia, Girard, et al. (2018) handles this regime by extrapolating an intermediate-level expectile estimator with a Weissman-type power factor. Direct LAWS or direct QB estimation at τ_n' is therefore not the object of the chapter: the stable statistical anchor remains the intermediate level τ_n, and the very-extreme target is reached by extrapolation.

The pooling protocol is unchanged. Each sample still communicates only the two primitives inherited from Daouia, Padoan, et al. (2024): the local Hill estimator $\hat{\gamma}_j(k_j)$ and the local Weissman quantile input $\hat{q}_j^\star(\,\cdot \mid k_j)$. The asymptotic covariance matrix, or its plug-in estimate in feasible procedures, is used to choose weights and standard errors; it is not an additional expectile-specific primitive. No raw observations and no expectile-specific scalar, such as a local LAWS estimate, are introduced at this stage. Consequently the intermediate-level anchor available for extrapolation is precisely the canonical pooled QB estimator

$$\widehat{\xi}^{\text{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) = \psi\big(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)\big)\widehat{q}^{\star}_n(\tau_n \mid \boldsymbol{\omega}_q)$$

from Equation (3.8). This is a consequence of the Paper-3 communication protocol, not a claim that QB extrapolation is intrinsically preferable to every possible expectile-specific procedure.

The role of Proposition 2.14, the thesis-level plug-in lemma adapted from Davison et al. (2023, Proposition C.6), is exactly to make this transfer modular. Once an intermediate estimator and a tail-index estimator satisfy a joint root-rate limit, their Weissman-extrapolated expectile will have an extreme-level limit governed only by the tail-index component. Chapter 3 supplies that joint limit for $\big(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma), \widehat{\xi}^{\text{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)\big)$. The remaining sections of the present chapter therefore do not develop a new expectile primitive; they specify the extrapolation regime, define the pooled Weissman-type lift, and read the limiting distribution and optimal weights from the pooled Hill component.

4.2 Extrapolation regime and input limit

Throughout this chapter we keep the setup of Section 3.1. The number of samples m is fixed; the samples are independent and have common marginal distribution function F; $n = \sum_{j=1}^m n_j$ and $k = \sum_{j=1}^m k_j$ with $n_j \to \infty$, $k_j \to \infty$ and $k_j/n_j \to 0$; and the proportionality conditions (2.12) hold. The common margin satisfies $\mathcal{C}_2(\gamma, \rho, A)$, with $\gamma \in (0,1)$, and obeys the finite first-moment condition

$$\mathbb{E}\, X^- < \infty.$$

The per-sample bias-rate conditions $\sqrt{k_j}A(n_j/k_j) \to \lambda_j \in \mathbb{R}$ are in force. The intermediate anchor level satisfies

$$L_j(\tau_n) = \log \frac{k_j}{n_j(1-\tau_n)} \;\longrightarrow\; L_j^\star \in \mathbb{R}, \qquad j = 1, \ldots, m,$$

and the moment-negligibility condition is

$$\frac{\sqrt{k}}{Q_X(\tau_n)} \;\longrightarrow\; 0. \tag{4.1}$$

All logarithms of Hill, Weissman and QB quantities are understood on the high-probability positivity event described in Section 3.2; arbitrary definitions on its complement do not affect the weak limits. The only strengthening relative to the intermediate chapter is that the second-order index is now assumed to satisfy

$$\rho < 0. \tag{4.2}$$

This strict version is part of the extrapolation theory in Davison et al. (2023, Proposition C.6), adapted here as Proposition 2.14, and is the same strict second-order condition that appears in the pooled Weissman theorem Theorem 2.19. All results below are therefore statements in the independent common-margin distributed specialisation. References to Theorem 2.19 are analogies within this setting, not extensions to the full tail-copula framework of Section 2.5.

The new target level is a very-extreme sequence

$$\tau_n' \uparrow 1, \qquad n(1-\tau_n') \to c \in [0,\infty), \qquad \frac{1-\tau_n'}{1-\tau_n} \to 0. \tag{4.3}$$

Write the associated Weissman log-rate as

$$\ell_n \;:=\; \log\left(\frac{1-\tau_n}{1-\tau_n'}\right). \tag{4.4}$$

Then $\ell_n \to \infty$, and the extrapolation is not allowed to be so far into the tail that the Weissman normalisation overtakes the available root-k information:

$$\frac{\sqrt{k}}{\ell_n} \;\longrightarrow\; \infty. \tag{4.5}$$

The finite-log condition $L_j(\tau_n) \to L_j^\star$ already implies $n(1-\tau_n) \to \infty$, because $n_j(1-\tau_n)$ has the same order as k_j for every j. Hence (4.3) is exactly the pair of regimes required by Proposition 2.14: τ_n is intermediate and τ_n' is very-extreme. Moreover, if $\alpha_j = \lim n_j/n \in (0,1)$, then the proportionality conditions give

$$\alpha_j = \frac{b_j^{-1}}{\sum_{i=1}^m b_i^{-1}}, \qquad n_j(1-\tau_n') \to \alpha_j c.$$

The very-extreme level is therefore global, but it has the expected local interpretation under the proportional-growth assumptions.

The logarithmic rate used here is also aligned with the pooled Weissman rate in Theorem 2.19. The finite-log condition and proportionality assumptions imply

$$\log\frac{k}{n(1-\tau_n)} = O(1),$$

and hence

$$\log\frac{k}{n(1-\tau_n')} = \ell_n + \log\frac{k}{n(1-\tau_n)} = \ell_n + O(1). \tag{4.6}$$

Since $\ell_n \to \infty$, the two logarithmic normalisers are asymptotically equivalent on the very-extreme scale.

For the application of Proposition 2.14 we take

$$a_n = \sqrt{k}, \qquad \hat{\gamma}_n = \hat{\gamma}_n(\boldsymbol{\omega}_\gamma), \qquad \widehat{\xi}^{\,I}_{\tau_n} = \widehat{\xi}^{\,\mathrm{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q).$$

The non-Gaussian hypotheses of the proposition are then verified as follows. Since $k \to \infty$, $a_n \to \infty$. The preceding paragraph gives $n(1-\tau_n) \to \infty$, while (4.3) gives $n(1-\tau_n') \to c$ and

$(1-\tau_n')/(1-\tau_n) \to 0$. The Weissman-rate condition is (4.5). The second-order limit

$$a_n A((1-\tau_n)^{-1}) = \sqrt{k}\, A((1-\tau_n)^{-1}) \;\longrightarrow\; \Lambda^\bullet \in \mathbb{R}$$

is Equation (3.18), and (4.1) gives $a_n/Q_X(\tau_n) \to 0$. Finally, $\mathbb{P}(\widehat{\xi}^I_{\tau_n} > 0) \to 1$ because $\widehat{q}^\star_n(\tau_n \mid \boldsymbol{\omega}_q) > 0$ on the local Weissman positivity event and $\psi(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)) > 0$ on the event on which the pooled Hill estimator lies in $(0,1)$.

Lemma 4.1 (Deterministic very-extreme ratio control)**.** *Under the tail and rate assumptions of this section,*

$$\frac{\sqrt{k}}{\ell_n}\left\{\log \xi_{\tau_n'} - \log \xi_{\tau_n} - \gamma \ell_n\right\} \;\longrightarrow\; 0.$$

This lemma records, in the present notation, the deterministic population control used in Daouia, Girard, et al. (2020, Theorem 5) and in the proof of Davison et al. (2023, Proposition C.6); it is not a new contribution of the thesis.

Proof. Put $t_n = (1-\tau_n)^{-1}$ and $y_n = (1-\tau_n)/(1-\tau_n')$, so $y_n \to \infty$ and $\ell_n = \log y_n$. The source proofs use, under $\rho < 0$, the sequential Weissman form of the second-order quantile control,

$$\log U(t_n y_n) - \log U(t_n) - \gamma \log y_n = O(\,|A(t_n)|\,). \tag{4.7}$$

This is the point at which the strict condition $\rho < 0$ enters the deterministic extrapolation step: the fixed-y statement of Condition 2.3 alone is not the form being used along the diverging sequence $y_n \to \infty$.

Decompose

$$\begin{aligned}
&\log \xi_{\tau_n'} - \log \xi_{\tau_n} - \gamma \ell_n \\
&\quad = \left[\log Q_X(\tau_n') - \log Q_X(\tau_n) - \gamma \ell_n\right] \\
&\qquad + \left[\log\{\xi_{\tau_n'}/Q_X(\tau_n')\} - \log\{\xi_{\tau_n}/Q_X(\tau_n)\}\right].
\end{aligned}$$

The first bracket is exactly (4.7). The second bracket is controlled by the expectile–quantile expansion (2.7). Since $A \in RV_\rho$ with $\rho < 0$, Potter bounds give $|A(t_n y_n)| = O(|A(t_n)|)$ along $y_n \geq 1$, and the monotonicity of the upper quantile gives $Q_X(\tau_n') \geq Q_X(\tau_n)$ eventually. Hence

$$\begin{aligned}
&\log \xi_{\tau_n'} - \log \xi_{\tau_n} - \gamma \ell_n \\
&\quad = O\!\left(\left|A((1-\tau_n)^{-1})\right|\right) + O(Q_X(\tau_n)^{-1}) + o\!\left(\left|A((1-\tau_n)^{-1})\right|\right) + o(Q_X(\tau_n)^{-1}).
\end{aligned}$$

Multiplying by $\sqrt{k}/\ell_n$ gives the result, because $\sqrt{k}A((1-\tau_n)^{-1})$ converges to the finite limit $\Lambda^\bullet$, $\sqrt{k}/Q_X(\tau_n) \to 0$, and $\ell_n \to \infty$. $\square$

It remains to record the exact intermediate Gaussian input that will be fed into the plug-in extrapolation lemma. Put

$$\boldsymbol{D}^\star := \operatorname{diag}(\boldsymbol{L}^\star), \qquad \boldsymbol{L}^\star = (L_1^\star, \ldots, L_m^\star)^\top.$$

Proposition 4.2 (Joint input for the extrapolation step)**.** *Under the assumptions stated above, for any deterministic affine weight vectors* $\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q \in \mathbb{R}^m$ *satisfying* $\boldsymbol{\omega}_\gamma^\top \mathbf{1} = \boldsymbol{\omega}_q^\top \mathbf{1} = 1$*,*

$$\sqrt{k}\begin{pmatrix} \hat{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma \\ \widehat{\xi}^{\text{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)/\xi_{\tau_n} - 1 \end{pmatrix} \xrightarrow{d} \mathcal{N}\left(\begin{pmatrix} \boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c} \\ \mu^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \end{pmatrix}, \begin{pmatrix} \boldsymbol{\omega}_\gamma^\top \boldsymbol{V_c} \boldsymbol{\omega}_\gamma & s_{\gamma,\xi}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \\ s_{\gamma,\xi}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) & \sigma^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2 \end{pmatrix}\right), \quad (4.8)$$

where μ^ξ *and* σ^ξ *are given in* (3.22) *and* (3.21)*, and*

$$s_{\gamma,\xi}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) = m(\gamma)\, \boldsymbol{\omega}_\gamma^\top \boldsymbol{V_c} \boldsymbol{\omega}_\gamma + \boldsymbol{\omega}_\gamma^\top \boldsymbol{D}^\star \boldsymbol{V_c} \boldsymbol{\omega}_q. \quad (4.9)$$

The same convergence holds with random weights $(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q)$ *satisfying*

$$\hat{\boldsymbol{\omega}}_\gamma^\top \mathbf{1} = \hat{\boldsymbol{\omega}}_q^\top \mathbf{1} = 1 \quad \textit{exactly}, \qquad \hat{\boldsymbol{\omega}}_\gamma \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_\gamma, \qquad \hat{\boldsymbol{\omega}}_q \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_q.$$

In this estimated-weight version, the mean and covariance on the right-hand side of (4.8) *are evaluated at the deterministic probability limits* $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$*.*

Proof. Proposition 3.2 gives the joint Gaussian limit of $\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)$ and $\log \widehat{q}^\star_n(\tau_n \mid \boldsymbol{\omega}_q)$. The first-order Taylor expansion of $\log \psi$ around γ, combined with (3.19), gives

$$\begin{aligned} &\sqrt{k} \log \frac{\widehat{\xi}^{\text{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)}{\xi_{\tau_n}} \\ &\quad = m(\gamma)\sqrt{k}\{\hat{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma\} + \sqrt{k}\{\log \widehat{q}^\star_n(\tau_n \mid \boldsymbol{\omega}_q) - \log Q_X(\tau_n)\} + b^{\xi/Q} + o_p(1). \end{aligned}$$

Thus the second coordinate is the image of the joint limit in Proposition 3.2 under the row $(m(\gamma), 1)$, plus the deterministic offset $b^{\xi/Q}$. This yields the mean and variance (3.22)–(3.21). Its covariance with the first coordinate is the covariance of the pooled-Hill component with $m(\gamma)$ times itself plus the pooled log-Weissman component, namely

$$m(\gamma)\, \boldsymbol{\omega}_\gamma^\top \boldsymbol{V_c} \boldsymbol{\omega}_\gamma + \boldsymbol{\omega}_\gamma^\top \boldsymbol{D}^\star \boldsymbol{V_c} \boldsymbol{\omega}_q,$$

by (3.12). Passing from the centred log-ratio to the relative error, and from its covariance with the Hill component, does not change the limit. Indeed, with

$$R_n = \log \frac{\widehat{\xi}^{\text{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)}{\xi_{\tau_n}}, \qquad R_n = O_p(k^{-1/2}),$$

we have

$$\sqrt{k}\left[\left\{\frac{\widehat{\xi}^{\text{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)}{\xi_{\tau_n}} - 1\right\} - R_n\right] = \sqrt{k}\, O_p(R_n^2) = o_p(1).$$

For estimated weights, exact affine normalisation removes the deterministic centring. For example,

$$\sqrt{k}\{\hat{\gamma}_n(\hat{\boldsymbol{\omega}}_\gamma) - \gamma\} = \hat{\boldsymbol{\omega}}_\gamma^\top \sqrt{k}(\hat{\boldsymbol{\gamma}}_n - \gamma \mathbf{1}).$$

Since the vector on the right is tight and $\hat{\boldsymbol{\omega}}_\gamma \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_\gamma$, this differs by $o_p(1)$ from the same display with $\boldsymbol{\omega}_\gamma$. The weighted-geometric Weissman input obeys the identical argument on the log scale, using $\hat{\boldsymbol{\omega}}_q^\top \mathbf{1} = 1$ and $\hat{\boldsymbol{\omega}}_q \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_q$. Substituting these two Slutsky steps into the Taylor expansion above gives the stated estimated-weight extension, with limiting mean and covariance computed at the deterministic probability limits. $\square$

Proposition 4.2 is more information than the final very-extreme theorem will visibly retain. The extrapolated limit uses only the first component, the pooled Hill estimator, because Proposition 2.14 divides by the diverging log-rate ℓ_n. The joint statement is nevertheless the right object to state here: it is exactly the hypothesis of the plug-in theorem, with

$$\Gamma \sim \mathcal{N}(\boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c},\, \boldsymbol{\omega}_\gamma^\top \boldsymbol{V_c} \boldsymbol{\omega}_\gamma), \qquad \Delta \text{ equal to the second component of (4.8).}$$

Together with the hypothesis verification above, this is precisely the input required by Proposition 2.14.

4.3 The pooled extrapolated expectile estimator

The extrapolated estimator is obtained by applying the Weissman-type lift of Equation (2.10) to the pooled intermediate anchor of Equation (3.8). For affine weights $\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q \in \mathbb{R}^m$ satisfying $\boldsymbol{\omega}_\gamma^\top \mathbf{1} = \boldsymbol{\omega}_q^\top \mathbf{1} = 1$, define

$$\widehat{\xi}^{\,\text{pool},\star}_{\tau_n'}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \;:=\; \left(\frac{1-\tau_n'}{1-\tau_n}\right)^{-\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)} \widehat{\xi}^{\,\text{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q). \tag{4.10}$$

Although $\boldsymbol{\omega}_q$ affects the finite-sample anchor and the intermediate-level component of the joint input limit, it disappears from the first-order very-extreme CLT after normalisation by $\sqrt{k}/\ell_n$. It remains part of the definition because different choices of $\boldsymbol{\omega}_q$ give different finite-sample estimators and higher-order terms. Equivalently, using Equation (3.8),

$$\widehat{\xi}^{\,\text{pool},\star}_{\tau_n'}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) = \exp\{\ell_n \hat{\gamma}_n(\boldsymbol{\omega}_\gamma)\}\, \psi(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma))\, \widehat{q}^{\,\star}_n(\tau_n \mid \boldsymbol{\omega}_q),$$

where ℓ_n is the log-rate in (4.4). Thus the same two weight vectors used at the intermediate level remain visible: $\boldsymbol{\omega}_\gamma$ acts on the pooled Hill input, both in the QB factor $\psi(\hat{\gamma}_n)$ and in the extrapolation power, while $\boldsymbol{\omega}_q$ acts only on the weighted-geometric quantile input that forms the intermediate anchor. As in Section 3.2, all log-scale and real-power expressions are read on the high-probability positivity event for the local Weissman factors and on the event where the pooled Hill estimator entering ψ lies in $(0, 1)$. This event also gives $\mathbb{P}\{\widehat{\xi}^{\,\text{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) > 0\} \to 1$, the positivity hypothesis in Proposition 2.14; arbitrary definitions on the complementary event do not alter the weak limits.

The construction targets the same expectile level τ_n' that appears in the population quantity $\xi_{\tau_n'}$. No adjusted quantile level is introduced. The reason is that the expectile–quantile bridge of Proposition 2.10 is a same-level equivalence, $\xi_\tau / Q_X(\tau) \to \psi(\gamma)$, and the Weissman step in Proposition 2.14 extrapolates the intermediate expectile itself from τ_n to τ_n'.

At matched weights, (4.10) is exactly the QB plug-in applied to the pooled Weissman quantile at the very-extreme level. Indeed, by the local Weissman identity,

$$\hat{q}_j^\star(\tau_n' \mid k_j) = \left(\frac{1-\tau_n'}{1-\tau_n}\right)^{-\hat{\gamma}_j(k_j)} \hat{q}_j^\star(\tau_n \mid k_j),$$

and therefore, for every affine $\boldsymbol{\omega}$,

$$\widehat{q}_n^\star(\tau_n' \mid \boldsymbol{\omega}) = \left(\frac{1-\tau_n'}{1-\tau_n}\right)^{-\hat{\gamma}_n(\boldsymbol{\omega})} \widehat{q}_n^\star(\tau_n \mid \boldsymbol{\omega}). \tag{4.11}$$

Combining (4.11) with (4.10) gives the exact matched-weight identity

$$\widehat{\xi}_{\tau_n'}^{\text{pool},\star}(\boldsymbol{\omega}, \boldsymbol{\omega}) = \psi(\hat{\gamma}_n(\boldsymbol{\omega}))\, \widehat{q}_n^\star(\tau_n' \mid \boldsymbol{\omega}). \tag{4.12}$$

Thus, on the matched diagonal, the Chapter 4 estimator is not a new third primitive: it is the same pooled-Hill / pooled-Weissman QB pipeline as in Chapter 3, evaluated after the Weissman quantile input has been extrapolated to τ_n'.

Remark 4.3 (Matched local-QB equivalence). There is also a local-QB analogue of the construction, parallel to Equation (3.7). Define the marginal extrapolated QB expectile on sample j by

$$\widehat{\xi}_{\tau_n',j}^\star \;:=\; \left(\frac{1-\tau_n'}{1-\tau_n}\right)^{-\hat{\gamma}_j(k_j)} \psi(\hat{\gamma}_j(k_j))\, \hat{q}_j^\star(\tau_n \mid k_j) = \psi(\hat{\gamma}_j(k_j))\, \hat{q}_j^\star(\tau_n' \mid k_j),$$

and geometrically pool these m scalars:

$$\widehat{\xi}_{\tau_n'}^{\text{loc},\star}(\boldsymbol{\omega}) \;:=\; \prod_{j=1}^m \left[\widehat{\xi}_{\tau_n',j}^\star\right]^{\omega_j}. \tag{4.13}$$

The comparison with (4.12) is identical to Proposition 3.1, because the very-extreme Weissman factor is common to both estimators. More precisely,

$$\log \frac{\widehat{\xi}_{\tau_n'}^{\text{pool},\star}(\boldsymbol{\omega}, \boldsymbol{\omega})}{\widehat{\xi}_{\tau_n'}^{\text{loc},\star}(\boldsymbol{\omega})} = \log \psi(\hat{\gamma}_n(\boldsymbol{\omega})) - \sum_{j=1}^m \omega_j \log \psi(\hat{\gamma}_j(k_j)),$$

so the unscaled log-ratio is $O_p(k^{-1})$ and in particular is negligible both on the intermediate $\sqrt{k}$ scale and on the extreme $\sqrt{k}/\ell_n$ scale. The canonical estimator (4.10) is therefore consistent with the local-QB interpretation whenever the weights are matched.

This matched-weight qualification is important. If $\boldsymbol{\omega}_\gamma \neq \boldsymbol{\omega}_q$, the alternative estimator

$$\psi(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma))\, \widehat{q}_n^\star(\tau_n' \mid \boldsymbol{\omega}_q)$$

is generally first-order different from the object analysed in this chapter; it is not an equivalent rewrite of (4.10). By (4.11), it replaces the extrapolation power $\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)$ by $\hat{\gamma}_n(\boldsymbol{\omega}_q)$, and their log-ratio is

$$\ell_n\{\hat{\gamma}_n(\boldsymbol{\omega}_q) - \hat{\gamma}_n(\boldsymbol{\omega}_\gamma)\}.$$

After multiplication by the extreme normalisation $\sqrt{k}/\ell_n$, this term contributes $\sqrt{k}\{\hat\gamma_n(\boldsymbol{\omega}_q) - \hat\gamma_n(\boldsymbol{\omega}_\gamma)\}$, which is generally non-negligible. Hence the two-weight estimator in (4.10) is the object to which the plug-in CLT will be applied; replacing its extrapolation exponent changes the first-order extreme limit unless the two Hill weightings agree asymptotically.

4.4 Main extrapolation theorem

We can now state the chapter's main limit theorem. It is a direct application of the plug-in extrapolation result Proposition 2.14 to the pooled intermediate input of Proposition 4.2. The theorem is intentionally short: all expectile-specific work has already been done at the intermediate level, and the very-extreme scale keeps only the pooled-Hill component.

Theorem 4.4 (Pooled Weissman extrapolation for extreme expectiles)**.** *Assume the setup and rate conditions of Section 4.2; in particular, $\rho < 0$, $\sqrt{k}/Q_X(\tau_n) \to 0$, $\ell_n \to \infty$ and $\sqrt{k}/\ell_n \to \infty$. Use the positivity and log-domain convention of Section 4.3, and the deterministic ratio control verified in Lemma 4.1. Let $\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q \in \mathbb{R}^m$ be deterministic affine weight vectors satisfying $\boldsymbol{\omega}_\gamma^\top \mathbf{1} = \boldsymbol{\omega}_q^\top \mathbf{1} = 1$, and define $\widehat{\xi}^{\text{pool},\star}_{\tau_n'}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$ by (4.10). Then*

$$\frac{\sqrt{k}}{\ell_n}\left(\frac{\widehat{\xi}^{\text{pool},\star}_{\tau_n'}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)}{\xi_{\tau_n'}} - 1\right) \xrightarrow{d} \mathcal{N}\Big(\boldsymbol{\omega}_\gamma^\top \boldsymbol{B}_{\boldsymbol{c}},\ \boldsymbol{\omega}_\gamma^\top \boldsymbol{V}_{\boldsymbol{c}} \boldsymbol{\omega}_\gamma\Big), \tag{4.14}$$

where $\ell_n = \log\{(1-\tau_n)/(1-\tau_n')\}$.

The same conclusion holds with estimated weights $(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q)$, provided

$$\hat{\boldsymbol{\omega}}_\gamma^\top \mathbf{1} = \hat{\boldsymbol{\omega}}_q^\top \mathbf{1} = 1 \quad \textit{exactly}, \qquad \hat{\boldsymbol{\omega}}_\gamma \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_\gamma, \qquad \hat{\boldsymbol{\omega}}_q \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_q.$$

Proof. Section 4.2 verifies the non-Gaussian hypotheses of Proposition 2.14 with $a_n = \sqrt{k}$, $\hat\gamma_n = \hat\gamma_n(\boldsymbol{\omega}_\gamma)$ and $\widehat{\xi}^{I}_{\tau_n} = \widehat{\xi}^{\text{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$. The required joint input is precisely Proposition 4.2; its first coordinate has limiting law

$$\Gamma \sim \mathcal{N}\Big(\boldsymbol{\omega}_\gamma^\top \boldsymbol{B}_{\boldsymbol{c}},\ \boldsymbol{\omega}_\gamma^\top \boldsymbol{V}_{\boldsymbol{c}} \boldsymbol{\omega}_\gamma\Big).$$

Applying Proposition 2.14 therefore gives (4.14). We spell out the log-scale mechanism, because it is the structural reason why $\boldsymbol{\omega}_q$ vanishes from the first-order limit.

On the high-probability positivity event fixed in Section 4.3,

$$\begin{aligned}\log \frac{\widehat{\xi}^{\text{pool},\star}_{\tau_n'}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)}{\xi_{\tau_n'}} &= \ell_n\{\hat\gamma_n(\boldsymbol{\omega}_\gamma) - \gamma\} + \log \frac{\widehat{\xi}^{\text{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)}{\xi_{\tau_n}} \\ &\quad - \{\log \xi_{\tau_n'} - \log \xi_{\tau_n} - \gamma \ell_n\}.\end{aligned} \tag{4.15}$$

Multiplication by $\sqrt{k}/\ell_n$ sends the first term to

$$\sqrt{k}\{\hat\gamma_n(\boldsymbol{\omega}_\gamma) - \gamma\} \xrightarrow{d} \mathcal{N}\Big(\boldsymbol{\omega}_\gamma^\top \boldsymbol{B}_{\boldsymbol{c}},\ \boldsymbol{\omega}_\gamma^\top \boldsymbol{V}_{\boldsymbol{c}} \boldsymbol{\omega}_\gamma\Big),$$

the pooled-Hill Gaussian limit of Theorem 2.16. For the second term, Proposition 4.2 gives

$$\sqrt{k}\left\{\frac{\widehat{\xi}^{\text{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)}{\xi_{\tau_n}}-1\right\}=O_p(1),$$

and the corresponding log-ratio is also $O_p(k^{-1/2})$ because the relative error is $o_p(1)$. Since $\ell_n\to\infty$, this term is $o_p(1)$ after multiplication by $\sqrt{k}/\ell_n$. The deterministic log-ratio condition in Proposition 2.14 holds by Lemma 4.1; therefore the third term in (4.15) is $o(\ell_n/\sqrt{k})$. Hence

$$\frac{\sqrt{k}}{\ell_n}\log\frac{\widehat{\xi}^{\text{pool},\star}_{\tau_n'}(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)}{\xi_{\tau_n'}}\xrightarrow{d}\Gamma.$$

Let

$$Z_n=\log\frac{\widehat{\xi}^{\text{pool},\star}_{\tau_n'}(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)}{\xi_{\tau_n'}}.$$

The preceding display gives $(\sqrt{k}/\ell_n)Z_n\xrightarrow{d}\Gamma$, hence $Z_n=O_p(\ell_n/\sqrt{k})=o_p(1)$ because $\ell_n/\sqrt{k}\to 0$. Therefore

$$\exp(Z_n)-1-Z_n=O_p(Z_n^2),\qquad \frac{\sqrt{k}}{\ell_n}\{\exp(Z_n)-1-Z_n\}=O_p\left(\frac{\ell_n}{\sqrt{k}}\right)=o_p(1).$$

Thus the relative-error normalisation has the same limit as the log-ratio normalisation, proving (4.14).

For estimated weights, use the estimated-weight part of Proposition 4.2. Exact affine normalisation removes any deterministic centring error, and $\hat{\boldsymbol{\omega}}_\gamma\xrightarrow{\mathbb{P}}\boldsymbol{\omega}_\gamma$ carries the pooled-Hill mean and variance to the deterministic limits in (4.14) by Slutsky's theorem. The convergence of $\hat{\boldsymbol{\omega}}_q$ affects only the intermediate-anchor term, which has already been shown to vanish on the $\sqrt{k}/\ell_n$ scale. □

Theorem 4.4 is the first-order expectile analogue of Theorem 2.19 in the independent common-margin specialisation used in the contribution chapters. In both cases, extrapolation to a very-extreme level multiplies the tail-index estimation error by a diverging logarithmic distance, while the intermediate scale estimate is too small to survive the normalisation. The only role of the expectile construction is to provide a valid positive intermediate anchor and to verify the deterministic expectile-ratio control needed by Proposition 2.14. The QB derivative $m(\gamma)$ and the intermediate expectile-bias terms $\boldsymbol{B}_{\boldsymbol{c}}^\star$ and $b^{\xi/Q}$ are not zero in general; they are lower order on the very-extreme normalisation because they enter only through the intermediate anchor.

4.5 First-order optimal Hill weights for extrapolation

The limiting law in Theorem 4.4 has the same optimal-weight structure as the pooled Hill and pooled Weissman limits of Theorems 2.16 and 2.19. Indeed, after the normalisation by $\sqrt{k}/\ell_n$, the intermediate anchor

$$\widehat{\xi}^{\text{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)$$

contributes only an $o_p(1)$ term. The first-order mean and variance are therefore

$$\mu^{\xi,\star}(\boldsymbol{\omega}_\gamma) = \boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c}, \qquad \sigma^{\xi,\star}(\boldsymbol{\omega}_\gamma)^2 = \boldsymbol{\omega}_\gamma^\top \boldsymbol{V_c}\boldsymbol{\omega}_\gamma. \tag{4.16}$$

This is the same first-order quadratic structure as in Proposition 2.17, and it is also the structure that Daouia, Padoan, et al. (2024, Theorem 2) obtain for the weighted-geometric pooled Weissman quantile. Within the independent common-margin setting of this chapter, the only difference is interpretive: here the extrapolated target is the common expectile $\xi_{\tau_n'}$ rather than the common extreme quantile. The first-order criterion therefore identifies only the extrapolation exponent, that is, the Hill-weight vector $\boldsymbol{\omega}_\gamma$; it does not identify the anchor weight $\boldsymbol{\omega}_q$.

As in Proposition 2.17, weights are real affine weights; no non-negativity constraint is imposed. If convex weights are required, the solution is the corresponding constrained quadratic programme and need not equal the closed forms below.

Proposition 4.5 (First-order optimal Hill weights for extrapolated expectiles)**.** *Assume the hypotheses of Theorem 4.4, and suppose that $\boldsymbol{V_c}$ is positive definite. Among affine Hill weights $\boldsymbol{\omega}_\gamma^\top \mathbf{1} = 1$, the variance-optimal extrapolation weight is*

$$\boldsymbol{\omega}_\gamma^{\xi,\mathrm{Var},\star} = \boldsymbol{\omega}^{\mathrm{Var}} = \frac{\boldsymbol{V_c}^{-1}\mathbf{1}}{\mathbf{1}^\top \boldsymbol{V_c}^{-1}\mathbf{1}}, \tag{4.17}$$

and the minimum first-order variance is

$$\sigma^{\xi,\star}(\boldsymbol{\omega}_\gamma^{\xi,\mathrm{Var},\star})^2 = \frac{1}{\mathbf{1}^\top \boldsymbol{V_c}^{-1}\mathbf{1}}. \tag{4.18}$$

If the limiting bias is retained, the first-order AMSE criterion is

$$\mathrm{AMSE}^{\xi,\star}(\boldsymbol{\omega}_\gamma) = \left(\boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c}\right)^2 + \boldsymbol{\omega}_\gamma^\top \boldsymbol{V_c}\boldsymbol{\omega}_\gamma, \qquad \boldsymbol{\omega}_\gamma^\top \mathbf{1} = 1, \tag{4.19}$$

whose unique minimiser is

$$\boldsymbol{\omega}_\gamma^{\xi,\mathrm{AMSE},\star} = \boldsymbol{\omega}^{\mathrm{AMSE}} = \frac{(1 + \boldsymbol{B_c}^\top \boldsymbol{V_c}^{-1}\boldsymbol{B_c})\,\boldsymbol{V_c}^{-1}\mathbf{1} - (\mathbf{1}^\top \boldsymbol{V_c}^{-1}\boldsymbol{B_c})\,\boldsymbol{V_c}^{-1}\boldsymbol{B_c}}{(1 + \boldsymbol{B_c}^\top \boldsymbol{V_c}^{-1}\boldsymbol{B_c})(\mathbf{1}^\top \boldsymbol{V_c}^{-1}\mathbf{1}) - (\mathbf{1}^\top \boldsymbol{V_c}^{-1}\boldsymbol{B_c})^2}. \tag{4.20}$$

The corresponding minimum AMSE is

$$\mathrm{AMSE}^{\xi,\star}(\boldsymbol{\omega}_\gamma^{\xi,\mathrm{AMSE},\star}) = \frac{1 + \boldsymbol{B_c}^\top \boldsymbol{V_c}^{-1}\boldsymbol{B_c}}{(1 + \boldsymbol{B_c}^\top \boldsymbol{V_c}^{-1}\boldsymbol{B_c})(\mathbf{1}^\top \boldsymbol{V_c}^{-1}\mathbf{1}) - (\mathbf{1}^\top \boldsymbol{V_c}^{-1}\boldsymbol{B_c})^2}. \tag{4.21}$$

The denominator in (4.20)–(4.21) is strictly positive. The quantile-anchor weight $\boldsymbol{\omega}_q$ is not identified by either first-order criterion: for the variance objective, every pair $(\boldsymbol{\omega}^{\mathrm{Var}}, \boldsymbol{\nu})$ with $\boldsymbol{\nu}^\top \mathbf{1} = 1$ has the same first-order variance, and for the AMSE objective the same is true of every pair $(\boldsymbol{\omega}^{\mathrm{AMSE}}, \boldsymbol{\nu})$.

Proof. Proposition 2.14, applied in Theorem 4.4, gives the first-order mean and variance (4.16).

The variance problem is therefore the same constrained quadratic programme as Proposition 2.17(i), giving (4.17)–(4.18).

For AMSE, set $\boldsymbol{M} = \boldsymbol{V_c} + \boldsymbol{B_c}\boldsymbol{B_c}^\top$. Since $\boldsymbol{V_c}$ is positive definite, so is $\boldsymbol{M}$. The constrained quadratic programme

$$\min_{\boldsymbol{\omega}_\gamma^\top \mathbf{1}=1} \boldsymbol{\omega}_\gamma^\top \boldsymbol{M} \boldsymbol{\omega}_\gamma$$

has solution

$$\boldsymbol{\omega}_\gamma = \frac{\boldsymbol{M}^{-1}\mathbf{1}}{\mathbf{1}^\top \boldsymbol{M}^{-1}\mathbf{1}}.$$

Applying the Sherman–Morrison identity,

$$\boldsymbol{M}^{-1} = \boldsymbol{V_c}^{-1} - \frac{\boldsymbol{V_c}^{-1}\boldsymbol{B_c}\boldsymbol{B_c}^\top \boldsymbol{V_c}^{-1}}{1 + \boldsymbol{B_c}^\top \boldsymbol{V_c}^{-1}\boldsymbol{B_c}},$$

yields (4.20). The minimum value is $(\mathbf{1}^\top \boldsymbol{M}^{-1}\mathbf{1})^{-1}$, which gives (4.21). Positivity follows from positive definiteness of $\boldsymbol{M}$. Since neither (4.16) nor (4.19) contains $\boldsymbol{\omega}_q$, the whole affine hyperplane $\{\boldsymbol{\nu} \in \mathbb{R}^m : \boldsymbol{\nu}^\top \mathbf{1} = 1\}$ is first-order equivalent for the anchor weight once $\boldsymbol{\omega}_\gamma$ is fixed at the relevant optimum. □

In the independent common-distribution distributed setting used throughout the contribution chapters, $R_{j,\ell} \equiv 0$ for $j \neq \ell$ and Equation (2.14) gives a diagonal covariance matrix. Consequently (4.17) reduces to

$$\omega_{\gamma,j}^{\xi,\mathrm{Var},\star} = \left(\sum_{i=1}^{m} c_i^{-1}\right)^{-1} c_j^{-1} \sim \frac{k_j}{k}, \qquad j = 1, \ldots, m. \tag{4.22}$$

Thus, at very-extreme expectile levels, the variance-optimal distributed rule is again effective-sample-size weighting. This is the same simplification as in the pooled extreme-quantile case of Equation (2.15). If the second-order bias vector is collinear with $\mathbf{1}$ — in particular under the undersmoothing calibration that makes $\boldsymbol{B_c} = \mathbf{0}$, or under the equal-effective-fraction common-distribution regime, where the aggregate-scale entries $\sqrt{c_j S}\lambda_j/(1-\rho)$ of $\boldsymbol{B_c}$, with $S = \sum_{i=1}^m c_i^{-1}$, are sample-free — then $\boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c}$ is constant on the affine constraint set. In that case the AMSE criterion (4.19) differs from the variance criterion only by an additive constant, so $\boldsymbol{\omega}_\gamma^{\xi,\mathrm{AMSE},\star} = \boldsymbol{\omega}_\gamma^{\xi,\mathrm{Var},\star}$.

Proposition 4.5 also explains why the very-extreme chapter differs from Section 3.5. At intermediate levels the delta method through the QB map retains the full pair of primitive contributions, and the expectile variance and AMSE involve $m(\gamma)$, $\boldsymbol{L}^\star$, $\boldsymbol{B_c}^\star$ and $b^{\xi/Q}$; see (3.21)–(3.22). At very-extreme levels those quantities live in the intermediate component Δ of Proposition 2.14. The division by the diverging extrapolation log-rate ℓ_n removes that component from the first-order law, leaving only the pooled-Hill mean and variance in (4.16). Hence the Chapter 4 AMSE weights are not the Chapter 3 expectile AMSE weights; they are the Daouia–Padoan–Stupfler Hill, and therefore extreme-quantile, weights read through the expectile extrapolation theorem.

The non-identification of $\boldsymbol{\omega}_q$ is only a first-order statement. The anchor weight still changes the finite-sample point estimator in (4.10), and it may affect higher-order accuracy through the intermediate estimator. A natural canonical representative is therefore the matched choice

$$\boldsymbol{\omega}_q = \boldsymbol{\omega}_\gamma,$$

with $\boldsymbol{\omega}_\gamma$ chosen as either (4.17) or (4.20). This is an implementation convention rather than a consequence of the first-order optimality criterion; a higher-order or finite-sample criterion would be needed to rank different affine choices of $\boldsymbol{\omega}_q$. The matched convention keeps the extrapolated expectile in the same weighted-geometric Weissman form as Equation (4.12): for matched weights, the estimator is exactly $\psi(\hat\gamma_n(\boldsymbol{\omega}_\gamma))\,\widehat{q}_n^{\star}(\tau_n' \mid \boldsymbol{\omega}_\gamma)$. It also gives a single default implementation for the confidence intervals of Section 4.7, whose first-order standard errors depend only on $\boldsymbol{\omega}_\gamma$.

4.6 Bias-reduction

The limit in Theorem 4.4 retains, on the $\sqrt{k}/\ell_n$ scale, the same asymptotic bias as the pooled Hill estimator:

$$\mu^{\xi,\star}(\boldsymbol{\omega}_\gamma) = \boldsymbol{\omega}_\gamma^\top \boldsymbol{B}_{\boldsymbol{c}}.$$

This is the only first-order bias visible after extrapolation to τ_n'. The intermediate QB bias terms that were central in Chapter 3 enter the second coordinate of the joint input in Proposition 4.2; Proposition 2.14 removes that coordinate by dividing by the diverging log-rate ℓ_n. Consequently, bias reduction at the very-extreme level is inherited directly from the tail-index correction of Daouia, Padoan, et al. (2024, §2.2).

Assume that a consistent estimator of the limiting Hill-bias vector is available,

$$\widehat{\boldsymbol{B}}_{\boldsymbol{c}} \xrightarrow{\mathbb{P}} \boldsymbol{B}_{\boldsymbol{c}}. \tag{4.23}$$

The following result is therefore conditional on a feasible second-order bias-estimation step; under the abstract condition $\mathcal{C}_2(\gamma,\rho,A)$ alone, (4.23) is not automatic. For any affine Hill-weight vector $\boldsymbol{\omega}_\gamma^\top \mathbf{1} = 1$, define the bias-reduced pooled tail-index estimator

$$\overline{\gamma}_n(\boldsymbol{\omega}_\gamma) \ :=\ \hat\gamma_n(\boldsymbol{\omega}_\gamma) - \frac{\boldsymbol{\omega}_\gamma^\top \widehat{\boldsymbol{B}}_{\boldsymbol{c}}}{\sqrt{k}}. \tag{4.24}$$

The associated bias-reduced extrapolated expectile keeps the same intermediate anchor as (4.10), but replaces the extrapolation exponent by (4.24):

$$\widehat{\xi}^{\,\mathrm{pool},\star,\mathrm{bc}}_{\tau_n'}(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q) \ :=\ \left(\frac{1-\tau_n'}{1-\tau_n}\right)^{-\overline{\gamma}_n(\boldsymbol{\omega}_\gamma)} \widehat{\xi}^{\,\mathrm{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q). \tag{4.25}$$

For feasible estimated weights satisfying the exact affine constraints we use the corresponding plug-in convention

$$\overline{\gamma}_n(\hat{\boldsymbol{\omega}}_\gamma) = \hat\gamma_n(\hat{\boldsymbol{\omega}}_\gamma) - \frac{\hat{\boldsymbol{\omega}}_\gamma^\top \widehat{\boldsymbol{B}}_{\boldsymbol{c}}}{\sqrt{k}},$$

and define

$$\widehat{\xi}^{\text{pool},\star,\text{bc}}_{\tau_n'}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q) = \left(\frac{1-\tau_n'}{1-\tau_n}\right)^{-\overline{\gamma}_n(\hat{\boldsymbol{\omega}}_\gamma)} \widehat{\xi}^{\text{pool}}_{\tau_n}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q).$$

Only the exponent is corrected because only the exponent contributes at the $\sqrt{k}/\ell_n$ scale. Replacing the occurrence of $\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)$ inside the intermediate QB factor $\psi(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma))$ by $\overline{\gamma}_n(\boldsymbol{\omega}_\gamma)$, on the same high-probability compact event used for Taylor expansion of $\log \psi$, would change (4.25) by a log-ratio of order $O_p(k^{-1/2})$, hence by $o_p(1)$ after multiplication by $\sqrt{k}/\ell_n$. The minimal correction above is therefore enough for the first-order very-extreme theorem. The correction introduces no new log-domain restriction: the extrapolation base is deterministic and positive, so the only random positivity requirement remains that of the intermediate anchor.

Corollary 4.6 (Bias-reduced extrapolated expectile CLT)**.** *Assume the hypotheses of Theorem 4.4, and assume (4.23); in particular $\ell_n \to \infty$ and $\sqrt{k}/\ell_n \to \infty$, and the positivity convention of Section 4.3 is in force. Let $\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q \in \mathbb{R}^m$ be deterministic affine weight vectors satisfying $\boldsymbol{\omega}_\gamma^\top \mathbf{1} = \boldsymbol{\omega}_q^\top \mathbf{1} = 1$. Then the estimator (4.25) satisfies*

$$\frac{\sqrt{k}}{\ell_n}\left(\frac{\widehat{\xi}^{\text{pool},\star,\text{bc}}_{\tau_n'}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)}{\xi_{\tau_n'}} - 1\right) \xrightarrow{d} \mathcal{N}\Big(0,\ \boldsymbol{\omega}_\gamma^\top \boldsymbol{V_c} \boldsymbol{\omega}_\gamma\Big)\,. \tag{4.26}$$

The same conclusion holds with estimated weights $(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q)$ satisfying the exact affine constraints $\hat{\boldsymbol{\omega}}_\gamma^\top \mathbf{1} = \hat{\boldsymbol{\omega}}_q^\top \mathbf{1} = 1$ and $\hat{\boldsymbol{\omega}}_\gamma \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_\gamma$, $\hat{\boldsymbol{\omega}}_q \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_q$.

Proof. By Theorem 2.16 and (4.23),

$$\sqrt{k}\{\overline{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma\} = \sqrt{k}\{\hat{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma\} - \boldsymbol{\omega}_\gamma^\top \widehat{\boldsymbol{B}}_{\boldsymbol{c}} \xrightarrow{d} \mathcal{N}\Big(0, \boldsymbol{\omega}_\gamma^\top \boldsymbol{V_c} \boldsymbol{\omega}_\gamma\Big)\,.$$

Moreover,

$$\sqrt{k}\{\overline{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma\} = \sqrt{k}\{\hat{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma\} - \boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c} + o_p(1).$$

Combining this recentering with Proposition 4.2 gives the joint input

$$\sqrt{k}\begin{pmatrix} \overline{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma \\ \widehat{\xi}^{\text{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)/\xi_{\tau_n} - 1 \end{pmatrix} \xrightarrow{d} \mathcal{N}\left(\begin{pmatrix} 0 \\ \mu^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \end{pmatrix}, \begin{pmatrix} \boldsymbol{\omega}_\gamma^\top \boldsymbol{V_c} \boldsymbol{\omega}_\gamma & s_{\gamma,\xi}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \\ s_{\gamma,\xi}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) & \sigma^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2 \end{pmatrix}\right).$$

Subtracting the consistent estimate of the deterministic bias shifts only the first-coordinate mean and leaves the covariance matrix unchanged. Now repeat the log-decomposition (4.15), with $\overline{\gamma}_n(\boldsymbol{\omega}_\gamma)$ in place of $\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)$. The intermediate-anchor term is still $o_p(1)$ on the $\sqrt{k}/\ell_n$ scale by Proposition 4.2, and the deterministic extrapolation error is still negligible by Lemma 4.1. Thus the log-ratio limit is the centred Gaussian display above. Since $\ell_n/\sqrt{k} \to 0$, the final passage from log-ratio to relative error is the same exponential Taylor argument used in the proof of Theorem 4.4.

For estimated weights, $\hat{\boldsymbol{\omega}}_\gamma^\top \widehat{\boldsymbol{B}}_{\boldsymbol{c}} \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c}$ by the assumed convergences, and the estimated-weight part of Theorem 4.4 supplies the corresponding Slutsky step for the pooled Hill component.

With the explicit estimated-weight definition above,

$$\sqrt{k}\{\overline{\gamma}_n(\hat{\boldsymbol{\omega}}_\gamma) - \gamma\} = \sqrt{k}\{\hat{\gamma}_n(\hat{\boldsymbol{\omega}}_\gamma) - \gamma\} - \hat{\boldsymbol{\omega}}_\gamma^\top \widehat{\boldsymbol{B}}_{\boldsymbol{c}} \xrightarrow{d} \mathcal{N}\Big(0, \boldsymbol{\omega}_\gamma^\top \boldsymbol{V}_{\boldsymbol{c}} \boldsymbol{\omega}_\gamma\Big).$$

The anchor weight remains first-order invisible on the very-extreme scale, as in Theorem 4.4. □

Corollary 4.6 is deliberately a first-order statement. It should not be read as saying that the intermediate expectile-bias terms $\boldsymbol{B}_{\boldsymbol{c}}^\star$ and $b^{\xi/Q}$ disappear from finite-sample behaviour; they disappear only from the leading very-extreme normalisation. A higher-order expansion would again have to keep track of the anchor and of the choice of $\boldsymbol{\omega}_q$.

The practical cost is the same one already present in Daouia, Padoan, et al. (2024): (4.23) requires information about the second-order component of the tail. In their feasible implementation this is obtained under a Hall-type working model, $A(t) = \gamma\beta t^\rho$, by estimating the second-order parameters and substituting them into the formula for $\boldsymbol{B}_{\boldsymbol{c}}$; their supplementary calculations collect standard estimators of ρ and β. This extra layer is not a consequence of the abstract condition $\mathcal{C}_2(\gamma, \rho, A)$ alone. It is an additional modelling and estimation step, and it inherits the usual instability of second-order bias estimation noted both by Daouia, Padoan, et al. (2024, Remark 6) and, for expectile estimation, by Daouia, Girard, et al. (2018, §3.1.1).

Once (4.25) is used, the first-order objective is variance rather than AMSE: the limiting mean in (4.26) is zero, and the remaining variance is $\boldsymbol{\omega}_\gamma^\top \boldsymbol{V}_{\boldsymbol{c}} \boldsymbol{\omega}_\gamma$. Thus the natural companion weight is the variance-optimal $\boldsymbol{\omega}^{\mathrm{Var}}$ of (4.17). Using the AMSE weight from (4.20) is still asymptotically valid, but after bias correction it is no longer first-order variance-optimal unless it coincides with $\boldsymbol{\omega}^{\mathrm{Var}}$. No additional first-order variance term appears from $\widehat{\boldsymbol{B}}_{\boldsymbol{c}}$, because its estimation error enters the normalised log ratio as $-\boldsymbol{\omega}_\gamma^\top(\widehat{\boldsymbol{B}}_{\boldsymbol{c}} - \boldsymbol{B}_{\boldsymbol{c}}) = o_p(1)$.

4.7 Asymptotic confidence intervals

The limiting theorem above is most naturally used on the logarithmic scale. This is the same scale used by Daouia, Padoan, et al. (2024, Corollary 4) for pooled Weissman quantiles, and it is also the scale on which Davison et al. (2023, §B.1) formulate their expectile confidence intervals. In the present chapter the standard error has a particularly simple form: the intermediate anchor is first-order invisible, and the standard error is the pooled-Hill standard error multiplied by the extrapolation distance ℓ_n. Equivalently, the log-scale variance is scaled by ℓ_n^2/k.

For estimated Hill weights define

$$\hat{\sigma}_{\gamma,n}^{\star\,2} \;:=\; \hat{\boldsymbol{\omega}}_\gamma^\top \widehat{\boldsymbol{V}}_{\boldsymbol{c}} \hat{\boldsymbol{\omega}}_\gamma, \qquad \ell_n = \log\left(\frac{1-\tau_n}{1-\tau_n'}\right). \tag{4.27}$$

The anchor weight $\hat{\boldsymbol{\omega}}_q$ enters the point estimator but not the first-order standard error.

Corollary 4.7 (Variance-only confidence interval for $\xi_{\tau_n'}$)**.** *Assume the setting of Theorem 4.4.*

Let $(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q)$ be estimated weights satisfying the affine constraints

$$\hat{\boldsymbol{\omega}}_\gamma^\top \mathbf{1} = \hat{\boldsymbol{\omega}}_q^\top \mathbf{1} = 1 \quad \textit{exactly}, \qquad \hat{\boldsymbol{\omega}}_\gamma \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_\gamma, \qquad \hat{\boldsymbol{\omega}}_q \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_q,$$

for deterministic affine limits $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$. Assume also that

$$\widehat{\boldsymbol{V}}_{\boldsymbol{c}} \xrightarrow{\mathbb{P}} \boldsymbol{V}_{\boldsymbol{c}}, \qquad \boldsymbol{\omega}_\gamma^\top \boldsymbol{V}_{\boldsymbol{c}} \boldsymbol{\omega}_\gamma > 0, \qquad \hat{\sigma}_{\gamma,n}^{\star 2} > 0 \textit{ with probability tending to one.}$$

Let $\widehat{\zeta}_{\tau_n'}$ denote either

$$\widehat{\xi}_{\tau_n'}^{\mathrm{pool},\star}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q)$$

under the centring condition $\boldsymbol{\omega}_\gamma^\top \boldsymbol{B}_{\boldsymbol{c}} = 0$ — in particular under the undersmoothing calibration (3.40) — or

$$\widehat{\xi}_{\tau_n'}^{\mathrm{pool},\star,\mathrm{bc}}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q)$$

under the bias-reduction assumptions of Corollary 4.6. Under the positivity convention of Section 4.3 and the expectile–quantile equivalence, both $\widehat{\zeta}_{\tau_n'}$ and $\xi_{\tau_n'}$ are positive with probability tending to one. Then, for every $\alpha \in (0,1)$, with

$$r_{n,\star} \;:=\; z_{1-\alpha/2}\,\ell_n \sqrt{\frac{\hat{\sigma}_{\gamma,n}^{\star 2}}{k}}, \qquad I_{n,\star}^{\log} \;:=\; \big[\log \widehat{\zeta}_{\tau_n'} - r_{n,\star},\, \log \widehat{\zeta}_{\tau_n'} + r_{n,\star}\big], \tag{4.28}$$

the log-scale interval $I_{n,\star}^{\log}$ and its exponential image

$$I_{n,\star} \;:=\; \exp(I_{n,\star}^{\log}) = [\exp(a_n), \exp(b_n)] \quad \textit{when } I_{n,\star}^{\log} = [a_n, b_n],$$

are asymptotic $(1-\alpha)$ confidence intervals for $\log \xi_{\tau_n'}$ and $\xi_{\tau_n'}$ respectively:

$$\mathbb{P}\big(\log \xi_{\tau_n'} \in I_{n,\star}^{\log}\big) \;\to\; 1-\alpha, \qquad \mathbb{P}\big(\xi_{\tau_n'} \in I_{n,\star}\big) \;\to\; 1-\alpha.$$

Proof. The proof of Theorem 4.4 gives the log-scale convergence

$$\frac{\sqrt{k}}{\ell_n} \left\{ \log \widehat{\xi}_{\tau_n'}^{\mathrm{pool},\star}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q) - \log \xi_{\tau_n'} \right\} \;\xrightarrow{d}\; \mathcal{N}\Big(\boldsymbol{\omega}_\gamma^\top \boldsymbol{B}_{\boldsymbol{c}},\, \boldsymbol{\omega}_\gamma^\top \boldsymbol{V}_{\boldsymbol{c}} \boldsymbol{\omega}_\gamma\Big),$$

with the same estimated-weight Slutsky step as in the theorem. Under $\boldsymbol{\omega}_\gamma^\top \boldsymbol{B}_{\boldsymbol{c}} = 0$ this limit is centred. Under the bias-reduced construction, Corollary 4.6 gives the identical centred log-scale limit with the same variance. Consequently, in either case in the statement,

$$\frac{\sqrt{k}}{\ell_n} \left\{ \log \widehat{\zeta}_{\tau_n'} - \log \xi_{\tau_n'} \right\} \xrightarrow{d} \mathcal{N}\Big(0,\, \boldsymbol{\omega}_\gamma^\top \boldsymbol{V}_{\boldsymbol{c}} \boldsymbol{\omega}_\gamma\Big).$$

The consistency of $\widehat{\boldsymbol{V}}_{\boldsymbol{c}}$ and $\hat{\boldsymbol{\omega}}_\gamma$ implies

$$\hat{\sigma}_{\gamma,n}^{\star 2} = \hat{\boldsymbol{\omega}}_\gamma^\top \widehat{\boldsymbol{V}}_{\boldsymbol{c}} \hat{\boldsymbol{\omega}}_\gamma \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_\gamma^\top \boldsymbol{V}_{\boldsymbol{c}} \boldsymbol{\omega}_\gamma,$$

so self-normalisation gives a standard normal limit. The coverage of (4.28) follows, and the original-scale statement is the image of the same event under the strictly increasing map

$x \mapsto \exp(x)$. □

Remark 4.8 (Bias-centred interval). If the asymptotic bias in Theorem 4.4 is retained rather than removed, the variance-only interval (4.28) is not centred correctly unless $\boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c} = 0$. Suppose instead that

$$\widehat{\boldsymbol{B}_{\boldsymbol{c}}} \xrightarrow{\mathbb{P}} \boldsymbol{B_c}, \qquad \hat{\mu}^\star_{\gamma,n} \;:=\; \hat{\boldsymbol{\omega}}_\gamma^\top \widehat{\boldsymbol{B}_{\boldsymbol{c}}} \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c}.$$

The subscript γ emphasises that, on the very-extreme scale, the surviving first-order bias is the Hill-bias component. With the same weights and the same $\widehat{\boldsymbol{B}_{\boldsymbol{c}}}$, this centering is algebraically identical to using the minimal exponent-corrected estimator in (4.28), since

$$\log \widehat{\xi}^{\text{pool},\star,\text{bc}}_{\tau_n'}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q) = \log \widehat{\xi}^{\text{pool},\star}_{\tau_n'}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q) - \hat{\mu}^\star_{\gamma,n}\frac{\ell_n}{\sqrt{k}}.$$

Then, for every $\alpha \in (0,1)$, using the radius $r_{n,\star}$ from (4.28), the log-scale bias-centred interval

$$C_{n,\star,\text{bcen}} \;:=\; \log \widehat{\xi}^{\text{pool},\star}_{\tau_n'}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q) - \hat{\mu}^\star_{\gamma,n}\frac{\ell_n}{\sqrt{k}}, \qquad I^{\log}_{n,\star,\text{bcen}} \;:=\; [C_{n,\star,\text{bcen}} - r_{n,\star},\, C_{n,\star,\text{bcen}} + r_{n,\star}] \tag{4.29}$$

has the same asymptotic coverage as (4.28), and so does its exponential image. This is only a feasible procedure when the Hill-bias vector can be estimated consistently. As in Daouia, Padoan, et al. (2024), such a step typically requires an additional second-order working model, for example a Hall-type form for A. The same practical caveat appears in the iid expectile discussion of Daouia, Girard, et al. (2018, §3.1.1): bias reduction is available in principle, but it imports the variability and modelling choices associated with estimating second-order tail parameters.

The width of (4.28) is

$$2z_{1-\alpha/2}\,\ell_n\sqrt{\hat{\boldsymbol{\omega}}_\gamma^\top \widehat{\boldsymbol{V}_{\boldsymbol{c}}} \hat{\boldsymbol{\omega}}_\gamma / k}.$$

Thus the interval expands linearly with the extrapolation distance from τ_n to τ_n' and contracts with the effective root-k information in the pooled Hill estimator. The standard error does not involve $m(\gamma)$, $\boldsymbol{L}^\star$, $\boldsymbol{B}^\star_{\boldsymbol{c}}$ or $b^{\xi/Q}$; those quantities affect the intermediate anchor but not the first-order very-extreme normalisation. For the same reason there is no special restriction at the root $m(\gamma) = 0$ in this section.

In the distributed common-distribution case, the natural variance-dominated choice is to pair Corollary 4.7 with the variance-optimal effective-sample-size weights of (4.22), using the matched convention $\hat{\boldsymbol{\omega}}_q = \hat{\boldsymbol{\omega}}_\gamma$ for the point estimator. Under undersmoothing, or more generally whenever $\boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c} = 0$, this yields the variance-only interval (4.28) without estimating $\boldsymbol{B_c}$. Outside such a centred calibration, the same variance-only interval is not asymptotically centred; valid inference then requires either the bias-reduced estimator of (4.25) or, equivalently, the bias-centred log interval (4.29). Both require a consistent estimator of the Hill-bias vector and therefore belong to the second-order-calibrated regime.

Chapter 5

Finite-sample performance

5.1 Purpose and scope

The asymptotic results of Chapters 3 and 4 are first-order statements. They identify the limiting distribution of the pooled QB expectile estimator, the weight vectors that minimise the leading variance or AMSE, and the log-scale confidence intervals that follow from the corresponding Gaussian approximations. A finite-sample study is useful for a different question: whether those first-order rules already behave sensibly at sample sizes comparable to the distributed extreme-value examples of Daouia, Padoan, et al. (2024).

The study in this chapter is simulation-only. It deliberately remains inside the thesis setting: observations are iid within each machine and independent across machines, all machines have the same marginal distribution, and only the communicated primitives

$$(\hat{\gamma}_j(k_j), \hat{q}_j^{\star}(\,\cdot \mid k_j)), \qquad j = 1, \ldots, m,$$

are used by the pooled procedures. No local LAWS expectile, raw observation, time-series feature, or multivariate observation is introduced. This keeps the numerical experiment aligned with the distributed-inference specialisation of Section 2.5.6, rather than the more general tail-dependent pooling framework of Daouia, Padoan, et al. (2024, §2).

5.2 Simulation design

The simulation has two targets. The intermediate target is ξ_{τ_n} with $\tau_n = 0.99$. The very-extreme target is $\xi_{\tau_n'}$ with $\tau_n' = 1 - 1/n$, estimated by the Weissman-style extrapolation of Chapter 4 from the intermediate anchor. The true expectiles are computed by deterministic root solving of

$$\tau\ \mathbb{E}[(X - x)_+] = (1 - \tau)\ \mathbb{E}[(x - X)_+],$$

using numerical integration of the distribution function and survival function of the data-generating model.

Six heavy-tailed marginal distributions are used. The first four lie in the finite-variance range $\gamma < 1/2$, where comparison with direct LAWS theory is meaningful. The final two are finite-mean but infinite-variance stress cases with $\gamma = 0.6$; they are included to check the QB estimator in the wider range allowed by the theory of Theorem 2.12.

Model	Parameters	Tail index
Pareto	survival $x^{-1/\gamma}$, $x \geq 1$	$\gamma = 0.25$
Fréchet	$F(x) = \exp(-x^{-1/\gamma})$	$\gamma = 0.25$
Burr	$F(x) = 1 - (1 + x^{-\rho/\gamma})^{1/\rho}$, $\rho = -1$	$\gamma = 0.25$
Absolute Student	$X = \lvert T_3 \rvert$	$\gamma = 1/3$
ParetoHeavy	survival $x^{-1/\gamma}$, $x \geq 1$	$\gamma = 0.6$
BurrHeavy	$F(x) = 1 - (1 + x^{-\rho/\gamma})^{1/\rho}$, $\rho = -1$	$\gamma = 0.6$

All six distributions are nonnegative, so the lower-tail integrability condition is automatic. The two heavier cases deliberately fall outside the direct-LAWS finite-variance regime while remaining inside the finite-mean QB regime $0 < \gamma < 1$.

The machine dimension is $m \in \{5, 10, 20\}$. For each m, the final grid includes balanced, moderately unbalanced and strongly unbalanced sample allocations. The main effective-sample rule is the equal-fraction choice $k_j/n_j = k/n$ with $k/n = 0.05$, the regime in which the variance-optimal expectile weights collapse to the Daouia–Padoan–Stupfler distributed weights. A sensitivity block varies k/n over $\{0.02, 0.05, 0.10, 0.15\}$ in the strongly unbalanced regime, and a separate stress block keeps k_j equal across machines under strongly unbalanced n_j to make the unequal-fraction AMSE effects visible.

5.3 Estimators and metrics

For each generated dataset, the local Hill and Weissman primitives are first computed on every machine. The feasible pooled estimators are:

(i) the naive pooled QB estimator with equal weights;

(ii) the variance-weighted estimator with $\omega_j^{\mathrm{Var}} = k_j/k$;

(iii) the plug-in AMSE estimator obtained by inserting estimated second-order quantities into the Chapter 3 stacked AMSE formula.

The plug-in AMSE row is exploratory. It follows the implementation strategy of Daouia, Padoan, et al. (2024, §2.2 and §3.2): when the R package `evt0` is available, the simulation scripts attempt to obtain second-order estimates from its MOP routines and insert them into the bias vector. This feasibility does not follow from the abstract $\mathcal{C}_2(\gamma, \rho, A)$ condition alone; it relies on a working second-order calibration. Because such estimation is often unstable in finite samples, two diagnostic comparators are also reported: an oracle AMSE estimator, using the known second-order constants of the data-generating model, and an infeasible full-sample QB benchmark computed from the combined raw sample.

Performance is evaluated on the log-relative-error scale

$$\log(\widehat{\xi}/\xi),$$

which matches the log-scale form of the confidence intervals in Sections 3.6 and 4.7. For each estimator and target the scripts report bias, variance, MSE, empirical coverage of nominal 95% confidence intervals, average confidence-interval length, the frequency with which AMSE weights are negative, and the average maximum absolute AMSE weight. For the heavier DGPs, a finite-sample Hill estimate may occasionally fall outside the QB domain $(0, 1)$. The scripts do not truncate such estimates. They record the corresponding estimator as invalid, compute log-error and coverage summaries over valid replications only, and report the invalid-domain rate separately.

5.4 Reproducibility

The simulation code lives in the repository under `simulation/`. The command

```
Rscript simulation/run_simulation.R -pilot
```

runs a 300-replication debugging grid. The command

```
Rscript simulation/run_simulation.R -final
```

runs the 5,000-replication publication grid. Results are stored as RDS files under `simulation/results/`; rendered tables and figures are generated by

```
Rscript simulation/render_results.R -final.
```

The random seed is fixed at 20260602. If `evt0` is not installed, the feasible plug-in AMSE rows are skipped and the scripts still report the naive, variance-optimal, oracle-AMSE and full-sample benchmarks. The package `ExtremeRisks` is used only as an external validation reference for standard expectile routines when it is available; the pooled expectile estimators themselves are implemented directly from the formulas of Chapters 3 and 4.

5.5 Generated summaries

The main comparison is between the variance-weighted distributed estimator and the infeasible full-sample benchmark. Under equal effective fractions, Proposition 3.4 predicts that the variance-weighted rule should inherit the same first-order efficiency logic as the quantile-pooling estimator of Daouia, Padoan, et al. (2024). The unequal-fraction stress block is designed to probe the complementary Chapter 3 message: AMSE calibration becomes target-dependent once the effective-fraction collapse is broken.

DGP	Regime	m	Estimator	Bias	MSE	Coverage
AbsStudent3	strong	10	amse_oracle	-0.048	0.004	0.813
AbsStudent3	strong	10	amse_plugin	-0.048	0.004	0.813
AbsStudent3	strong	10	naive	-0.050	0.006	0.867
AbsStudent3	strong	10	variance	-0.048	0.004	0.813
Burr	strong	10	amse_oracle	-0.127	0.017	0.004
Burr	strong	10	amse_plugin	-0.127	0.017	0.004
Burr	strong	10	naive	-0.127	0.017	0.033
Burr	strong	10	variance	-0.127	0.017	0.004
BurrHeavy	strong	10	amse_oracle	-0.006	0.018	0.946
BurrHeavy	strong	10	amse_plugin	-0.006	0.018	0.946
BurrHeavy	strong	10	naive	-0.009	0.030	0.947
BurrHeavy	strong	10	variance	-0.006	0.018	0.947
Frechet	strong	10	amse_oracle	-0.143	0.021	0.000
Frechet	strong	10	amse_plugin	-0.143	0.021	0.000
Frechet	strong	10	naive	-0.144	0.022	0.011
Frechet	strong	10	variance	-0.143	0.021	0.000
Pareto	strong	10	amse_oracle	-0.158	0.025	0.000
Pareto	strong	10	amse_plugin	-0.157	0.026	0.000
Pareto	strong	10	naive	-0.159	0.026	0.004
Pareto	strong	10	variance	-0.158	0.025	0.000
ParetoHeavy	strong	10	amse_oracle	-0.062	0.020	0.889
ParetoHeavy	strong	10	amse_plugin	-0.062	0.020	0.889
ParetoHeavy	strong	10	naive	-0.063	0.031	0.910
ParetoHeavy	strong	10	variance	-0.062	0.020	0.888

Table 5.1: Monte Carlo summary for the intermediate expectile target. The displayed rows are produced automatically from the latest simulation run; the final thesis version should be regenerated from the `-final` grid.

DGP	Estimator	Invalid rate	Valid reps
AbsStudent3	amse_oracle	0.0000	5000
AbsStudent3	amse_plugin	0.0000	5000
AbsStudent3	full_sample	0.0000	5000
AbsStudent3	naive	0.0000	5000
AbsStudent3	variance	0.0000	5000
Burr	amse_oracle	0.0000	5000
Burr	amse_plugin	0.0000	5000
Burr	full_sample	0.0000	5000
Burr	naive	0.0000	5000
Burr	variance	0.0000	5000
BurrHeavy	amse_oracle	0.0000	5000
BurrHeavy	amse_plugin	0.0000	5000
BurrHeavy	full_sample	0.0000	5000
BurrHeavy	naive	0.0000	5000
BurrHeavy	variance	0.0000	5000
Frechet	amse_oracle	0.0000	5000
Frechet	amse_plugin	0.0000	5000
Frechet	full_sample	0.0000	5000
Frechet	naive	0.0000	5000
Frechet	variance	0.0000	5000
Pareto	amse_oracle	0.0000	5000
Pareto	amse_plugin	0.0010	4995
Pareto	full_sample	0.0000	5000
Pareto	naive	0.0000	5000
Pareto	variance	0.0000	5000
ParetoHeavy	amse_oracle	0.0000	5000
ParetoHeavy	amse_plugin	0.0004	4998
ParetoHeavy	full_sample	0.0000	5000
ParetoHeavy	naive	0.0000	5000
ParetoHeavy	variance	0.0000	5000

Table 5.2: QB domain-validity diagnostic for the intermediate target. Invalid rows correspond to finite-sample pooled or full-sample Hill estimates outside $(0, 1)$; they are not truncated and are excluded from the bias, MSE and coverage summaries.

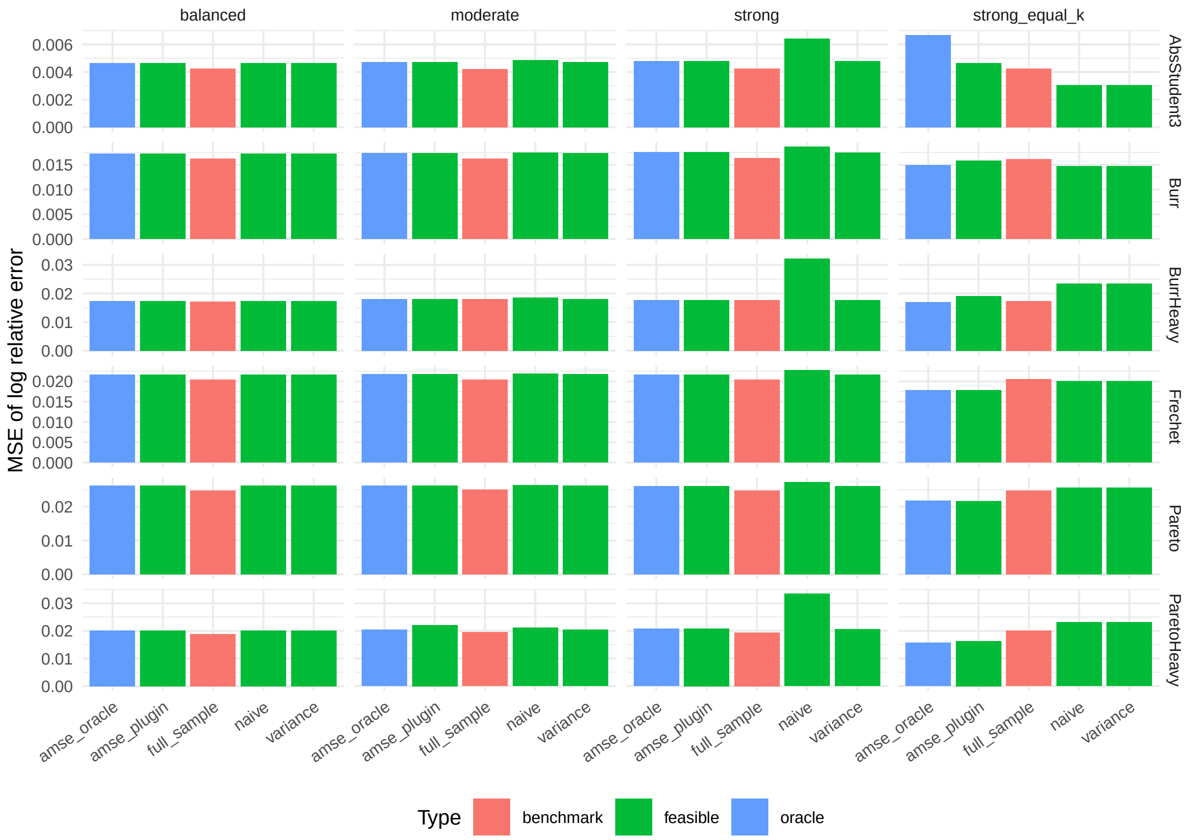

Figure 5.1: MSE of log-relative error across estimators, DGPs and sample allocation regimes.

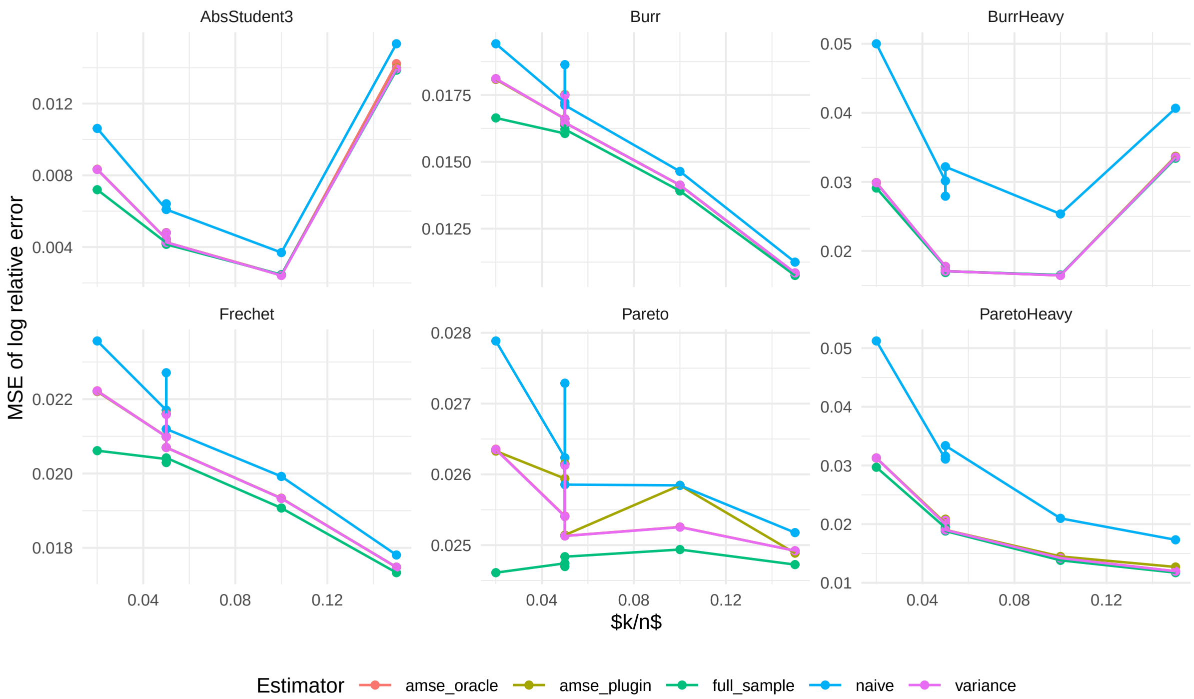

Figure 5.2: Sensitivity of the intermediate-expectile MSE to the effective sample fraction k/n in the strongly unbalanced regime.

DGP	Regime	m	Estimator	Neg. rate	Avg. max $\lvert w \rvert$
AbsStudent3	balanced	10	amse_oracle	0.000	0.100
AbsStudent3	balanced	10	amse_plugin	0.000	0.100
AbsStudent3	moderate	10	amse_oracle	0.000	0.136
AbsStudent3	moderate	10	amse_plugin	0.000	0.134
AbsStudent3	strong	10	amse_oracle	0.000	0.247
AbsStudent3	strong	10	amse_plugin	0.000	0.246
AbsStudent3	strong_equal_k	10	amse_oracle	1.000	0.468
AbsStudent3	strong_equal_k	10	amse_plugin	1.000	0.316
Burr	balanced	10	amse_oracle	0.000	0.100
Burr	balanced	10	amse_plugin	0.000	0.100
Burr	moderate	10	amse_oracle	0.000	0.134
Burr	moderate	10	amse_plugin	0.000	0.134
Burr	strong	10	amse_oracle	0.000	0.256
Burr	strong	10	amse_plugin	0.001	0.254
Burr	strong_equal_k	10	amse_oracle	1.000	0.348
Burr	strong_equal_k	10	amse_plugin	0.998	0.281
BurrHeavy	balanced	10	amse_oracle	0.000	0.100
BurrHeavy	balanced	10	amse_plugin	0.000	0.100
BurrHeavy	moderate	10	amse_oracle	0.000	0.135
BurrHeavy	moderate	10	amse_plugin	0.000	0.135
BurrHeavy	strong	10	amse_oracle	0.000	0.248
BurrHeavy	strong	10	amse_plugin	0.000	0.247
BurrHeavy	strong_equal_k	10	amse_oracle	1.000	0.451
BurrHeavy	strong_equal_k	10	amse_plugin	1.000	0.463
Frechet	balanced	10	amse_oracle	0.000	0.100
Frechet	balanced	10	amse_plugin	0.000	0.100
Frechet	moderate	10	amse_oracle	0.000	0.134
Frechet	moderate	10	amse_plugin	0.000	0.134
Frechet	strong	10	amse_oracle	0.000	0.257
Frechet	strong	10	amse_plugin	0.007	0.258
Frechet	strong_equal_k	10	amse_oracle	1.000	0.891
Frechet	strong_equal_k	10	amse_plugin	1.000	0.904
Pareto	balanced	10	amse_oracle	0.000	0.100
Pareto	balanced	10	amse_plugin	0.000	0.100
Pareto	moderate	10	amse_oracle	0.000	0.134
Pareto	moderate	10	amse_plugin	0.005	0.148
Pareto	strong	10	amse_oracle	0.000	0.258
Pareto	strong	10	amse_plugin	0.017	0.270
Pareto	strong_equal_k	10	amse_oracle	1.000	1.318
Pareto	strong_equal_k	10	amse_plugin	1.000	1.495
ParetoHeavy	balanced	10	amse_oracle	0.000	0.100
ParetoHeavy	balanced	10	amse_plugin	0.000	0.100
ParetoHeavy	moderate	10	amse_oracle	0.000	0.135
ParetoHeavy	moderate	10	amse_plugin	0.002	0.138
ParetoHeavy	strong	10	amse_oracle	0.000	0.249
ParetoHeavy	strong	10	amse_plugin	0.004	0.260
ParetoHeavy	strong_equal_k	10	amse_oracle	1.000	0.457
ParetoHeavy	strong_equal_k	10	amse_plugin	1.000	0.457

Table 5.3: AMSE-weight stability diagnostics. Negative weights are not forbidden by the affine pooling theory, but high negative-weight rates or large maximum absolute weights indicate finite-sample instability.

Chapter 6

Discussion and future work

6.1 Summary of contributions

[**TODO:** Restate the contributions of Chapters 3 and 4 in the language of Section 1.5. Emphasise the modular architecture: the pooled inputs $(\hat{\gamma}_n, \widehat{q}_n^\star)$ of Daouia, Padoan, et al. (2024) compose cleanly with the expectile–quantile asymptotic equivalence via the plug-in CLT of Davison et al. (2023, Prop. C.6), so the only expectile-specific computation is the delta-method calculus around $\psi(\gamma) = (\gamma^{-1} - 1)^{-\gamma}$.]

6.2 Connections to Expected Shortfall

[**TODO:** For a heavy-tailed loss with index $\gamma \in (0, 1)$ there is an asymptotic relation $\xi_\tau \sim \mathrm{ES}_{\tau'}$ for a specific matching level $\tau' = \tau'(\tau, \gamma)$ that can be expressed in closed form (cf. Daouia, Girard, et al. (2018)). Thus pooled expectile inference yields, as an immediate corollary, pooled inference on the Expected Shortfall, sidestepping the non-elicitability of ES. Sketch the level-matching identity and the corresponding plug-in estimator.]

6.3 Limitations

[**TODO:** (i) Tail-homogeneity across samples is assumed when pooling extreme quantiles geometrically (cf. assumption $(\mathcal{H})$ in Daouia, Padoan, et al. (2024, §2.3)) and is also needed for pooling expectiles. The test of Section 3.7 should be applied first.
(ii) The choice of effective sample sizes k_j is data-driven and introduces variability that the asymptotic theory does not account for.
(iii) The quantile-based route remains theoretically available for finite-mean tails $\gamma \in [1/2, 1)$, but standard direct-LAWS finite-variance CLTs no longer apply there; simulation evidence in that range should therefore be interpreted as QB stress testing rather than LAWS comparison.
(iv) The framework is iid within and across samples. Time-series dependence within a sample and dependence across samples are excluded.]

6.4 Future work

[**TODO:** (i) *Time-series dependence within each sample.* Extending the framework to β-mixing observations within a sample, in the spirit of Davison et al. (2023), would require inflating the within-sample asymptotic variance by a serial-dependence factor. The pooled inference architecture itself is preserved.
(ii) *Multivariate observations.* Each sample could be d-dimensional, with the goal of pooling marginal-component-wise expectiles. The joint expectile machinery of Padoan and Stupfler (2022) provides the within-sample tools; the cross-sample step is identical to the univariate case.
(iii) *Tail-heteroscedasticity.* A simple parametric model for $q_j(\tau)/q_\ell(\tau) \to \kappa_{j,\ell}$ would allow pooling even when assumption $(\mathcal{H})$ fails.
(iv) *Empirical applications.* Distributed-inference settings in operational risk (cross-bank loss data), insurance (multi-line claims), and climate (multi-station extremes) are natural use cases.
(v) *Bias-robust pooled estimators.* Use bias-reduced tail-index estimators (e.g. corrected Hill) in place of the plain Hill estimator in the pooled construction.]

Appendix A

Proofs

This appendix collects the detailed proofs of the named results of Chapters 2 to 4 whose proof sketches in the main text omit routine bookkeeping. Each section is self-contained and re-uses only the notation and named results of the body of the thesis.

A.1 Proof of Proposition 3.1

We prove the $\sqrt{k}$-asymptotic equivalence of the two pooled QB intermediate-expectile estimators $\widehat{\xi}^{(A)}_{\tau_n}$ and $\widehat{\xi}^{(B)}_{\tau_n}$ introduced in Section 3.2, applied to a common, *deterministic* weight vector $\boldsymbol{\omega} \in \mathbb{R}^m$ with $\boldsymbol{\omega}^\top \mathbf{1} = 1$, the number of samples m being fixed as $n \to \infty$. The estimated-weight extension stated in the proposition is proved after the deterministic case.

Reduction to a ψ-difference. From the definitions (3.6) and (3.7),

$$\begin{aligned}
\widehat{\xi}^{(A)}_{\tau_n}(\boldsymbol{\omega}, \boldsymbol{\omega}) &= \psi(\hat{\gamma}_n(\boldsymbol{\omega}))\, \widehat{q}^{\star}_n(\tau_n \mid \boldsymbol{\omega}), \\
\widehat{\xi}^{(B)}_{\tau_n}(\boldsymbol{\omega}) &= \Big(\prod_{j=1}^{m} \psi(\hat{\gamma}_j(k_j))^{\omega_j} \Big)\, \widehat{q}^{\star}_n(\tau_n \mid \boldsymbol{\omega}).
\end{aligned}$$

The Weissman factor is identical in the two estimators, so the ratio reduces to a function of the Hill estimators only:

$$\log \frac{\widehat{\xi}^{(A)}_{\tau_n}(\boldsymbol{\omega}, \boldsymbol{\omega})}{\widehat{\xi}^{(B)}_{\tau_n}(\boldsymbol{\omega})} = \log \psi(\hat{\gamma}_n(\boldsymbol{\omega})) - \sum_{j=1}^{m} \omega_j \log \psi(\hat{\gamma}_j(k_j)). \tag{A.1}$$

The proposition therefore reduces to

$$\sqrt{k} \Big[\log \psi(\hat{\gamma}_n(\boldsymbol{\omega})) - \sum_{j} \omega_j \log \psi(\hat{\gamma}_j(k_j)) \Big] \xrightarrow{\mathbb{P}} 0. \tag{A.2}$$

The log-geometric Weissman factors are defined on the high-probability positivity event described after (3.5). They cancel exactly in (A.1), so no Weissman-specific stochastic term enters the remainder of this proof.

Good event supporting the Taylor expansion. The map $g \mapsto \log\psi(g) = -g\log(g^{-1}-1)$ is defined on the open interval $(0,1)$ and singular at its endpoints, so the Taylor expansion below is carried out on a compact subinterval of $(0,1)$. Choose $\eta \in (0, \min\{\gamma, 1-\gamma\})$, so that $\gamma \in (\eta, 1-\eta)$. By Theorem 2.6, $\hat\gamma_j(k_j) \xrightarrow{\mathbb{P}} \gamma$ for every j. The pooled estimator also satisfies $\hat\gamma_n(\boldsymbol{\omega}) = \sum_j \omega_j\, \hat\gamma_j(k_j) \xrightarrow{\mathbb{P}} \gamma$, because it is a finite weighted sum of consistent terms with deterministic weights summing to one. Hence the compact good event

$$E_{n,\eta} \;\coloneqq\; \{\hat\gamma_1(k_1), \dots, \hat\gamma_m(k_m), \hat\gamma_n(\boldsymbol{\omega}) \in [\eta, 1-\eta]\}$$

satisfies $\mathbb{P}(E_{n,\eta}) \to 1$. It suffices to establish the claimed $O_p(k^{-1})$ bound on $E_{n,\eta}$: modifying a sequence on an event of probability tending to zero does not affect convergence in probability or $O_p(\cdot)$ bounds. Throughout the deterministic part of the proof we therefore work on $E_{n,\eta}$ without further comment.

Taylor expansion with Lagrange remainder. The map $g \mapsto \log\psi(g) = -g\log(g^{-1}-1)$ is C^∞ on the open interval $(0,1)$, with first derivative $m(g) = (1-g)^{-1} - \log(g^{-1}-1)$ (the function introduced in (3.15)) and second derivative

$$m'(g) \;=\; \frac{1}{(1-g)^2} + \frac{1}{1-g} + \frac{1}{g} \;>\; 0, \qquad g \in (0,1).$$

By Taylor's theorem with Lagrange remainder, for each $j \in \{1, \dots, m\}$ there exists a random $\tilde\gamma_j$ lying between $\hat\gamma_j(k_j)$ and γ such that

$$\log\psi(\hat\gamma_j(k_j)) - \log\psi(\gamma) \;=\; m(\gamma)\,(\hat\gamma_j(k_j) - \gamma) + \tfrac{1}{2}\,m'(\tilde\gamma_j)\,\big(\hat\gamma_j(k_j) - \gamma\big)^2, \tag{A.3}$$

and similarly there exists $\tilde\gamma_\star$ between $\hat\gamma_n(\boldsymbol{\omega})$ and γ with

$$\log\psi(\hat\gamma_n(\boldsymbol{\omega})) - \log\psi(\gamma) \;=\; m(\gamma)\,(\hat\gamma_n(\boldsymbol{\omega}) - \gamma) + \tfrac{1}{2}\,m'(\tilde\gamma_\star)\,\big(\hat\gamma_n(\boldsymbol{\omega}) - \gamma\big)^2. \tag{A.4}$$

Subtract the $\boldsymbol{\omega}$-weighted sum of (A.3) across j from (A.4). The constant terms $\log\psi(\gamma)$ cancel because $\sum_j \omega_j = 1$, and the linear terms cancel because the pooled Hill estimator is the weighted linear combination $\hat\gamma_n(\boldsymbol{\omega}) = \sum_j \omega_j\hat\gamma_j(k_j)$, which implies $\sum_j \omega_j(\hat\gamma_j - \gamma) = \hat\gamma_n(\boldsymbol{\omega}) - \gamma$. We obtain

$$\log\psi(\hat\gamma_n(\boldsymbol{\omega})) - \sum_j \omega_j \log\psi(\hat\gamma_j(k_j)) \;=\; \tfrac{1}{2}\,m'(\tilde\gamma_\star)\,\big(\hat\gamma_n(\boldsymbol{\omega}) - \gamma\big)^2 - \tfrac{1}{2}\sum_j \omega_j\, m'(\tilde\gamma_j)\,\big(\hat\gamma_j(k_j) - \gamma\big)^2. \tag{A.5}$$

Order of the remainder terms. By Theorem 2.6, $\sqrt{k_j}(\hat\gamma_j(k_j) - \gamma) \xrightarrow{d} \mathcal{N}(\lambda_j/(1-\rho), \gamma^2)$ for every j, hence $\hat\gamma_j(k_j) - \gamma = O_p(k_j^{-1/2})$. The proportionality condition (2.12), $k_1/k_j \to c_j \in (0,\infty)$, gives $k_j \asymp k$ uniformly in j, so

$$\hat\gamma_j(k_j) - \gamma \;=\; O_p(k^{-1/2}), \qquad \big(\hat\gamma_j(k_j) - \gamma\big)^2 \;=\; O_p(k^{-1}), \qquad j = 1, \dots, m. \tag{A.6}$$

The pooled Hill estimator inherits the rate as a finite weighted combination of $\sqrt{k}$-consistent terms:

$$\hat{\gamma}_n(\boldsymbol{\omega}) - \gamma \;=\; O_p(k^{-1/2}), \qquad \big(\hat{\gamma}_n(\boldsymbol{\omega}) - \gamma\big)^2 \;=\; O_p(k^{-1}). \tag{A.7}$$

For the Lagrange-remainder pre-factors, all intermediate points $\tilde{\gamma}_j$ and $\tilde{\gamma}_\star$ lie in $[\eta, 1-\eta]$ on $E_{n,\eta}$. Since m' is continuous on this compact interval,

$$C_\eta \;:=\; \sup_{g\in[\eta,1-\eta]} \big|m'(g)\big| \;<\; \infty.$$

Thus the Lagrange factors are deterministically bounded on the good event.

Inserting these orders into (A.5): each summand on the right-hand side is the product of an $O_p(1)$ factor ($m'(\tilde{\gamma}_\cdot)$) and an $O_p(k^{-1})$ factor (a squared deviation), so each is $O_p(k^{-1})$. Since the sum on the right has $m+1$ terms and m is fixed, the right-hand side of (A.5) is itself $O_p(k^{-1})$.

Conclusion for deterministic weights. Combining the bound in (A.5) with the reduction (A.1) gives

$$\log \frac{\widehat{\xi}^{(A)}_{\tau_n}(\boldsymbol{\omega},\boldsymbol{\omega})}{\widehat{\xi}^{(B)}_{\tau_n}(\boldsymbol{\omega})} \;=\; O_p(k^{-1}),$$

hence

$$\sqrt{k}\,\log \frac{\widehat{\xi}^{(A)}_{\tau_n}(\boldsymbol{\omega},\boldsymbol{\omega})}{\widehat{\xi}^{(B)}_{\tau_n}(\boldsymbol{\omega})} \;=\; O_p(k^{-1/2}) \;=\; o_p(1),$$

which proves the convergence in probability claimed by Proposition 3.1. Since $e^x - 1 \sim x$ as $x \to 0$ and our log-ratio is $O_p(k^{-1})$, the continuous mapping theorem applied to exp further yields

$$\frac{\widehat{\xi}^{(A)}_{\tau_n}(\boldsymbol{\omega},\boldsymbol{\omega})}{\widehat{\xi}^{(B)}_{\tau_n}(\boldsymbol{\omega})} - 1 \;=\; O_p(k^{-1}).$$

The k^{-1} rate is one full order below the $k^{-1/2}$ scale of Theorem 3.3: the two constructions agree asymptotically not only at the CLT scale but at a strictly finer scale, so the two estimators are interchangeable to leading order in every quantity downstream of the joint Gaussian limit.

Estimated-weight extension. Let now $\hat{\boldsymbol{\omega}}$ be a random weight vector satisfying $\hat{\boldsymbol{\omega}} \xrightarrow{\mathbb{P}} \boldsymbol{\omega}$ for a deterministic $\boldsymbol{\omega}$ with $\boldsymbol{\omega}^\top \mathbf{1} = 1$. Then $\|\hat{\boldsymbol{\omega}}\| = O_p(1)$. If $\hat{\boldsymbol{\omega}}^\top \mathbf{1} = 1$ exactly, the previous proof applies with $\boldsymbol{\omega}$ replaced by $\hat{\boldsymbol{\omega}}$ on the joint high-probability event where $\|\hat{\boldsymbol{\omega}}\|$ is bounded and all local and pooled Hill estimators lie in $[\eta, 1-\eta]$. The weighted Hill deviation remains $O_p(k^{-1/2})$, because

$$\left|\sum_{j=1}^{m} \hat{\omega}_j\big(\hat{\gamma}_j(k_j) - \gamma\big)\right| \;\le\; \|\hat{\boldsymbol{\omega}}\|\,\|\hat{\boldsymbol{\gamma}}_n - \gamma\mathbf{1}\| \;=\; O_p(k^{-1/2}),$$

and the weighted sum of squared local deviations remains $O_p(k^{-1})$, since m is fixed, $\|\hat{\boldsymbol{\omega}}\| = O_p(1)$, and each $(\hat{\gamma}_j(k_j) - \gamma)^2 = O_p(k^{-1})$. Hence the unscaled log-ratio is still $O_p(k^{-1})$.

If exact normalisation is relaxed, write $s_n = \hat{\boldsymbol{\omega}}^\top \mathbf{1}$ and $f(g) = \log \psi(g)$. Taylor expansion around

γ gives

$$f\left(\sum_{j=1}^{m}\hat{\omega}_j\hat{\gamma}_j(k_j)\right)-\sum_{j=1}^{m}\hat{\omega}_j f(\hat{\gamma}_j(k_j))=(s_n-1)\{\gamma f'(\gamma)-f(\gamma)\}+O_p(k^{-1})+o_p(k^{-1/2}),$$

where the remainders use $\|\hat{\boldsymbol{\omega}}\|=O_p(1)$, $\hat{\gamma}_j(k_j)-\gamma=O_p(k^{-1/2})$ uniformly over fixed j, and $\sqrt{k}\,|s_n-1|\xrightarrow{\mathbb{P}}0$. Multiplication by $\sqrt{k}$ therefore still yields $o_p(1)$, which proves the stated estimated-weight $\sqrt{k}$-equivalence. □

A.2 Proof of Proposition 3.2

We establish the joint $\sqrt{k}$-Gaussian limit (3.11) of the pair $\big(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma),\,\log\widehat{q}_n^{\star}(\tau_n\mid\boldsymbol{\omega}_q)\big)$ under the standing assumptions of Section 3.1: m independent samples sharing the common marginal F and the common tail index $\gamma\in(0,1)$, the proportionality conditions (2.12), the second-order condition $\mathcal{C}_2(\gamma,\rho,A)$ with the per-sample bias rate $\sqrt{k_j}\,A(n_j/k_j)\to\lambda_j\in\mathbb{R}$, an intermediate target level $\tau_n\to 1$ with $n(1-\tau_n)\to\infty$, and the chapter's log-rate convergence $L_j(\tau_n)\to L_j^{\star}\in\mathbb{R}$ for every j. The number of samples m is fixed.

The proof composes four steps: (i) the within-sample joint CLT for the Hill estimator and the intermediate log-order statistic; (ii) the linear lift of (i) to the (Hill, log-Weissman) pair via the exact decomposition (3.9); (iii) cross-sample independence, which renders the $2m$-vector of all per-sample inputs jointly Gaussian with block-diagonal covariance on the common $\sqrt{k}$ scale; (iv) the linear pooling map, which collapses the $2m$-vector onto the 2-vector (3.11) and yields the covariance (3.12).

Step (i): within-sample joint CLT for Hill and intermediate log-order statistic. Fix $j\in\{1,\dots,m\}$ and write $\tau_{n_j}:=1-k_j/n_j$ for the sample-j intermediate level. The joint $\sqrt{k_j}$-asymptotics of the Hill estimator and its companion intermediate log-order statistic is a standard tail-empirical-process result (Haan and Ferreira 2006, Theorem 2.4.8 and §5.1); in the form we use it is the input to the iid QB plug-in CLT reproduced as Theorem 2.12. Under $\mathcal{C}_2(\gamma,\rho,A)$ with $k_j\to\infty$, $k_j/n_j\to 0$, and $\sqrt{k_j}\,A(n_j/k_j)\to\lambda_j\in\mathbb{R}$,

$$\sqrt{k_j}\begin{pmatrix}\hat{\gamma}_j(k_j)-\gamma\\ \log X_{n_j-k_j:n_j,j}-\log Q_X(\tau_{n_j})\end{pmatrix}\xrightarrow{d}\mathcal{N}\left(\begin{pmatrix}b_j^H\\0\end{pmatrix},\,\gamma^2\boldsymbol{I}_2\right),\tag{A.8}$$

where $b_j^H=\lambda_j/(1-\rho)$ is the Hill bias and $\boldsymbol{I}_2$ is the 2×2 identity matrix. The *zero* second component is the zero asymptotic bias of the intermediate log-order statistic on the $\sqrt{k_j}$ scale, centred at its own population quantile $\log Q_X(\tau_{n_j})=\log U(n_j/k_j)$. In the tail-quantile-process representation, evaluating the threshold process at $s=1$ removes the deterministic second-order term because $\Psi_\rho(1)=0$ for the second-order kernel (3.14); hence no $A(n_j/k_j)$-bias remains in the second coordinate (Haan and Ferreira 2006, Theorem 2.4.8). This is in contrast to the Hill estimator, whose bias b_j^H arises precisely from averaging the second-order tail-quantile term across the top k_j order statistics, where it does not vanish.

The load-bearing structural fact in (A.8) is the *zero off-diagonal entry*. The same Brownian representation, with W_j a standard Brownian motion arising as the tail-empirical-process limit for sample j, writes the centred threshold fluctuation as $\gamma W_j(1)$ and the centred Hill fluctuation as

$$\gamma \int_0^1 \{s^{-1} W_j(s) - W_j(1)\}\, ds$$

plus the deterministic Hill bias. Therefore

$$\operatorname{Cov}\Big(W_j(1), \int_0^1 \{s^{-1} W_j(s) - W_j(1)\}\, ds\Big) = \int_0^1 \{s^{-1} \operatorname{Cov}(W_j(1), W_j(s)) - 1\}\, ds = 0,$$

which gives the asymptotic independence of the two random components and their common variance γ^2. The diagonal form $\gamma^2 \boldsymbol{I}_2$ is therefore essential for the simple $\boldsymbol{V_c}$-and-diag$(\boldsymbol{L}^\star)$ covariance structure of (3.12) used in the rest of the chapter.

Step (ii): linear lift to (Hill, log-Weissman). The Weissman decomposition (3.9), exact for every n, reads

$$\log \hat{q}_j^\star(\tau_n \mid k_j) \;-\; \log Q_X(\tau_n) \;=\; L_j(\tau_n)\,\big(\hat{\gamma}_j(k_j) - \gamma\big) \;+\; \big(\log X_{n_j-k_j:n_j,\,j} - \log Q_X(\tau_{n_j})\big) \;+\; r_j(\tau_n), \tag{A.9}$$

where $r_j(\tau_n) \coloneqq \gamma\, L_j(\tau_n) + \log Q_X(\tau_{n_j}) - \log Q_X(\tau_n)$ is a deterministic remainder. Identity (A.9) follows by inserting the definition $\hat{q}_j^\star(\tau_n \mid k_j) = (k_j/(n_j(1-\tau_n)))^{\hat{\gamma}_j(k_j)} X_{n_j-k_j:n_j,\,j}$ and adding and subtracting $\gamma\, L_j(\tau_n)$ and $\log Q_X(\tau_{n_j})$.

The remainder $r_j(\tau_n)$ measures the deterministic gap between the population log-quantile at the chapter level τ_n and its first-order regular-variation approximation pivoted at τ_{n_j}. By the second-order condition $\mathcal{C}_2(\gamma, \rho, A)$ applied to $U = (1/\overline{F})^{\leftarrow}$ at base point $t = n_j/k_j$ and ratio $y = k_j/(n_j(1-\tau_n))$, one has $\log U(t\,y) - \log U(t) - \gamma \log y = A(t)\, \Psi_\rho(y) + o(A(t))$ uniformly on compact y-sets, with

$$\Psi_\rho(y) = \begin{cases} (y^\rho - 1)/\rho, & \rho < 0, \\ \log y, & \rho = 0, \end{cases}$$

the endpoint being read in the usual second-order convention. Therefore

$$r_j(\tau_n) \;=\; -A(n_j/k_j)\, \Psi_\rho\big(k_j/(n_j(1-\tau_n))\big) \;+\; o\big(A(n_j/k_j)\big), \tag{A.10}$$

and the per-sample bias rate $\sqrt{k_j}\, A(n_j/k_j) \to \lambda_j$ together with $L_j(\tau_n) \to L_j^\star \in \mathbb{R}$ yields $\sqrt{k_j}\, r_j(\tau_n) \to -\lambda_j\, \Psi_\rho(e^{L_j^\star}) =: b_j^R$, a finite deterministic shift. Since $r_j(\tau_n)$ is deterministic, it shifts only the limiting mean and does not enter the covariance: the asymptotic covariance below is determined entirely by the two random inputs $\hat{\gamma}_j(k_j) - \gamma$ and $\log X_{n_j-k_j:n_j,\,j} - \log Q_X(\tau_{n_j})$ of (A.8).

Write (A.9) in matrix form,

$$\sqrt{k_j}\begin{pmatrix} \hat{\gamma}_j(k_j) - \gamma \\ \log \hat{q}_j^\star(\tau_n \mid k_j) - \log Q_X(\tau_n) \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 \\ L_j(\tau_n) & 1 \end{pmatrix}}_{=: \boldsymbol{M}_j(\tau_n)} \sqrt{k_j}\begin{pmatrix} \hat{\gamma}_j(k_j) - \gamma \\ \log X_{n_j - k_j : n_j, j} - \log Q_X(\tau_{n_j}) \end{pmatrix} + \begin{pmatrix} 0 \\ \sqrt{k_j}\, r_j(\tau_n) \end{pmatrix}.$$

The deterministic matrix $\boldsymbol{M}_j(\tau_n) \to \boldsymbol{M}_j^\star := (1, 0;\, L_j^\star, 1)$ entrywise, and the deterministic shift converges to $(0, b_j^R)^\top$. The continuous mapping theorem and Slutsky combine (A.8) with these deterministic convergences to deliver

$$\sqrt{k_j}\begin{pmatrix} \hat{\gamma}_j(k_j) - \gamma \\ \log \hat{q}_j^\star(\tau_n \mid k_j) - \log Q_X(\tau_n) \end{pmatrix} \xrightarrow{d} \mathcal{N}\left(\boldsymbol{\mu}_j^\star,\ \gamma^2 \begin{pmatrix} 1 & L_j^\star \\ L_j^\star & 1 + (L_j^\star)^2 \end{pmatrix}\right), \tag{A.11}$$

with mean $\boldsymbol{\mu}_j^\star = \boldsymbol{M}_j^\star (b_j^H, 0)^\top + (0, b_j^R)^\top = \left(b_j^H,\ L_j^\star\, b_j^H + b_j^R\right)^\top$, the zero order-statistic asymptotic bias of (A.8) carrying through. The covariance is $\boldsymbol{M}_j^\star\, (\gamma^2 \boldsymbol{I}_2)\, (\boldsymbol{M}_j^\star)^\top = \gamma^2 (1, L_j^\star;\, L_j^\star, 1 + (L_j^\star)^2)$, which is precisely (3.10).

Step (iii): cross-sample independence and rescaling to $\sqrt{k}$. The m samples are independent, so the $2m$ random variables $\left(\hat{\gamma}_j(k_j) - \gamma,\ \log \hat{q}_j^\star(\tau_n \mid k_j) - \log Q_X(\tau_n)\right)_{j=1}^m$ are organised as m independent 2-blocks. Stacking them into the column vector

$$\boldsymbol{Z}_n := \left(\hat{\gamma}_1 - \gamma, \ldots, \hat{\gamma}_m - \gamma,\ \log \hat{q}_1^\star - \log Q_X(\tau_n), \ldots, \log \hat{q}_m^\star - \log Q_X(\tau_n)\right)^\top \in \mathbb{R}^{2m}$$

(Hill components stacked first, log-Weissman components second), the joint CLT of $\sqrt{k_j}\boldsymbol{Z}_n^{(j)}$ from (A.11) aggregates across j under independence to a joint $2m$-dimensional Gaussian limit. To express the limit on the common $\sqrt{k}$ scale, multiply each per-sample $\sqrt{k_j}$ input by the deterministic factor $\sqrt{k/k_j}$, which converges to a finite limit:

$$\frac{k}{k_j} = \frac{\sum_{i=1}^m k_i}{k_j} = \sum_{i=1}^m \frac{k_1/k_j}{k_1/k_i} \longrightarrow c_j \sum_{i=1}^m c_i^{-1},$$

using (2.12). The per-sample within-block covariance (A.11) scales by the factor k/k_j, so its diagonal entry $(\sqrt{k}/\sqrt{k_j})^2 \cdot \gamma^2 \to \gamma^2\, c_j \sum_i c_i^{-1} = (\boldsymbol{V_c})_{j,j}$, the diagonal entry of the diagonal pooled-Hill covariance (2.14) for the independent-samples specialisation. We therefore obtain

$$\sqrt{k}\, \boldsymbol{Z}_n \xrightarrow{d} \mathcal{N}(\boldsymbol{\mu}_Z, \boldsymbol{\Sigma}_Z), \tag{A.12}$$

where, with $\boldsymbol{D}^\star := \operatorname{diag}(\boldsymbol{L}^\star)$, the $2m \times 2m$ covariance $\boldsymbol{\Sigma}_Z$ has the block-structured form

$$\boldsymbol{\Sigma}_Z = \begin{pmatrix} \boldsymbol{V_c} & \boldsymbol{D}^\star \boldsymbol{V_c} \\ \boldsymbol{V_c} \boldsymbol{D}^\star & \boldsymbol{V_c} + \boldsymbol{D}^\star \boldsymbol{V_c} \boldsymbol{D}^\star \end{pmatrix}, \tag{A.13}$$

expressed in $m \times m$ blocks. Each block is diagonal: the upper-left block is $\boldsymbol{V_c}$ itself (with diagonal entries $(\boldsymbol{V_c})_{j,j} = \gamma^2 c_j \sum_i c_i^{-1}$); the off-diagonal blocks have jth diagonal entry $L_j^\star (\boldsymbol{V_c})_{j,j}$; the lower-right block has jth diagonal entry $(1 + (L_j^\star)^2)(\boldsymbol{V_c})_{j,j}$. All non-diagonal entries within each block vanish because of cross-sample independence. The mean $\boldsymbol{\mu}_Z$ is the corresponding linear functional of the per-sample bias parameters b_j^H and b_j^R, rescaled to the $\sqrt{k}$ scale: explicitly, $(\boldsymbol{\mu}_Z)_j = \sqrt{c_j \sum_i c_i^{-1}}\, b_j^H = (\boldsymbol{B_c})_j$ for $j = 1, \dots, m$, and $(\boldsymbol{\mu}_Z)_{m+j} = \sqrt{c_j \sum_i c_i^{-1}}\, (L_j^\star b_j^H + b_j^R) =: (\boldsymbol{B}_{\boldsymbol{c}}^\star)_j$, the order-statistic bias being absent by (A.8). The first half of $\boldsymbol{\mu}_Z$ is exactly the pooled-Hill bias vector $\boldsymbol{B_c}$ of Theorem 2.16; the second half is the pooled-Weissman bias vector $\boldsymbol{B}_{\boldsymbol{c}}^\star$ of (3.14), specialised to the intermediate-level target.

Step (iv): linear pooling map. By definitions (3.4) and (3.5),

$$\begin{aligned} \hat{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma \;&=\; \sum_{j=1}^m \omega_{\gamma,j}\, (\hat{\gamma}_j(k_j) - \gamma), \\ \log \widehat{q}_n^\star(\tau_n \mid \boldsymbol{\omega}_q) - \log Q_X(\tau_n) \;&=\; \sum_{j=1}^m \omega_{q,j}\, \big(\log \hat{q}_j^\star - \log Q_X(\tau_n)\big), \end{aligned}$$

where the second identity uses $\boldsymbol{\omega}_q^\top \mathbf{1} = 1$ so that subtracting the common term $\log Q_X(\tau_n)$ commutes with the weighted sum. Both pooled coordinates are linear in $\boldsymbol{Z}_n$, and the relation is exactly

$$\begin{pmatrix} \hat{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma \\ \log \widehat{q}_n^\star(\tau_n \mid \boldsymbol{\omega}_q) - \log Q_X(\tau_n) \end{pmatrix} \;=\; \boldsymbol{A}\,\boldsymbol{Z}_n, \qquad \boldsymbol{A} \;:=\; \begin{pmatrix} \boldsymbol{\omega}_\gamma^\top & \mathbf{0}_m^\top \\ \mathbf{0}_m^\top & \boldsymbol{\omega}_q^\top \end{pmatrix} \in \mathbb{R}^{2\times 2m}.$$

Applying $\boldsymbol{A}$ to (A.12),

$$\sqrt{k}\,\boldsymbol{A}\,\boldsymbol{Z}_n \;\xrightarrow{d}\; \mathcal{N}(\boldsymbol{A}\,\boldsymbol{\mu}_Z,\; \boldsymbol{A}\,\boldsymbol{\Sigma}_Z\,\boldsymbol{A}^\top).$$

The mean follows from the block structure of $\boldsymbol{\mu}_Z$ identified in Step (iii):

$$\boldsymbol{A}\,\boldsymbol{\mu}_Z = (\boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c},\; \boldsymbol{\omega}_q^\top \boldsymbol{B}_{\boldsymbol{c}}^\star)^\top,$$

which is the chapter's $\boldsymbol{\mu}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$.

For the covariance, partition $\boldsymbol{A}$ along the same Hill / log-Weissman split as $\boldsymbol{\Sigma}_Z$ and compute block-by-block, using (A.13):

$$\begin{aligned} \boldsymbol{A}\,\boldsymbol{\Sigma}_Z\,\boldsymbol{A}^\top\big|_{1,1} \;&=\; \boldsymbol{\omega}_\gamma^\top\,\boldsymbol{V_c}\,\boldsymbol{\omega}_\gamma, \\ \boldsymbol{A}\,\boldsymbol{\Sigma}_Z\,\boldsymbol{A}^\top\big|_{1,2} \;&=\; \boldsymbol{\omega}_\gamma^\top\,\boldsymbol{D}^\star\boldsymbol{V_c}\,\boldsymbol{\omega}_q, \\ \boldsymbol{A}\,\boldsymbol{\Sigma}_Z\,\boldsymbol{A}^\top\big|_{2,2} \;&=\; \boldsymbol{\omega}_q^\top\,(\boldsymbol{V_c} + \boldsymbol{D}^\star\boldsymbol{V_c}\boldsymbol{D}^\star)\,\boldsymbol{\omega}_q, \end{aligned}$$

with the $(2,1)$ block equal to the $(1,2)$ block by symmetry of $\boldsymbol{\Sigma}_Z$. The diagonality of $\boldsymbol{V_c}$ (under independent sampling, (2.14)) ensures that $\boldsymbol{V_c}$ and $\boldsymbol{D}^\star$ commute, so the off-diagonal block can equivalently be written $\boldsymbol{\omega}_\gamma^\top\,\boldsymbol{V_c}\,\boldsymbol{D}^\star\,\boldsymbol{\omega}_q$ and is the entrywise sum $\sum_j \omega_{\gamma,j}\,\omega_{q,j}\,L_j^\star\,(\boldsymbol{V_c})_{j,j}$. This recovers the covariance (3.12) of the proposition.

Combining the two displays, the pooled 2-vector converges in distribution to a Gaussian with mean $\boldsymbol{\mu}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$ and covariance $\boldsymbol{\Sigma}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$, which is precisely (3.11). $\square$

A.3 Proof of Theorem 3.3

We prove the pooled QB intermediate-expectile CLT (3.20) under the standing assumptions of Section 3.1: m independent samples with common marginal F, common tail index $\gamma \in (0,1)$, the proportionality conditions (2.12), the second-order condition $\mathcal{C}_2(\gamma, \rho, A)$ with per-sample bias rate $\sqrt{k_j}\, A(n_j/k_j) \to \lambda_j \in \mathbb{R}$, an intermediate target $\tau_n \to 1$ with $n(1-\tau_n) \to \infty$, and the log-rate convergence $L_j(\tau_n) \to L_j^\star \in \mathbb{R}$ for every j. The number of samples m is fixed. We work for arbitrary deterministic weight vectors $\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q \in \mathbb{R}^m$ satisfying $\boldsymbol{\omega}_\gamma^\top \mathbf{1} = \boldsymbol{\omega}_q^\top \mathbf{1} = 1$; the extension to estimated weights is collected in Corollary 3.7.

The proof composes four steps: (i) a deterministic identity on the log scale that decomposes the expectile log-ratio into a random Hill increment, a random log-Weissman increment, and a deterministic expectile–quantile gap; (ii) a Taylor linearisation of $\log \psi$ at γ that converts the Hill increment into a linear functional of $\hat{\gamma}_n - \gamma$ up to a $\sqrt{k}$-negligible remainder; (iii) the second-order expectile–quantile expansion that controls the deterministic gap; (iv) the multivariate delta method applied to Proposition 3.2 to read off the asymptotic mean and covariance, followed by the continuous-mapping theorem applied to exp.

Step (i): the deterministic log-ratio decomposition. By the QB factorisation (3.8),

$$\log \widehat{\xi}^{\text{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \;=\; \log \psi\big(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)\big) \;+\; \log \widehat{q}^{\star}_n(\tau_n \mid \boldsymbol{\omega}_q).$$

Subtract $\log \xi_{\tau_n}$ from both sides and insert $\pm \log \psi(\gamma) \pm \log Q_X(\tau_n)$:

$$\begin{aligned}
\log \widehat{\xi}^{\text{pool}}_{\tau_n} - \log \xi_{\tau_n} \;&=\; \underbrace{\big[\log \psi(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)) - \log \psi(\gamma)\big]}_{\text{(H)}} \\
&\quad+ \underbrace{\big[\log \widehat{q}^{\star}_n(\tau_n \mid \boldsymbol{\omega}_q) - \log Q_X(\tau_n)\big]}_{\text{(W)}} \\
&\quad+ \underbrace{\big[\log \psi(\gamma) + \log Q_X(\tau_n) - \log \xi_{\tau_n}\big]}_{\text{(G)}}.
\end{aligned} \tag{A.14}$$

The identity (A.14) is exact for every n. Terms (H) and (W) are random fluctuations around their population centering quantities; their $\sqrt{k}$-means are those supplied by Proposition 3.2 after the linearisation in Step (ii). Term (G) is deterministic and depends on the target level τ_n only through population quantities. The sign in (G) is chosen so that the rescaled $\sqrt{k}\,$(G) converges directly to $b^{\xi/Q}$ in (3.19), eliminating the bookkeeping of an intermediate minus sign in Step (iii).

Step (ii): Taylor linearisation of $\log \psi$. The argument is identical in spirit to that of Appendix A.1. The map $g \mapsto \log \psi(g) = -g \log(g^{-1} - 1)$ is C^∞ on $(0,1)$ with first derivative

$m(g)$ defined in (3.15) and second derivative

$$m'(g) \;=\; \frac{1}{(1-g)^2} + \frac{1}{1-g} + \frac{1}{g} \;>\; 0, \qquad g \in (0,1).$$

By Theorem 2.6 and (3.2), $\hat{\gamma}_n(\boldsymbol{\omega}_\gamma) \xrightarrow{\mathbb{P}} \gamma \in (0,1)$, so the event $E_n := \{\hat{\gamma}_n(\boldsymbol{\omega}_\gamma) \in (0,1)\}$ satisfies $\mathbb{P}(E_n) \to 1$ and we may work on it without affecting convergence in distribution. On E_n, Taylor's theorem with Lagrange remainder yields a random $\tilde{\gamma}_\star$ between $\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)$ and γ such that

$$\log\psi(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)) - \log\psi(\gamma) \;=\; m(\gamma)\,(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma) + \tfrac{1}{2}\, m'(\tilde{\gamma}_\star)\,(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma)^2. \tag{A.15}$$

Proposition 3.2 gives $\hat{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma = O_p(k^{-1/2})$, hence $(\hat{\gamma}_n - \gamma)^2 = O_p(k^{-1})$. The squeeze $\tilde{\gamma}_\star \xrightarrow{\mathbb{P}} \gamma$ together with continuity of m' on $(0,1)$ gives $m'(\tilde{\gamma}_\star) \xrightarrow{\mathbb{P}} m'(\gamma) = O_p(1)$. The Lagrange remainder in (A.15) is therefore $O_p(k^{-1}) = o_p(k^{-1/2})$, so

$$\sqrt{k}\,(\mathrm{H}) \;=\; m(\gamma)\,\sqrt{k}\,(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma) \;+\; o_p(1). \tag{A.16}$$

Step (iii): the deterministic expectile–quantile gap. The first-order equivalence of Proposition 2.10,

$$\xi_\tau / Q_X(\tau) \to \psi(\gamma), \qquad \tau \uparrow 1,$$

sharpens, under the second-order condition $\mathcal{C}_2(\gamma,\rho,A)$ with the finite first-moment condition in (3.2), via the second-order expansion

$$\begin{aligned}\log\xi_\tau - \log\psi(\gamma) - \log Q_X(\tau) = c(\gamma,\rho)\, A\big((1-\tau)^{-1}\big) + o\big(A((1-\tau)^{-1})\big) \\ + O\big(Q_X(\tau)^{-1}\big), \qquad \tau \uparrow 1.\end{aligned} \tag{A.17}$$

the logarithmic form of (2.7) of Daouia, Girard, et al. (2020), with the explicit constant $c(\gamma,\rho)$ recorded there. The additive log form follows by a Taylor expansion of $\log(1+x)$ around zero. The first-moment term $O(Q_X(\tau)^{-1})$ is displayed explicitly because it need not be $o(A((1-\tau)^{-1}))$ without an additional comparison of rates. Along the sequence τ_n, however, it is $\sqrt{k}$-negligible under (3.3). Since $\sqrt{k}\,A((1-\tau_n)^{-1}) = O(1)$ by the argument below, the $o(A)$ term is also $o(k^{-1/2})$. Consequently,

$$\sqrt{k}\Big[\log\xi_{\tau_n} - \log\psi(\gamma) - \log Q_X(\tau_n)\Big] = c(\gamma,\rho)\sqrt{k}\,A\big((1-\tau_n)^{-1}\big) + o(1). \tag{A.18}$$

Only the finiteness of $c(\gamma,\rho)$ for $\gamma \in (0,1)$ and $\rho \le 0$ is used below.

To identify the deterministic shift in (A.18), we show that

$$\sqrt{k}\,A\big((1-\tau_n)^{-1}\big) \;\longrightarrow\; \Lambda^\bullet \;\in\; \mathbb{R}, \tag{A.19}$$

a finite limit determined by the chapter setup. Indeed, for any fixed j, the log-rate convergence

$L_j(\tau_n) \to L_j^\star$ gives the ratio

$$\frac{(1-\tau_n)^{-1}}{n_j/k_j} \;=\; \frac{k_j}{n_j(1-\tau_n)} \;=\; e^{L_j(\tau_n)} \;\longrightarrow\; e^{L_j^\star} \;\in\; (0,\infty),$$

so by Condition 2.3, whose auxiliary function A is eventually of constant sign and has signed ratio limit y^ρ,

$$\frac{A((1-\tau_n)^{-1})}{A(n_j/k_j)} \;\longrightarrow\; (e^{L_j^\star})^\rho.$$

Multiplying by $\sqrt{k_j}\,A(n_j/k_j) \to \lambda_j$ and then by $\sqrt{k/k_j} \to \sqrt{c_j \sum_i c_i^{-1}}$, the limit (A.19) exists and equals

$$\Lambda^\bullet \;=\; \sqrt{c_j \textstyle\sum_{i=1}^m c_i^{-1}}\;\lambda_j\,(e^{L_j^\star})^\rho \qquad \text{for every } j = 1,\dots,m. \tag{A.20}$$

The right-hand side of (A.20) appears to depend on j, but the left-hand side $\sqrt{k}\,A((1-\tau_n)^{-1})$ does not: hence the per-j limits must coincide whenever they exist. This is a consistency identity implicit in the chapter setup — not an additional hypothesis — expressing the compatibility of the per-sample bias rates $\sqrt{k_j}\,A(n_j/k_j) \to \lambda_j$ and log-rate limits $L_j(\tau_n) \to L_j^\star$ across j under a single global constant-sign auxiliary function A. Inserting (A.19) into the sequence-level expansion (A.18) and using the sign convention of (G) chosen in (A.14),

$$\sqrt{k}\,(\mathrm{G}) \;=\; \sqrt{k}\,[\log\psi(\gamma) + \log Q_X(\tau_n) - \log\xi_{\tau_n}] \;\longrightarrow\; -c(\gamma,\rho)\,\Lambda^\bullet \;=\; b^{\xi/Q}, \tag{A.21}$$

which is the finite deterministic shift announced in (3.19).

Step (iv): the delta method and the limit. Insert (A.16) and the trivial identity $\sqrt{k}\,(\mathrm{W}) = \sqrt{k}\,(\mathrm{W})$ into the $\sqrt{k}$-rescaled version of (A.14), and recall (A.21):

$$\sqrt{k}\,[\log\widehat{\xi}^{\,\mathrm{pool}}_{\tau_n} - \log\xi_{\tau_n}] \;=\; (m(\gamma),\,1)\,\sqrt{k}\begin{pmatrix} \hat\gamma_n(\boldsymbol{\omega}_\gamma) - \gamma \\ \log\widehat{q}^{\,\star}_n(\tau_n \mid \boldsymbol{\omega}_q) - \log Q_X(\tau_n)\end{pmatrix} \;+\; \sqrt{k}\,(\mathrm{G}) \;+\; o_p(1). \tag{A.22}$$

By Proposition 3.2 and the continuous-mapping theorem applied to the deterministic linear functional $(g,x) \mapsto m(\gamma)\,g + x$, the leading term on the right of (A.22) converges in distribution to a univariate Gaussian with mean $(m(\gamma),1)\,\boldsymbol{\mu}(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)$ and variance $(m(\gamma),1)\,\boldsymbol{\Sigma}(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)\,(m(\gamma),1)^\top$. Reading off from (3.13) and (3.12),

$$(m(\gamma),1)\,\boldsymbol{\mu}(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q) \;=\; m(\gamma)\,\boldsymbol{\omega}_\gamma^\top\boldsymbol{B}_{\boldsymbol{c}} \;+\; \boldsymbol{\omega}_q^\top\boldsymbol{B}_{\boldsymbol{c}}^\star,$$

and the quadratic form expands to

$$\begin{aligned}(m(\gamma),1)\,\boldsymbol{\Sigma}(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)\,(m(\gamma),1)^\top \;&=\; m(\gamma)^2\,\boldsymbol{\omega}_\gamma^\top\boldsymbol{V}_{\boldsymbol{c}}\,\boldsymbol{\omega}_\gamma \;+\; 2\,m(\gamma)\,\boldsymbol{\omega}_\gamma^\top\,\mathrm{diag}(\boldsymbol{L}^\star)\,\boldsymbol{V}_{\boldsymbol{c}}\,\boldsymbol{\omega}_q \\ &\quad+\; \boldsymbol{\omega}_q^\top(\boldsymbol{V}_{\boldsymbol{c}} + \mathrm{diag}(\boldsymbol{L}^\star)^2\,\boldsymbol{V}_{\boldsymbol{c}})\boldsymbol{\omega}_q \;=\; \sigma^\xi(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)^2.\end{aligned}$$

The deterministic shift $\sqrt{k}\,(\mathrm{G})$ in (A.22) converges to $b^{\xi/Q}$ by (A.21). Combining via Slutsky's theorem,

$$\sqrt{k}\,[\log\widehat{\xi}^{\,\mathrm{pool}}_{\tau_n} - \log\xi_{\tau_n}] \;\xrightarrow{d}\; \mathcal{N}(\mu^\xi(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q),\,\sigma^\xi(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)^2), \tag{A.23}$$

with $\mu^{\xi}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) = m(\gamma)\, \boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c} + \boldsymbol{\omega}_q^\top \boldsymbol{B_c^\star} + b^{\xi/Q}$ as in (3.22): the random part contributes $m(\gamma)\, \boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c} + \boldsymbol{\omega}_q^\top \boldsymbol{B_c^\star}$ to the mean and the deterministic shift contributes $+b^{\xi/Q}$, both with the same sign as in (A.22).

To convert the log-ratio statement (A.23) into the ratio statement (3.20), observe that (A.23) forces $\sqrt{k}\,\log(\widehat{\xi}^{\text{pool}}_{\tau_n}/\xi_{\tau_n}) = O_p(1)$, equivalently $\log(\widehat{\xi}^{\text{pool}}_{\tau_n}/\xi_{\tau_n}) = o_p(1)$. A first-order Taylor expansion of $u \mapsto e^u - 1$ around $u = 0$ then gives the asymptotic relation

$$\frac{\widehat{\xi}^{\text{pool}}_{\tau_n}}{\xi_{\tau_n}} - 1 \;=\; \log\frac{\widehat{\xi}^{\text{pool}}_{\tau_n}}{\xi_{\tau_n}} \cdot \big(1 + o_p(1)\big), \tag{A.24}$$

since the next term in the expansion is squared and thus $O_p(1/k) = o_p(1/\sqrt{k})$. Multiplying through by $\sqrt{k}$ and applying Slutsky's theorem,

$$\sqrt{k}\left(\frac{\widehat{\xi}^{\text{pool}}_{\tau_n}}{\xi_{\tau_n}} - 1\right) \;=\; \sqrt{k}\,\log\frac{\widehat{\xi}^{\text{pool}}_{\tau_n}}{\xi_{\tau_n}} \;+\; o_p(1) \;\xrightarrow{d}\; \mathcal{N}(\mu^{\xi},\, \sigma^{\xi,2}),$$

which is (3.20). □

A.4 Proof of Propositions 3.4 and 3.5

We prove Proposition 3.5 (the general case). The equal-fraction Proposition 3.4 is recovered as the specialisation $\boldsymbol{L}^\star = L^\star \boldsymbol{1}$ at the end of the section.

We minimise the asymptotic variance $\sigma^{\xi}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2$ of (3.21) over deterministic weight vectors $\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q \in \mathbb{R}^m$ satisfying $\boldsymbol{\omega}_\gamma^\top \boldsymbol{1} = \boldsymbol{\omega}_q^\top \boldsymbol{1} = 1$, under the standing assumptions of Section 3.1. Recall $\boldsymbol{D} \coloneqq \operatorname{diag}(\boldsymbol{L}^\star)$ and the scalars a, b, c, Δ defined in (3.28).

Step (i): the change of variables. The factorisation (3.25),

$$\sigma^{\xi}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2 \;=\; \boldsymbol{u}^\top \boldsymbol{V_c} \boldsymbol{u} + \boldsymbol{\omega}_q^\top \boldsymbol{V_c} \boldsymbol{\omega}_q, \qquad \boldsymbol{u} \coloneqq m(\gamma)\, \boldsymbol{\omega}_\gamma + \boldsymbol{D}\, \boldsymbol{\omega}_q,$$

relies only on the diagonality of $\boldsymbol{V_c}$ and is verified by direct expansion. For $m(\gamma) \neq 0$, the linear map $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \mapsto (\boldsymbol{u}, \boldsymbol{\omega}_q)$ is a bijection of $\mathbb{R}^m \times \mathbb{R}^m$ onto itself, with inverse $\boldsymbol{\omega}_\gamma = m(\gamma)^{-1}(\boldsymbol{u} - \boldsymbol{D}\, \boldsymbol{\omega}_q)$. The sum-to-one constraints translate accordingly: from $\boldsymbol{\omega}_\gamma^\top \boldsymbol{1} = 1$ and $\boldsymbol{D}\boldsymbol{1} = \boldsymbol{L}^\star$,

$$\boldsymbol{u}^\top \boldsymbol{1} - \boldsymbol{\omega}_q^\top \boldsymbol{L}^\star \;=\; m(\gamma), \qquad \boldsymbol{\omega}_q^\top \boldsymbol{1} \;=\; 1. \tag{A.25}$$

The optimisation (3.25) reduces to a strictly convex quadratic programme on $\mathbb{R}^{2m}$, with the block-diagonal positive-definite Hessian $\operatorname{diag}(\boldsymbol{V_c}, \boldsymbol{V_c})$ and the two linear constraints (A.25).

Step (ii): the dual variables. Introduce Lagrange multipliers $\mu, \nu \in \mathbb{R}$ and form

$$\mathcal{L}(\boldsymbol{u}, \boldsymbol{\omega}_q; \mu, \nu) \;=\; \boldsymbol{u}^\top \boldsymbol{V_c} \boldsymbol{u} + \boldsymbol{\omega}_q^\top \boldsymbol{V_c} \boldsymbol{\omega}_q - \mu\, (\boldsymbol{u}^\top \boldsymbol{1} - \boldsymbol{\omega}_q^\top \boldsymbol{L}^\star - m(\gamma)) - \nu\, (\boldsymbol{\omega}_q^\top \boldsymbol{1} - 1).$$

The first-order conditions read

$$2\boldsymbol{V_c}\,\boldsymbol{u} \;=\; \mu\,\mathbf{1}, \qquad 2\boldsymbol{V_c}\,\boldsymbol{\omega}_q \;=\; \nu\,\mathbf{1} - \mu\,\boldsymbol{L}^{\star}. \tag{A.26}$$

Both equations admit the unique solutions

$$\boldsymbol{u} \;=\; (\mu/2)\,\boldsymbol{V_c}^{-1}\mathbf{1}, \qquad \boldsymbol{\omega}_q \;=\; (\nu/2)\,\boldsymbol{V_c}^{-1}\mathbf{1} - (\mu/2)\,\boldsymbol{V_c}^{-1}\boldsymbol{L}^{\star}, \tag{A.27}$$

displaying $\boldsymbol{u}$ in $\operatorname{span}\{\boldsymbol{V_c}^{-1}\mathbf{1}\}$ and $\boldsymbol{\omega}_q$ in $\operatorname{span}\{\boldsymbol{V_c}^{-1}\mathbf{1}, \boldsymbol{V_c}^{-1}\boldsymbol{L}^{\star}\}$ at the optimum.

Step (iii): solving for the multipliers. Substitute (A.27) into the constraints (A.25). Recall $a = \mathbf{1}^\top\boldsymbol{V_c}^{-1}\mathbf{1}$, $b = \mathbf{1}^\top\boldsymbol{V_c}^{-1}\boldsymbol{L}^{\star} = (\boldsymbol{L}^{\star})^\top\boldsymbol{V_c}^{-1}\mathbf{1}$ (using symmetry of $\boldsymbol{V_c}^{-1}$), and $c = (\boldsymbol{L}^{\star})^\top\boldsymbol{V_c}^{-1}\boldsymbol{L}^{\star}$:

$$\boldsymbol{u}^\top\mathbf{1} = (\mu/2)\,a, \qquad \boldsymbol{\omega}_q^\top\boldsymbol{L}^{\star} = (\nu/2)\,b - (\mu/2)\,c, \qquad \boldsymbol{\omega}_q^\top\mathbf{1} = (\nu/2)\,a - (\mu/2)\,b.$$

The constraints (A.25) become the linear system

$$\begin{pmatrix} a+c & -\,b \\ -\,b & a \end{pmatrix}\begin{pmatrix} \mu/2 \\ \nu/2 \end{pmatrix} \;=\; \begin{pmatrix} m(\gamma) \\ 1 \end{pmatrix}. \tag{A.28}$$

The coefficient matrix has determinant $a(a+c) - b^2 = \Delta$. Cauchy–Schwarz in the $\boldsymbol{V_c}^{-1}$-inner product gives $b^2 \le ac$, with equality iff $\boldsymbol{L}^{\star} \in \operatorname{span}\{\mathbf{1}\}$, that is, iff $\boldsymbol{L}^{\star} = L^{\star}\mathbf{1}$ for some scalar $L^{\star}$. Therefore $ac - b^2 \ge 0$ and, since $a > 0$,

$$\Delta \;=\; a^2 + (ac - b^2) \;>\; 0, \tag{A.29}$$

strictly, with $\Delta = a^2$ exactly under the equal-effective-fraction specialisation $\boldsymbol{L}^{\star} = L^{\star}\mathbf{1}$. Cramer's rule applied to (A.28) yields

$$\mu/2 \;=\; \frac{m(\gamma)\,a + b}{\Delta}, \qquad \nu/2 \;=\; \frac{a + c + m(\gamma)\,b}{\Delta}. \tag{A.30}$$

Step (iv): the primal solution. Inserting (A.30) into (A.27) gives the optimal $\boldsymbol{\omega}_q$ of (3.29). For $\boldsymbol{\omega}_\gamma$, invert the change of variables: $\boldsymbol{\omega}_\gamma^\star = m(\gamma)^{-1}(\boldsymbol{u}^\star - \boldsymbol{D}\,\boldsymbol{\omega}_q^\star)$. Using the diagonality identities $\boldsymbol{D}\boldsymbol{V_c}^{-1}\mathbf{1} = \boldsymbol{V_c}^{-1}\boldsymbol{L}^{\star}$ and $\boldsymbol{D}\boldsymbol{V_c}^{-1}\boldsymbol{L}^{\star} = \boldsymbol{V_c}^{-1}\boldsymbol{D}^2\mathbf{1}$ (both consequences of $\boldsymbol{V_c}, \boldsymbol{D}$ being diagonal and hence commuting), one obtains

$$\boldsymbol{u}^\star - \boldsymbol{D}\,\boldsymbol{\omega}_q^\star \;=\; \frac{m(\gamma)\,a + b}{\Delta}\,\boldsymbol{V_c}^{-1}\mathbf{1} \;-\; \frac{a + c + m(\gamma)\,b}{\Delta}\,\boldsymbol{V_c}^{-1}\boldsymbol{L}^{\star} \;+\; \frac{m(\gamma)\,a + b}{\Delta}\,\boldsymbol{V_c}^{-1}\boldsymbol{D}^2\mathbf{1},$$

which is (3.30) after the factor $m(\gamma)^{-1}$. The minimum variance follows from Lagrangian duality: at the optimum of a strictly convex quadratic minimised under linear constraints, the optimal value equals $\frac{1}{2}(\mu^\star\,d_1 + \nu^\star\,d_2)$ where $d_1 = m(\gamma)$, $d_2 = 1$ are the constraint right-hand sides. Substituting (A.30):

$$\sigma^\xi(\boldsymbol{\omega}_\gamma^{\xi,\mathrm{Var}}, \boldsymbol{\omega}_q^{\xi,\mathrm{Var}})^2 \;=\; \frac{m(\gamma)\,[m(\gamma)\,a + b] + [a + c + m(\gamma)\,b]}{\Delta} \;=\; \frac{(m(\gamma)^2 + 1)\,a + 2\,m(\gamma)\,b + c}{\Delta},$$

which is (3.31).

Step (v): the departure formula. A direct computation gives

$$\boldsymbol{\omega}_q^{\xi,\mathrm{Var}} - \boldsymbol{\omega}^{\mathrm{Var}} = \frac{(a+c+m(\gamma)\,b)\,\boldsymbol{V_c}^{-1}\mathbf{1} - (m(\gamma)\,a+b)\,\boldsymbol{V_c}^{-1}\boldsymbol{L}^\star}{\Delta} - \frac{\boldsymbol{V_c}^{-1}\mathbf{1}}{a}.$$

Bring to a common denominator $a\Delta$ and collect the $\boldsymbol{V_c}^{-1}\mathbf{1}$ coefficient: $a(a+c+m(\gamma)\,b)-\Delta = a^2+ac+m(\gamma)\,ab-a^2-ac+b^2 = b\,(m(\gamma)\,a+b)$. Hence

$$\boldsymbol{\omega}_q^{\xi,\mathrm{Var}} - \boldsymbol{\omega}^{\mathrm{Var}} = \frac{m(\gamma)\,a+b}{a\,\Delta}\left[b\,\boldsymbol{V_c}^{-1}\mathbf{1} - a\,\boldsymbol{V_c}^{-1}\boldsymbol{L}^\star\right] = -\frac{m(\gamma)\,a+b}{\Delta}\,\boldsymbol{V_c}^{-1}(\boldsymbol{L}^\star - \bar{L}^\star\mathbf{1}),$$

with $\bar{L}^\star = b/a$, which is (3.32).

Specialisation to equal effective fractions. Substituting $\boldsymbol{L}^\star = L^\star\mathbf{1}$ gives $b = L^\star a$, $c = (L^\star)^2 a$, $\Delta = a^2$. From (A.30):

$$\mu/2 = \frac{(m(\gamma)+L^\star)\,a}{a^2} = \frac{m(\gamma)+L^\star}{a},$$
$$\nu/2 = \frac{(1+L^{\star 2}+m(\gamma)\,L^\star)\,a}{a^2} = \frac{1+L^{\star 2}+m(\gamma)\,L^\star}{a},$$

and $\boldsymbol{\omega}_q^\star = (\nu/2-(\mu/2)L^\star)\,\boldsymbol{V_c}^{-1}\mathbf{1} = ((1+L^{\star 2}+m(\gamma)L^\star) - L^\star(m(\gamma)+L^\star))\boldsymbol{V_c}^{-1}\mathbf{1}/a = \boldsymbol{V_c}^{-1}\mathbf{1}/a = \boldsymbol{\omega}^{\mathrm{Var}}$. For the Hill-weight component, $\boldsymbol{u}^\star = (m(\gamma)+L^\star)\boldsymbol{V_c}^{-1}\mathbf{1}/a$ and $\boldsymbol{D}\boldsymbol{\omega}_q^\star = L^\star\boldsymbol{V_c}^{-1}\mathbf{1}/a$, so

$$\boldsymbol{\omega}_\gamma^\star = m(\gamma)^{-1}(\boldsymbol{u}^\star - \boldsymbol{D}\boldsymbol{\omega}_q^\star) = \boldsymbol{V_c}^{-1}\mathbf{1}/a = \boldsymbol{\omega}^{\mathrm{Var}}.$$

The minimum-variance value reduces to $[(m(\gamma)+L^\star)^2+1]/a$, recovering Proposition 3.4.

The case $m(\gamma)=0$. When $m(\gamma)=0$, the change-of-variables map is no longer a bijection: $\boldsymbol{u} = \boldsymbol{D}\boldsymbol{\omega}_q$ depends only on $\boldsymbol{\omega}_q$ and $\boldsymbol{\omega}_\gamma$ disappears from the variance. The factorisation reduces to $\sigma^\xi(\boldsymbol{\omega}_\gamma,\boldsymbol{\omega}_q)^2 = \boldsymbol{\omega}_q^\top\boldsymbol{V_c}(\boldsymbol{I}+\boldsymbol{D}^2)\boldsymbol{\omega}_q$, a strictly convex quadratic in $\boldsymbol{\omega}_q$ alone (since $\boldsymbol{V_c}(\boldsymbol{I}+\boldsymbol{D}^2)$ is the product of two positive-definite diagonal matrices, hence positive definite). Minimisation under $\boldsymbol{\omega}_q^\top\mathbf{1}=1$ gives, by the standard Lagrangian calculation for a single linear constraint,

$$\boldsymbol{\omega}_q^{\xi,\mathrm{Var}}\big|_{m(\gamma)=0} = \frac{(\boldsymbol{I}+\boldsymbol{D}^2)^{-1}\boldsymbol{V_c}^{-1}\mathbf{1}}{\mathbf{1}^\top(\boldsymbol{I}+\boldsymbol{D}^2)^{-1}\boldsymbol{V_c}^{-1}\mathbf{1}}, \tag{A.31}$$

with $\boldsymbol{\omega}_\gamma$ indeterminate. In the equal-effective-fraction case $\boldsymbol{D} = L^\star\boldsymbol{I}$, $(\boldsymbol{I}+\boldsymbol{D}^2)^{-1}$ is a scalar multiple of $\boldsymbol{I}$ and (A.31) reduces to $\boldsymbol{\omega}^{\mathrm{Var}}$; in general it differs from $\boldsymbol{\omega}^{\mathrm{Var}}$. □

A.5 Proof of Proposition 3.6

We prove existence, uniqueness, and the closed form (3.35) of the AMSE minimiser under the standing assumptions of Section 3.1 and $m(\gamma)\neq 0$. Recall the stacked notation $\boldsymbol{w} = (\boldsymbol{\omega}_\gamma^\top,\boldsymbol{\omega}_q^\top)^\top$, the bias gradient and variance Hessian $\boldsymbol{h},\boldsymbol{Q}$ of (3.34), the constraint matrix $\boldsymbol{A}\in\mathbb{R}^{2m\times 2}$ with columns $\mathbf{1}_g = (\mathbf{1}^\top, \mathbf{0}^\top)^\top$ and $\mathbf{1}_q = (\mathbf{0}^\top, \mathbf{1}^\top)^\top$, and the matrix $\boldsymbol{M} \coloneqq \boldsymbol{Q} + \boldsymbol{h}\boldsymbol{h}^\top$.

Step (i): rewriting the AMSE as a quadratic form. By construction $\mu^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) = \boldsymbol{h}^\top \boldsymbol{w} + b^{\xi/Q}$ and $\sigma^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2 = \boldsymbol{w}^\top \boldsymbol{Q} \boldsymbol{w}$, so

$$\begin{aligned} \mathrm{AMSE}^\xi(\boldsymbol{w}) &= (\boldsymbol{h}^\top \boldsymbol{w} + b^{\xi/Q})^2 + \boldsymbol{w}^\top \boldsymbol{Q} \boldsymbol{w} \\ &= \boldsymbol{w}^\top \boldsymbol{M}\, \boldsymbol{w} \;+\; 2\, b^{\xi/Q}\, \boldsymbol{h}^\top \boldsymbol{w} \;+\; (b^{\xi/Q})^2, \end{aligned}$$

recognising $(\boldsymbol{h}^\top \boldsymbol{w})^2 = \boldsymbol{w}^\top (\boldsymbol{h}\boldsymbol{h}^\top) \boldsymbol{w}$ and $\boldsymbol{M} = \boldsymbol{Q} + \boldsymbol{h}\boldsymbol{h}^\top$. The additive constant $(b^{\xi/Q})^2$ does not influence the argmin and is dropped below.

Step (ii): positive-definiteness of $\boldsymbol{M}$. $\boldsymbol{Q}$ is positive definite. Indeed, the variance factorisation $\boldsymbol{w}^\top \boldsymbol{Q} \boldsymbol{w} = \boldsymbol{u}^\top \boldsymbol{V_c} \boldsymbol{u} + \boldsymbol{\omega}_q^\top \boldsymbol{V_c} \boldsymbol{\omega}_q$ with $\boldsymbol{u} = m(\gamma)\boldsymbol{\omega}_\gamma + \boldsymbol{D}\boldsymbol{\omega}_q$ exhibits $\boldsymbol{Q}$ as the pull-back of the block-diagonal positive-definite matrix $\mathrm{diag}(\boldsymbol{V_c}, \boldsymbol{V_c})$ under the linear bijection $\boldsymbol{w} \mapsto (\boldsymbol{u}, \boldsymbol{\omega}_q)$ (valid for $m(\gamma) \neq 0$, see Appendix A.4). Hence $\boldsymbol{w}^\top \boldsymbol{Q} \boldsymbol{w} \geq 0$ with equality iff $\boldsymbol{u} = 0$ and $\boldsymbol{\omega}_q = 0$, which forces $\boldsymbol{w} = 0$. The rank-one positive semi-definite update $\boldsymbol{h}\boldsymbol{h}^\top$ preserves positive definiteness, so $\boldsymbol{M} = \boldsymbol{Q} + \boldsymbol{h}\boldsymbol{h}^\top$ is positive definite. Strict convexity of $\mathrm{AMSE}^\xi(\boldsymbol{w}) - (b^{\xi/Q})^2$ on $\mathbb{R}^{2m}$, and a fortiori on the closed affine constraint set $\{\boldsymbol{A}^\top \boldsymbol{w} = \mathbf{1}_2\}$, follows. Positive definiteness also makes the quadratic objective coercive, so a minimiser exists on this non-empty closed affine set.

Step (iii): existence and uniqueness via the Lagrangian. Form

$$\mathcal{L}(\boldsymbol{w}; \boldsymbol{\nu}) \;=\; \boldsymbol{w}^\top \boldsymbol{M}\, \boldsymbol{w} + 2\, b^{\xi/Q}\, \boldsymbol{h}^\top \boldsymbol{w} - \boldsymbol{\nu}^\top (\boldsymbol{A}^\top \boldsymbol{w} - \mathbf{1}_2),$$

with $\boldsymbol{\nu} \in \mathbb{R}^2$. The first-order conditions read

$$2\, \boldsymbol{M}\, \boldsymbol{w} + 2\, b^{\xi/Q}\, \boldsymbol{h} - \boldsymbol{A}\, \boldsymbol{\nu} \;=\; \mathbf{0}, \tag{A.32}$$

together with $\boldsymbol{A}^\top \boldsymbol{w} = \mathbf{1}_2$. From (A.32),

$$\boldsymbol{w} \;=\; \boldsymbol{M}^{-1} \Big[\tfrac{1}{2} \boldsymbol{A}\, \boldsymbol{\nu} - b^{\xi/Q}\, \boldsymbol{h} \Big]. \tag{A.33}$$

Imposing the constraint $\boldsymbol{A}^\top \boldsymbol{w} = \mathbf{1}_2$:

$$\tfrac{1}{2}\, \boldsymbol{A}^\top \boldsymbol{M}^{-1} \boldsymbol{A}\, \boldsymbol{\nu} - b^{\xi/Q}\, \boldsymbol{A}^\top \boldsymbol{M}^{-1} \boldsymbol{h} \;=\; \mathbf{1}_2.$$

The matrix $\boldsymbol{A}^\top \boldsymbol{M}^{-1} \boldsymbol{A} \in \mathbb{R}^{2\times 2}$ is positive definite: $\boldsymbol{M}^{-1}$ is PD, and $\boldsymbol{A}$ has full column rank since its two columns $\mathbf{1}_g, \mathbf{1}_q$ have disjoint nonzero supports. Hence

$$\boldsymbol{\nu} \;=\; 2\, (\boldsymbol{A}^\top \boldsymbol{M}^{-1} \boldsymbol{A})^{-1} (\mathbf{1}_2 \;+\; b^{\xi/Q}\, \boldsymbol{A}^\top \boldsymbol{M}^{-1} \boldsymbol{h}). \tag{A.34}$$

Substituting (A.34) back into (A.33) delivers the closed form (3.35). Strict convexity of the objective on the affine constraint set together with the linear FOC (A.32) forces this $\boldsymbol{w}$ to be the unique constrained minimiser.

Step (iv): equivalence with the block FOC system. Expand (A.32) block-by-block. The product $\boldsymbol{M}\boldsymbol{w} = \boldsymbol{Q}\boldsymbol{w} + (\boldsymbol{h}^\top \boldsymbol{w})\,\boldsymbol{h} = \boldsymbol{Q}\boldsymbol{w} + (\mu^\xi - b^{\xi/Q})\boldsymbol{h}$, using $\mu^\xi = \boldsymbol{h}^\top \boldsymbol{w} + b^{\xi/Q}$. Substituting, (A.32) becomes

$$2\,\boldsymbol{Q}\boldsymbol{w} \;+\; 2\,(\mu^\xi - b^{\xi/Q})\,\boldsymbol{h} \;+\; 2\,b^{\xi/Q}\,\boldsymbol{h} \;=\; \boldsymbol{A}\,\boldsymbol{\nu} \quad\Longleftrightarrow\quad \boldsymbol{Q}\boldsymbol{w} \;+\; \mu^\xi\,\boldsymbol{h} \;=\; \tfrac{1}{2}\,\boldsymbol{A}\,\boldsymbol{\nu}.$$

Writing $\boldsymbol{\nu} = (\lambda_\gamma, \lambda_q)^\top$ and reading off the upper and lower blocks of size m (these λ's are Lagrange multipliers, not the second-order bias-rate constants λ_j):

$$\begin{aligned} [\boldsymbol{Q}\boldsymbol{w}]_{1,\dots,m} &= m(\gamma)^2\,\boldsymbol{V_c}\,\boldsymbol{\omega}_\gamma + m(\gamma)\,\boldsymbol{D}\,\boldsymbol{V_c}\,\boldsymbol{\omega}_q,\\ [\boldsymbol{Q}\boldsymbol{w}]_{m+1,\dots,2m} &= m(\gamma)\,\boldsymbol{D}\,\boldsymbol{V_c}\,\boldsymbol{\omega}_\gamma + (\boldsymbol{V_c} + \boldsymbol{D}^2\boldsymbol{V_c})\,\boldsymbol{\omega}_q, \end{aligned}$$

$[\boldsymbol{h}]_{1,\dots,m} = m(\gamma)\boldsymbol{B_c}$, $[\boldsymbol{h}]_{m+1,\dots,2m} = \boldsymbol{B}^\star_{\boldsymbol{c}}$, $[\boldsymbol{A}\boldsymbol{\nu}]_{1,\dots,m} = \lambda_\gamma\boldsymbol{1}$, $[\boldsymbol{A}\boldsymbol{\nu}]_{m+1,\dots,2m} = \lambda_q\boldsymbol{1}$. The block decomposition reproduces (3.36). Uniqueness from strict convexity, combined with the consistency identity $\mu^\xi = m(\gamma)\,\boldsymbol{\omega}_\gamma^\top \boldsymbol{B_c} + \boldsymbol{\omega}_q^\top \boldsymbol{B}^\star_{\boldsymbol{c}} + b^{\xi/Q}$ at the minimiser, completes the equivalence. □

A.6 Proof of Corollary 3.7

We prove the asymptotic-coverage statement of Corollary 3.7 under the standing assumptions of Section 3.1, the undersmoothing condition (3.40), and the consistency assumptions (3.41) on the estimated weight pair $(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q)$, the plug-in covariance $\widehat{\boldsymbol{V}}_{\boldsymbol{c}}$, and the plug-in QB Jacobian $\hat{m}$. We retain the affine sum-to-one constraints $\hat{\boldsymbol{\omega}}_\gamma^\top\boldsymbol{1} = \hat{\boldsymbol{\omega}}_q^\top\boldsymbol{1} = 1$ *exactly*, which holds for the variance- and AMSE-optimal plug-ins of Propositions 2.17, 3.5 and 3.6 by construction. All logarithms and square roots below are understood on high-probability events on which they are well defined; the right-tail setup, the expectile–quantile equivalence, and the consistency of the variance plug-in make the complements of these events negligible.

The proof composes four steps: (i) recast the conclusion of Theorem 3.3 on the log scale; (ii) lift the deterministic-weight CLT to the estimated weights via the exact sum-to-one constraint and Slutsky; (iii) substitute the plug-in variance to obtain the asymptotic pivot under undersmoothing; (iv) transport coverage to the original scale by strict monotonicity of exp.

Step (i): log-scale recasting. By the QB factorisation (3.8),

$$\log \widehat{\xi}^{\,\mathrm{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \;=\; \log \psi\big(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)\big) \;+\; \log \widehat{q}^{\,\star}_n(\tau_n \mid \boldsymbol{\omega}_q). \tag{A.35}$$

The population expectile is positive eventually by the right-heavy-tailed setup and Proposition 2.10. The local Weissman factors, and hence their weighted-geometric pool, are positive with probability tending one, so (A.35) is legitimate on an event whose probability tends to one. Theorem 3.3 delivers $\sqrt{k}\,(\widehat{\xi}^{\,\mathrm{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)/\xi_{\tau_n} - 1) \xrightarrow{d} \mathcal{N}(\mu^\xi, \sigma^{\xi,2})$. The displayed log-CLT (A.23) in the proof of Theorem 3.3 (Appendix A.3) is the same statement on the log scale,

$$\sqrt{k}\,\big(\log \widehat{\xi}^{\,\mathrm{pool}}_{\tau_n}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) - \log \xi_{\tau_n}\big) \;\xrightarrow{d}\; \mathcal{N}\big(\mu^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q),\, \sigma^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2\big). \tag{A.36}$$

Under undersmoothing (3.40), every per-sample bias rate $\lambda_j = \lim \sqrt{k_j} A(n_j/k_j)$ is zero. The formulas (3.14) and Theorem 2.16 then give $\boldsymbol{B_c} = \mathbf{0}$ and $\boldsymbol{B_c^\star} = \mathbf{0}$. Moreover, the compatibility identity

$$\Lambda^\bullet = \sqrt{c_j S}\,\lambda_j e^{\rho L_j^\star}, \qquad j = 1, \dots, m,$$

forces $\Lambda^\bullet = 0$, and therefore $b^{\xi/Q} = -c(\gamma,\rho)\Lambda^\bullet = 0$ by (3.19). Hence $\mu^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) = 0$ and the limit in (A.36) reduces to $\mathcal{N}(0, \sigma^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2)$.

Step (ii): estimated-weight lift. We show that replacing $(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$ by $(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q)$ in (A.35) changes the left-hand side of (A.36) by $o_p(1)$. Decompose this change into a Hill and a Weissman contribution,

$$\begin{aligned}\Delta_n^\gamma &:= \log\psi(\hat\gamma_n(\hat{\boldsymbol{\omega}}_\gamma)) - \log\psi(\hat\gamma_n(\boldsymbol{\omega}_\gamma)),\\ \Delta_n^q &:= \log\widehat{q}_n^\star(\tau_n \mid \hat{\boldsymbol{\omega}}_q) - \log\widehat{q}_n^\star(\tau_n \mid \boldsymbol{\omega}_q).\end{aligned}$$

Hill contribution. Linearity of the pooled Hill in its weights ((3.4)) gives

$$\hat\gamma_n(\hat{\boldsymbol{\omega}}_\gamma) - \hat\gamma_n(\boldsymbol{\omega}_\gamma) \;=\; (\hat{\boldsymbol{\omega}}_\gamma - \boldsymbol{\omega}_\gamma)^\top \hat{\boldsymbol{\gamma}}_n \;=\; (\hat{\boldsymbol{\omega}}_\gamma - \boldsymbol{\omega}_\gamma)^\top (\hat{\boldsymbol{\gamma}}_n - \gamma\mathbf{1}),$$

the deterministic offset $\gamma\mathbf{1}$ vanishing by the exact sum-to-one identity $(\hat{\boldsymbol{\omega}}_\gamma - \boldsymbol{\omega}_\gamma)^\top\mathbf{1} = 1 - 1 = 0$ on both the plug-in and the limit weight vectors. The per-sample Hill CLT (within-sample component of (3.10)) gives $\sqrt{k_j}(\hat\gamma_j - \gamma) = O_p(1)$, hence $\sqrt{k}(\hat\gamma_j - \gamma) = \sqrt{k/k_j}\,O_p(1) = O_p(1)$ because $k/k_j \to c_j \sum_i c_i^{-1} < \infty$ by (2.12). Therefore $\sqrt{k}(\hat{\boldsymbol{\gamma}}_n - \gamma\mathbf{1}) = O_p(1)$, and the Cauchy–Schwarz inequality combined with $\hat{\boldsymbol{\omega}}_\gamma - \boldsymbol{\omega}_\gamma = o_p(1)$ yields

$$\sqrt{k}\,(\hat\gamma_n(\hat{\boldsymbol{\omega}}_\gamma) - \hat\gamma_n(\boldsymbol{\omega}_\gamma)) \;=\; (\hat{\boldsymbol{\omega}}_\gamma - \boldsymbol{\omega}_\gamma)^\top \sqrt{k}(\hat{\boldsymbol{\gamma}}_n - \gamma\mathbf{1}) \;=\; o_p(1)\cdot O_p(1) \;=\; o_p(1). \tag{A.37}$$

The map $\log\psi$ is continuously differentiable on $(0,1)$ with derivative $m(\,\cdot\,)$ defined in (3.15). Let

$$E_n^\gamma = \{\hat\gamma_n(\boldsymbol{\omega}_\gamma) \in (0,1),\ \hat\gamma_n(\hat{\boldsymbol{\omega}}_\gamma) \in (0,1)\}.$$

By consistency, $\hat\gamma_n(\hat{\boldsymbol{\omega}}_\gamma), \hat\gamma_n(\boldsymbol{\omega}_\gamma) \xrightarrow{\mathbb{P}} \gamma$, so $\mathbb{P}(E_n^\gamma) \to 1$. On this event, the mean value theorem yields a random $\tilde\gamma_n$ between the two pooled values such that

$$\Delta_n^\gamma \;=\; m(\tilde\gamma_n)\,[\hat\gamma_n(\hat{\boldsymbol{\omega}}_\gamma) - \hat\gamma_n(\boldsymbol{\omega}_\gamma)].$$

Squeeze gives $\tilde\gamma_n \xrightarrow{\mathbb{P}} \gamma$ and, by continuity of $m(\,\cdot\,)$, $m(\tilde\gamma_n) \xrightarrow{\mathbb{P}} m(\gamma)$. Combining with (A.37), $\sqrt{k}\,\Delta_n^\gamma = O_p(1)\cdot o_p(1) = o_p(1)$.

Weissman contribution. Weighted-geometric linearity of $\log\widehat{q}_n^\star$ in $\boldsymbol{\omega}_q$ ((3.5)) gives

$$\Delta_n^q \;=\; (\hat{\boldsymbol{\omega}}_q - \boldsymbol{\omega}_q)^\top\big(\log\hat{\boldsymbol{q}}_n^\star - \log Q_X(\tau_n)\mathbf{1}\big),$$

where the deterministic offset $\log Q_X(\tau_n)\mathbf{1}$ cancels via the sum-to-one identity $(\hat{\boldsymbol{\omega}}_q - \boldsymbol{\omega}_q)^\top\mathbf{1} = 0$. The within-sample Weissman component of (3.10) delivers $\sqrt{k_j}(\log\hat q_j^\star - \log Q_X(\tau_n)) = O_p(1)$

under $L_j(\tau_n) \to L_j^\star \in \mathbb{R}$, and the same $\sqrt{k/k_j} = O(1)$ rescaling promotes this to $\sqrt{k}(\log \hat{\boldsymbol{q}}_n^\star - \log Q_X(\tau_n)\mathbf{1}) = O_p(1)$. Hence

$$\sqrt{k}\,\Delta_n^q \;=\; (\hat{\boldsymbol{\omega}}_q - \boldsymbol{\omega}_q)^\top \sqrt{k}\big(\log \hat{\boldsymbol{q}}_n^\star - \log Q_X(\tau_n)\mathbf{1}\big) \;=\; o_p(1)\cdot O_p(1) \;=\; o_p(1). \tag{A.38}$$

Assembly. Combining (A.36) (under undersmoothing, limit-mean zero) with $\sqrt{k}\,\Delta_n^\gamma + \sqrt{k}\,\Delta_n^q = o_p(1)$ and Slutsky's theorem,

$$\sqrt{k}\,\big(\log \widehat{\xi}^{\text{pool}}_{\tau_n}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q) - \log \xi_{\tau_n}\big) \;\xrightarrow{d}\; \mathcal{N}(0,\, \sigma^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2). \tag{A.39}$$

Step (iii): self-normalisation. The variance (3.21) is a polynomial in $(\boldsymbol{V_c}, \boldsymbol{L}^\star, m, \boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$, and $\sigma^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2 > 0$ because $\boldsymbol{V_c}$ is positive definite, $\boldsymbol{\omega}_q^\top \mathbf{1} = 1$ forces $\boldsymbol{\omega}_q \neq \mathbf{0}$, and the structurally non-negative contribution $\boldsymbol{\omega}_q^\top \boldsymbol{V_c} \boldsymbol{\omega}_q$ to (3.21) is strictly positive. The plug-in substitutes $\widehat{\boldsymbol{V}}_{\boldsymbol{c}}, L_{j,n} := L_j(\tau_n), \hat{m}, \hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q$ for their population analogues. The log-rate plug-in $L_{j,n} = L_j(\tau_n) = \log(k_j/(n_j(1-\tau_n)))$ is deterministic and converges to $L_j^\star$ by the chapter setup of Section 3.1; the remaining four convergences are the assumptions (3.41). The continuous-mapping theorem applied first to the polynomial (3.21) and then to $\sqrt{\cdot}$ on $(0,\infty)$ yields

$$\hat{\sigma}^\xi(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q) \;\xrightarrow{\mathbb{P}}\; \sigma^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \;>\; 0.$$

In particular, $\mathbb{P}\{\hat{\sigma}^\xi(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q) > 0\} \to 1$, so the division below is well defined with probability tending one. Slutsky's theorem applied to the ratio of (A.39) and this consistent variance estimator produces the asymptotic pivot

$$Z_n \;:=\; \frac{\sqrt{k}\,\big(\log \widehat{\xi}^{\text{pool}}_{\tau_n}(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q) - \log \xi_{\tau_n}\big)}{\hat{\sigma}^\xi(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q)} \;\xrightarrow{d}\; \mathcal{N}(0,1). \tag{A.40}$$

For every $\alpha \in (0,1)$, $\mathbb{P}(|Z_n| \le z_{1-\alpha/2}) \to 1-\alpha$. Inverting the inequality $|Z_n| \le z_{1-\alpha/2}$ for $\log \xi_{\tau_n}$ recovers exactly the event $\log \xi_{\tau_n} \in I_n^{\log}$ with $I_n^{\log}$ defined in (3.42), so $\mathbb{P}(\log \xi_{\tau_n} \in I_n^{\log}) \to 1-\alpha$.

Step (iv): transport to the original scale. The map $x \mapsto e^x$ is strictly increasing and continuous on $\mathbb{R}$, so for the random closed interval $I_n^{\log} = [a_n, b_n]$,

$$\log \xi_{\tau_n} \in [a_n, b_n] \quad\Longleftrightarrow\quad \xi_{\tau_n} \in [e^{a_n}, e^{b_n}] \;=\; \exp(I_n^{\log}) \;=\; I_n.$$

The two events coincide pointwise on the underlying probability space, hence $\mathbb{P}(\xi_{\tau_n} \in I_n) = \mathbb{P}(\log \xi_{\tau_n} \in I_n^{\log}) \to 1-\alpha$. □

A.7 Proof of Theorem 3.9 and Corollary 3.10

We prove the high-expectile homogeneity diagnostic of Theorem 3.9 and the central calibration of Corollary 3.10 under the standing assumptions of Section 3.1, with $\hat{\boldsymbol{\Sigma}}_{\Xi,n} \xrightarrow{\mathbb{P}} \boldsymbol{\Sigma}_\Xi$ and $\boldsymbol{\Sigma}_\Xi$ positive definite. The argument composes six steps: (i) the per-sample QB log-expectile CLT, obtained by the single-sample specialisation of Appendix A.3; (ii) aggregation across the independent samples and rescaling to the common $\sqrt{k}$ scale, which yields the joint limit (3.44) with covariance (3.45) and mean (3.46); (iii) the GLS algebra reducing the statistic to a quadratic form in a single asymptotically Gaussian vector; (iv) the resulting noncentral $\chi^2_{m-1}(\delta)$ limit; (v) the two regimes in which $\delta = 0$; (vi) consistency under the alternative. Throughout, $S := \sum_{i=1}^m c_i^{-1}$, so that $(\boldsymbol{V_c})_{j,j} = \gamma^2\, c_j\, S$ by (2.14) and $k/k_j \to c_j\, S$ by (2.12), as established in Appendix A.2.

Step (i): per-sample QB log-expectile CLT. Fix $j \in \{1, \ldots, m\}$. The per-sample QB log-expectile (3.43) decomposes exactly, as in (A.14) with $m = 1$ and unit weight, into

$$\log \widehat{\xi}_{\tau_n, j} - \log \xi_{\tau_n} \;=\; \underbrace{\big[\log \psi(\hat{\gamma}_j(k_j)) - \log \psi(\gamma)\big]}_{(\mathrm{H})_j} \;+\; \underbrace{\big[\log \hat{q}^\star_j(\tau_n \mid k_j) - \log Q_X(\tau_n)\big]}_{(\mathrm{W})_j} \;+ (\mathrm{G}),$$

with $(\mathrm{G}) = \log \psi(\gamma) + \log Q_X(\tau_n) - \log \xi_{\tau_n}$ the deterministic expectile–quantile gap, which is common to all samples. The Taylor linearisation of Step (ii) of Appendix A.3 gives $(\mathrm{H})_j = m(\gamma)(\hat{\gamma}_j(k_j) - \gamma) + O_p(k_j^{-1})$, so on the $\sqrt{k_j}$ scale

$$\sqrt{k_j}\,\big(\log \widehat{\xi}_{\tau_n, j} - \log \xi_{\tau_n}\big) \;=\; (m(\gamma),\, 1)\,\sqrt{k_j} \begin{pmatrix} \hat{\gamma}_j(k_j) - \gamma \\ \log \hat{q}^\star_j(\tau_n \mid k_j) - \log Q_X(\tau_n) \end{pmatrix} \;+\; \sqrt{k_j}\,(\mathrm{G}) \;+\; o_p(1).$$

The within-sample 2-vector obeys the joint CLT (3.10) (proved as (A.11)), with mean $\boldsymbol{\mu}^\star_j = \big(b^H_j,\, L^\star_j\, b^H_j + b^R_j\big)^\top$, $b^R_j = -\lambda_j\, \Psi_\rho(e^{L^\star_j})$ (the order-statistic bias being zero by (A.8)), and covariance $\gamma^2\big(1, L^\star_j;\, L^\star_j, 1 + (L^\star_j)^2\big)$. The deterministic gap is controlled by the second-order expansion (A.17) and the regular variation of A used in (A.19). The moment remainder is $o(k_j^{-1/2})$ because $k_j \asymp k$ and (3.3) is in force. Consequently,

$$\sqrt{k_j}\,(\mathrm{G}) \;=\; -c(\gamma, \rho)\,\sqrt{k_j}\,A\big((1 - \tau_n)^{-1}\big) + o(1) \;\longrightarrow\; -c(\gamma, \rho)\,\lambda_j\,(e^{L^\star_j})^\rho \;=:\; g_j.$$

Reading the row $(m(\gamma), 1)$ through the within-sample covariance gives the scalar variance $\gamma^2[(m(\gamma) + L^\star_j)^2 + 1]$ (the computation of (3.21) at $m = 1$, unit weight), so Slutsky's theorem yields

$$\sqrt{k_j}\,\big(\log \widehat{\xi}_{\tau_n, j} - \log \xi_{\tau_n}\big) \;\xrightarrow{d}\; \mathcal{N}\Big(\beta^\circ_j,\; \gamma^2\big[(m(\gamma) + L^\star_j)^2 + 1\big]\Big), \qquad \beta^\circ_j \;=\; (m(\gamma) + L^\star_j)\, b^H_j + b^R_j + g_j. \tag{A.41}$$

Step (ii): aggregation and rescaling to $\sqrt{k}$. The m samples are independent (3.1), so the scalars in (A.41) are independent across j. Joint convergence follows by the Cramér–Wold device: for any fixed vector $\boldsymbol{a}$, the corresponding linear combination is a sum of independent per-sample terms whose characteristic functions factor, and the marginal limits in (A.41) therefore combine into the stated Gaussian law. Thus the vector $\sqrt{k}(\hat{\boldsymbol{\Xi}}_n - \log \xi_{\tau_n}\mathbf{1})$ has independent limiting coordinates obtained by multiplying (A.41) by $\sqrt{k/k_j} \to \sqrt{c_j S}$. The jth limiting variance is

$$c_j S\, \gamma^2\big[(m(\gamma) + L^\star_j)^2 + 1\big] \;=\; \big[(m(\gamma) + L^\star_j)^2 + 1\big]\,(\boldsymbol{V_c})_{j,j},$$

which is the jth diagonal entry of $\boldsymbol{\Sigma}_\Xi$ in (3.45); the off-diagonal entries vanish by independence, so the limiting covariance is exactly $\boldsymbol{\Sigma}_\Xi$. The jth limiting mean is $\sqrt{c_j S}\,\beta_j^\circ$. Splitting β_j° as in (A.41) and using the $\sqrt{k}$-rescaled bias identities of Appendix A.2,

$$\sqrt{c_j S}\, b_j^H = (\boldsymbol{B_c})_j, \qquad \sqrt{c_j S}\,(L_j^\star\, b_j^H + b_j^R) = (\boldsymbol{B_c^\star})_j,$$

the first from Theorem 2.16 and the second from (3.14), we obtain

$$\sqrt{c_j S}\,\beta_j^\circ \;=\; \sqrt{c_j S}\,[m(\gamma)\, b_j^H + (L_j^\star\, b_j^H + b_j^R)] \;+\; \sqrt{c_j S}\, g_j \;=\; m(\gamma)\,(\boldsymbol{B_c})_j + (\boldsymbol{B_c^\star})_j \;+\; \sqrt{c_j S}\, g_j.$$

For the last term, the per-j identity (A.20) reads $\sqrt{c_j S}\,\lambda_j\,(e^{L_j^\star})^\rho = \Lambda^\bullet$, so

$$\sqrt{c_j S}\, g_j \;=\; -c(\gamma,\rho)\,\sqrt{c_j S}\,\lambda_j\,(e^{L_j^\star})^\rho \;=\; -c(\gamma,\rho)\,\Lambda^\bullet \;=\; b^{\xi/Q}$$

by (3.19), *independently of j*. Hence $\sqrt{c_j S}\,\beta_j^\circ = m(\gamma)\,(\boldsymbol{B_c})_j + (\boldsymbol{B_c^\star})_j + b^{\xi/Q}$, i.e. the limiting mean vector is $\boldsymbol{\beta} = m(\gamma)\boldsymbol{B_c} + \boldsymbol{B_c^\star} + b^{\xi/Q}\mathbf{1}$ of (3.46). This proves (3.44).

Step (iii): GLS reduction to a quadratic form. Write $\boldsymbol{Y}_n := \sqrt{k}(\hat{\boldsymbol{\Xi}}_n - \log\xi_{\tau_n}\mathbf{1})$, so $\boldsymbol{Y}_n \xrightarrow{d} \boldsymbol{Y} \sim \mathcal{N}(\boldsymbol{\beta}, \boldsymbol{\Sigma}_\Xi)$ by Step (ii). Introduce the random oblique projection

$$\boldsymbol{P}_n \;:=\; \boldsymbol{I} - \frac{\mathbf{1}\,\mathbf{1}^\top\hat{\boldsymbol{\Sigma}}_{\Xi,n}^{-1}}{\mathbf{1}^\top\hat{\boldsymbol{\Sigma}}_{\Xi,n}^{-1}\mathbf{1}},$$

which satisfies $\boldsymbol{P}_n\mathbf{1} = \mathbf{0}$ and, from the definition of $\hat\nu_n$ in (3.47), $\hat{\boldsymbol{\Xi}}_n - \hat\nu_n\mathbf{1} = \boldsymbol{P}_n\hat{\boldsymbol{\Xi}}_n = \boldsymbol{P}_n(\hat{\boldsymbol{\Xi}}_n - \log\xi_{\tau_n}\mathbf{1}) = k^{-1/2}\,\boldsymbol{P}_n\boldsymbol{Y}_n$, the second equality because $\boldsymbol{P}_n\mathbf{1} = \mathbf{0}$ removes the common centring. Substituting into (3.47),

$$\Lambda_n^\xi \;=\; k\,\big(k^{-1/2}\boldsymbol{P}_n\boldsymbol{Y}_n\big)^\top\hat{\boldsymbol{\Sigma}}_{\Xi,n}^{-1}\big(k^{-1/2}\boldsymbol{P}_n\boldsymbol{Y}_n\big) \;=\; \boldsymbol{Y}_n^\top\,\boldsymbol{P}_n^\top\hat{\boldsymbol{\Sigma}}_{\Xi,n}^{-1}\boldsymbol{P}_n\,\boldsymbol{Y}_n. \tag{A.42}$$

By hypothesis $\hat{\boldsymbol{\Sigma}}_{\Xi,n} \xrightarrow{\mathbb{P}} \boldsymbol{\Sigma}_\Xi$, so $\boldsymbol{P}_n \xrightarrow{\mathbb{P}} \boldsymbol{P} := \boldsymbol{I} - \mathbf{1}\mathbf{1}^\top\boldsymbol{\Sigma}_\Xi^{-1}/(\mathbf{1}^\top\boldsymbol{\Sigma}_\Xi^{-1}\mathbf{1})$ and $\boldsymbol{P}_n^\top\hat{\boldsymbol{\Sigma}}_{\Xi,n}^{-1}\boldsymbol{P}_n \xrightarrow{\mathbb{P}} \boldsymbol{M} := \boldsymbol{P}^\top\boldsymbol{\Sigma}_\Xi^{-1}\boldsymbol{P}$, both by the continuous-mapping theorem. A direct computation, writing $\boldsymbol{w} := \boldsymbol{\Sigma}_\Xi^{-1}\mathbf{1}$ and $s := \mathbf{1}^\top\boldsymbol{\Sigma}_\Xi^{-1}\mathbf{1} = \mathbf{1}^\top\boldsymbol{w}$, gives the standard GLS identity

$$\boldsymbol{M} \;=\; \boldsymbol{P}^\top\boldsymbol{\Sigma}_\Xi^{-1}\boldsymbol{P} \;=\; \boldsymbol{\Sigma}_\Xi^{-1} \;-\; \frac{\boldsymbol{\Sigma}_\Xi^{-1}\mathbf{1}\,\mathbf{1}^\top\boldsymbol{\Sigma}_\Xi^{-1}}{\mathbf{1}^\top\boldsymbol{\Sigma}_\Xi^{-1}\mathbf{1}}, \tag{A.43}$$

because the two first-order cross terms in the expansion of $(\boldsymbol{I} - \boldsymbol{w}\mathbf{1}^\top/s)\boldsymbol{\Sigma}_\Xi^{-1}(\boldsymbol{I} - \mathbf{1}\boldsymbol{w}^\top/s)$ each contribute $-\boldsymbol{w}\boldsymbol{w}^\top/s$, while the second-order term contributes $+\boldsymbol{w}\boldsymbol{w}^\top/s$ (using $\mathbf{1}^\top\boldsymbol{w} = s$ and symmetry of $\boldsymbol{\Sigma}_\Xi^{-1}$), leaving $\boldsymbol{\Sigma}_\Xi^{-1} - \boldsymbol{w}\boldsymbol{w}^\top/s$. The matrix $\boldsymbol{M}$ is symmetric positive semi-definite and satisfies $\boldsymbol{M}\mathbf{1} = \boldsymbol{w} - \boldsymbol{w}(s/s) = \mathbf{0}$. By (A.42), Slutsky's theorem, and the continuous-mapping theorem applied to the (jointly continuous) quadratic-form map $(\boldsymbol{A}, \boldsymbol{y}) \mapsto \boldsymbol{y}^\top\boldsymbol{A}\boldsymbol{y}$,

$$\Lambda_n^\xi \;\xrightarrow{d}\; \boldsymbol{Y}^\top\boldsymbol{M}\,\boldsymbol{Y}, \qquad \boldsymbol{Y} \sim \mathcal{N}(\boldsymbol{\beta}, \boldsymbol{\Sigma}_\Xi). \tag{A.44}$$

Step (iv): the noncentral chi-square limit. Standardise $\boldsymbol{Z} := \boldsymbol{\Sigma}_{\Xi}^{-1/2}\boldsymbol{Y} \sim \mathcal{N}(\boldsymbol{\Sigma}_{\Xi}^{-1/2}\boldsymbol{\beta}, \boldsymbol{I}_m)$, where $\boldsymbol{\Sigma}_{\Xi}^{-1/2}$ is the symmetric positive-definite square root of $\boldsymbol{\Sigma}_{\Xi}^{-1}$. Then $\boldsymbol{Y}^\top \boldsymbol{M}\boldsymbol{Y} = \boldsymbol{Z}^\top \widetilde{\boldsymbol{M}}\boldsymbol{Z}$ with

$$\widetilde{\boldsymbol{M}} \;:=\; \boldsymbol{\Sigma}_{\Xi}^{1/2}\boldsymbol{M}\,\boldsymbol{\Sigma}_{\Xi}^{1/2} \;=\; \boldsymbol{I}_m - \frac{\boldsymbol{v}\boldsymbol{v}^\top}{\boldsymbol{v}^\top\boldsymbol{v}}, \qquad \boldsymbol{v} := \boldsymbol{\Sigma}_{\Xi}^{-1/2}\mathbf{1},$$

using (A.43) and $\boldsymbol{\Sigma}_{\Xi}^{1/2}\boldsymbol{w} = \boldsymbol{\Sigma}_{\Xi}^{-1/2}\mathbf{1} = \boldsymbol{v}$, $s = \boldsymbol{v}^\top\boldsymbol{v}$. The matrix $\widetilde{\boldsymbol{M}}$ is the orthogonal projection onto $\boldsymbol{v}^\perp$: it is symmetric, idempotent, and of rank $m-1$. A quadratic form $\boldsymbol{Z}^\top\widetilde{\boldsymbol{M}}\boldsymbol{Z}$ in a unit-covariance Gaussian $\boldsymbol{Z} \sim \mathcal{N}(\boldsymbol{\mu}_Z, \boldsymbol{I}_m)$ with $\widetilde{\boldsymbol{M}}$ a rank-r orthogonal projection is noncentral χ_r^2 with noncentrality $\boldsymbol{\mu}_Z^\top\widetilde{\boldsymbol{M}}\boldsymbol{\mu}_Z$. Here $r = m-1$ and $\boldsymbol{\mu}_Z = \boldsymbol{\Sigma}_{\Xi}^{-1/2}\boldsymbol{\beta}$, so

$$\Lambda_n^{\xi} \xrightarrow{d} \chi_{m-1}^2(\delta), \qquad \delta \;=\; \boldsymbol{\beta}^\top\boldsymbol{\Sigma}_{\Xi}^{-1/2}\widetilde{\boldsymbol{M}}\boldsymbol{\Sigma}_{\Xi}^{-1/2}\boldsymbol{\beta} \;=\; \boldsymbol{\beta}^\top\boldsymbol{M}\boldsymbol{\beta} \;=\; \boldsymbol{\beta}^\top\boldsymbol{\Sigma}_{\Xi}^{-1}\boldsymbol{\beta} - \frac{(\mathbf{1}^\top\boldsymbol{\Sigma}_{\Xi}^{-1}\boldsymbol{\beta})^2}{\mathbf{1}^\top\boldsymbol{\Sigma}_{\Xi}^{-1}\mathbf{1}},$$

the last equality by (A.43). This is (3.49). Since $\widetilde{\boldsymbol{M}}$ is positive semi-definite, $\delta \geq 0$, with $\delta = 0$ iff $\widetilde{\boldsymbol{M}}\boldsymbol{\Sigma}_{\Xi}^{-1/2}\boldsymbol{\beta} = \mathbf{0}$, i.e. $\boldsymbol{\Sigma}_{\Xi}^{-1/2}\boldsymbol{\beta} \in \operatorname{span}\{\boldsymbol{v}\} = \operatorname{span}\{\boldsymbol{\Sigma}_{\Xi}^{-1/2}\mathbf{1}\}$, equivalently $\boldsymbol{\beta} \in \operatorname{span}\{\mathbf{1}\}$.

Step (v): the two regimes with $\delta = 0$. *(i) Undersmoothing.* Under (3.40), $\lambda_j = 0$ for every j, so the per-sample $\sqrt{k_j}$-biases $b_j^H = \lambda_j/(1-\rho)$ and $b_j^R = -\lambda_j\Psi_\rho(e^{L_j^\star})$ both vanish (the order-statistic bias being identically zero), and $\Lambda^\bullet = 0$, whence $\boldsymbol{B_c} = \boldsymbol{B_c^\star} = \mathbf{0}$ and $b^{\xi/Q} = 0$. Therefore $\boldsymbol{\beta} = \mathbf{0} \in \operatorname{span}\{\mathbf{1}\}$ and $\delta = 0$.

(ii) Equal-fraction distributed inference. Here $L_j^\star = L^\star$ is sample-free. The two per-sample $\sqrt{k_j}$-biases are proportional to λ_j with coefficients depending only on the common parameters $(\gamma, \rho, L^\star)$: $b_j^H = \lambda_j/(1-\rho)$ and $b_j^R = -\lambda_j\,\Psi_\rho(e^{L^\star})$, the order-statistic bias being identically zero. The $\sqrt{k}$-rescaling factor satisfies, by (A.20) at $L_j^\star = L^\star$,

$$\sqrt{c_j S}\,\lambda_j \;=\; \Lambda^\bullet\,(e^{L^\star})^{-\rho}, \qquad \text{sample-free.}$$

Consequently $(\boldsymbol{B_c})_j = \sqrt{c_j S}\,b_j^H = \Lambda^\bullet(e^{L^\star})^{-\rho}/(1-\rho)$ and $(\boldsymbol{B_c^\star})_j = \sqrt{c_j S}\,(L^\star b_j^H + b_j^R) = \Lambda^\bullet(e^{L^\star})^{-\rho}\big[L^\star/(1-\rho) - \Psi_\rho(e^{L^\star})\big]$ are both sample-free, and $b^{\xi/Q}\mathbf{1}$ is collinear with $\mathbf{1}$ by construction. Hence every coordinate of $\boldsymbol{\beta} = m(\gamma)\boldsymbol{B_c} + \boldsymbol{B_c^\star} + b^{\xi/Q}\mathbf{1}$ equals the common constant

$$\beta_0 \;=\; \Lambda^\bullet(e^{L^\star})^{-\rho}\left[\frac{m(\gamma) + L^\star}{1-\rho} - \Psi_\rho(e^{L^\star})\right] + b^{\xi/Q},$$

so $\boldsymbol{\beta} = \beta_0\mathbf{1} \in \operatorname{span}\{\mathbf{1}\}$ and $\delta = 0$. This is the expectile-level counterpart of the bias-collinearity that renders the tail-index test Theorem 2.20 asymptotically central under the same equal-fraction distributed regime.

Step (vi): consistency under diagnostic alternatives. This final step proves the consistency assertion when (3.47) is read as a diagnostic outside the chapter's common-marginal null model. It therefore uses the additional alternative-side hypotheses stated in Theorem 3.9, rather than the common-target CLT of Steps (i)–(v). Write $\boldsymbol{\Xi}_n^\star := (\log\xi_{\tau_n,1}, \ldots, \log\xi_{\tau_n,m})^\top$ for the vector of *population* per-sample log-expectiles, and abbreviate $\boldsymbol{D}_n := \hat{\boldsymbol{\Sigma}}_{\Xi,n}^{-1}$, with induced norm $\|\boldsymbol{x}\|_{\boldsymbol{D}_n}^2 := \boldsymbol{x}^\top\boldsymbol{D}_n\boldsymbol{x}$. Assume $\hat{\boldsymbol{\Xi}}_n - \boldsymbol{\Xi}_n^\star = O_p(k^{-1/2})$, $\|\boldsymbol{D}_n\| = O_p(1)$, and $\mathbb{P}\{\lambda_{\min}(\boldsymbol{D}_n) > \eta\} \to 1$

for some $\eta > 0$. These conditions say, respectively, that the local log-expectile estimators remain root-k consistent for their own targets and that the plug-in precision matrix does not degenerate. Because the GLS residual $\hat{\boldsymbol{\Xi}}_n - \hat{\nu}_n \mathbf{1} = \boldsymbol{P}_n \hat{\boldsymbol{\Xi}}_n$ is, by definition of $\hat{\nu}_n$, the minimiser of $c \mapsto \|\hat{\boldsymbol{\Xi}}_n - c\mathbf{1}\|^2_{\boldsymbol{D}_n}$, the statistic is the squared GLS distance of $\hat{\boldsymbol{\Xi}}_n$ from $\operatorname{span}\{\mathbf{1}\}$:

$$\Lambda_n^{\xi} \;=\; k \min_{c\in\mathbb{R}} \|\hat{\boldsymbol{\Xi}}_n - c\mathbf{1}\|^2_{\boldsymbol{D}_n}.$$

On the event $\lambda_{\min}(\boldsymbol{D}_n) > \eta$, the norm $\|\cdot\|_{\boldsymbol{D}_n}$ is well defined and the triangle inequality gives, with $S_1 := \operatorname{span}\{\mathbf{1}\}$ and $d_{\boldsymbol{D}_n}(\boldsymbol{x}, S_1) := \inf_{\boldsymbol{s}\in S_1} \|\boldsymbol{x} - \boldsymbol{s}\|_{\boldsymbol{D}_n}$ (while d denotes Euclidean distance),

$$\sqrt{\Lambda_n^{\xi}} \;=\; \sqrt{k}\, d_{\boldsymbol{D}_n}(\hat{\boldsymbol{\Xi}}_n, S_1) \;\geq\; \sqrt{k}\, d_{\boldsymbol{D}_n}(\boldsymbol{\Xi}_n^{\star}, S_1) \;-\; \sqrt{k}\, \|\hat{\boldsymbol{\Xi}}_n - \boldsymbol{\Xi}_n^{\star}\|_{\boldsymbol{D}_n}.$$

Moreover $d_{\boldsymbol{D}_n}(\boldsymbol{\Xi}_n^{\star}, S_1) \geq \sqrt{\lambda_{\min}(\boldsymbol{D}_n)}\, d(\boldsymbol{\Xi}_n^{\star}, S_1)$, since the Rayleigh quotient bound applies to every vector $\boldsymbol{\Xi}_n^{\star} - c\mathbf{1}$. Therefore, on the same event,

$$\sqrt{\Lambda_n^{\xi}} \;\geq\; \lambda_{\min}(\boldsymbol{D}_n)^{1/2} \sqrt{k}\, \min_{c} \|\boldsymbol{\Xi}_n^{\star} - c\mathbf{1}\| \;-\; \sqrt{k}\, \|\hat{\boldsymbol{\Xi}}_n - \boldsymbol{\Xi}_n^{\star}\|_{\boldsymbol{D}_n}.$$

The subtracted term is $O_p(1)$ since $\sqrt{k}(\hat{\boldsymbol{\Xi}}_n - \boldsymbol{\Xi}_n^{\star}) = O_p(1)$ and $\|\boldsymbol{D}_n\| = O_p(1)$. Under the stated alternative $\sqrt{k}\, \min_c \|\boldsymbol{\Xi}_n^{\star} - c\mathbf{1}\| \to \infty$, while $\lambda_{\min}(\boldsymbol{D}_n)$ is bounded away from zero in probability. The first term on the right therefore diverges in probability and dominates the $O_p(1)$ correction, so $\Lambda_n^{\xi} \xrightarrow{\mathbb{P}} \infty$. □

A.8 Proof of Proposition 2.14

We prove the thesis-level plug-in CLT used in Chapter 4. The argument is the log-scale core of Davison et al. (2023, Proposition C.6), with the normalising sequence written as a generic a_n and the deterministic population extrapolation error imposed explicitly.

Let

$$\ell_n = \log\left(\frac{1-\tau_n}{1-\tau_n'}\right),$$

so $\ell_n \to \infty$ by $(1-\tau_n')/(1-\tau_n) \to 0$, and $\ell_n/a_n \to 0$ by the assumption $a_n/\ell_n \to \infty$. Work on the event on which $\widehat{\xi}^{I}_{\tau_n} > 0$; this event has probability tending one by assumption, and arbitrary definitions on its complement do not affect the weak limit. On this event,

$$\widehat{\xi}^{\star}_{\tau_n'} = \exp(\ell_n \hat{\gamma}_n)\, \widehat{\xi}^{I}_{\tau_n}.$$

Define the log-relative extrapolation error

$$Z_n := \log \frac{\widehat{\xi}^{\star}_{\tau_n'}}{\xi_{\tau_n'}}.$$

A direct decomposition gives

$$Z_n = \ell_n(\hat{\gamma}_n - \gamma) + \log \frac{\widehat{\xi}^{I}_{\tau_n}}{\xi_{\tau_n}} - \left\{\log \xi_{\tau_n'} - \log \xi_{\tau_n} - \gamma \ell_n\right\}. \tag{A.45}$$

Multiplying (A.45) by a_n/ℓ_n yields

$$\frac{a_n}{\ell_n} Z_n = a_n(\hat{\gamma}_n - \gamma) + \frac{a_n}{\ell_n} \log \frac{\widehat{\xi}^{I}_{\tau_n}}{\xi_{\tau_n}} - \frac{a_n}{\ell_n} \left\{ \log \xi_{\tau'_n} - \log \xi_{\tau_n} - \gamma \ell_n \right\}.$$

The first term converges in distribution to Γ by the joint input limit (2.11). The third term converges to zero by the deterministic extrapolation-error assumption. For the middle term, the same input limit gives

$$\frac{\widehat{\xi}^{I}_{\tau_n}}{\xi_{\tau_n}} - 1 = O_p(a_n^{-1}) = o_p(1).$$

Therefore

$$\log \frac{\widehat{\xi}^{I}_{\tau_n}}{\xi_{\tau_n}} = O_p(a_n^{-1}),$$

by the Taylor expansion $\log(1+x) = x + O(x^2)$ for $x = o_p(1)$, and so

$$\frac{a_n}{\ell_n} \log \frac{\widehat{\xi}^{I}_{\tau_n}}{\xi_{\tau_n}} = O_p(\ell_n^{-1}) = o_p(1).$$

Slutsky's theorem gives the log-scale limit

$$\frac{a_n}{\ell_n} \log \frac{\widehat{\xi}^{\star}_{\tau'_n}}{\xi_{\tau'_n}} \xrightarrow{d} \Gamma. \tag{A.46}$$

It remains only to pass from the log-relative error to the relative error in the statement of Proposition 2.14. From (A.46), $Z_n = O_p(\ell_n/a_n) = o_p(1)$. Hence

$$\exp(Z_n) - 1 - Z_n = O_p(Z_n^2), \qquad \frac{a_n}{\ell_n}\{\exp(Z_n) - 1 - Z_n\} = O_p\left(\frac{\ell_n}{a_n}\right) = o_p(1).$$

Since $\exp(Z_n) - 1 = \widehat{\xi}^{\star}_{\tau'_n}/\xi_{\tau'_n} - 1$, combining this display with (A.46) proves the stated relative-error convergence. □

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